DOP-C02

DEVOPS ENGINEER



COURSE SLIDES · WSQ

DevOps Engineering on AWS

WSQ Course Code: TGS-2023040474

Conducted by Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

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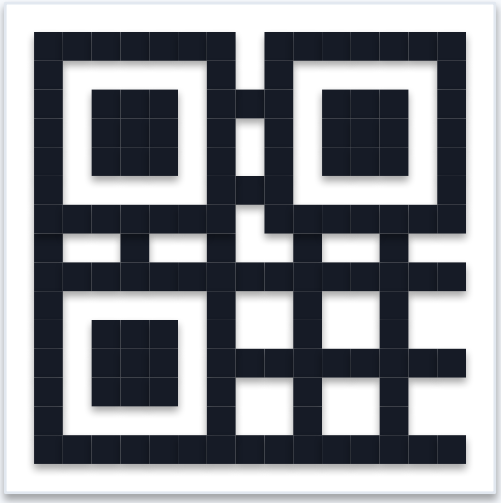
Digital Attendance (Mandatory)

Take the SSG digital attendance at three points on every funded day:

AM	PM	ASSESSMENT
Start of the morning session	Start of the afternoon session	Before the Day 5 assessment

1. The trainer/administrator displays the SSG digital-attendance QR code from the portal.
2. Scan the QR code with your mobile phone camera and submit your attendance.
3. It is mandatory for AM, PM and the Assessment sessions on WSQ-funded courses.

SCAN TO SUBMIT ATTENDANCE



SSG digital-attendance QR — displayed live from the portal by the trainer

Minimum 75% attendance on the SSG Digital Attendance record is required to be eligible for assessment and funding.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

QUALIFICATIONS
PhD; ACTA/DACE-certified adult educator.

EXPERTISE
AWS cloud architecture, DevOps, CI/CD automation, Infrastructure as Code and cloud security.

DELIVERS
WSQ courses on AWS, DevOps engineering, cloud and software engineering.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with AWS, DevOps or automation (if any).
- What you want to automate or improve on AWS after this course.

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HOUSEKEEPING

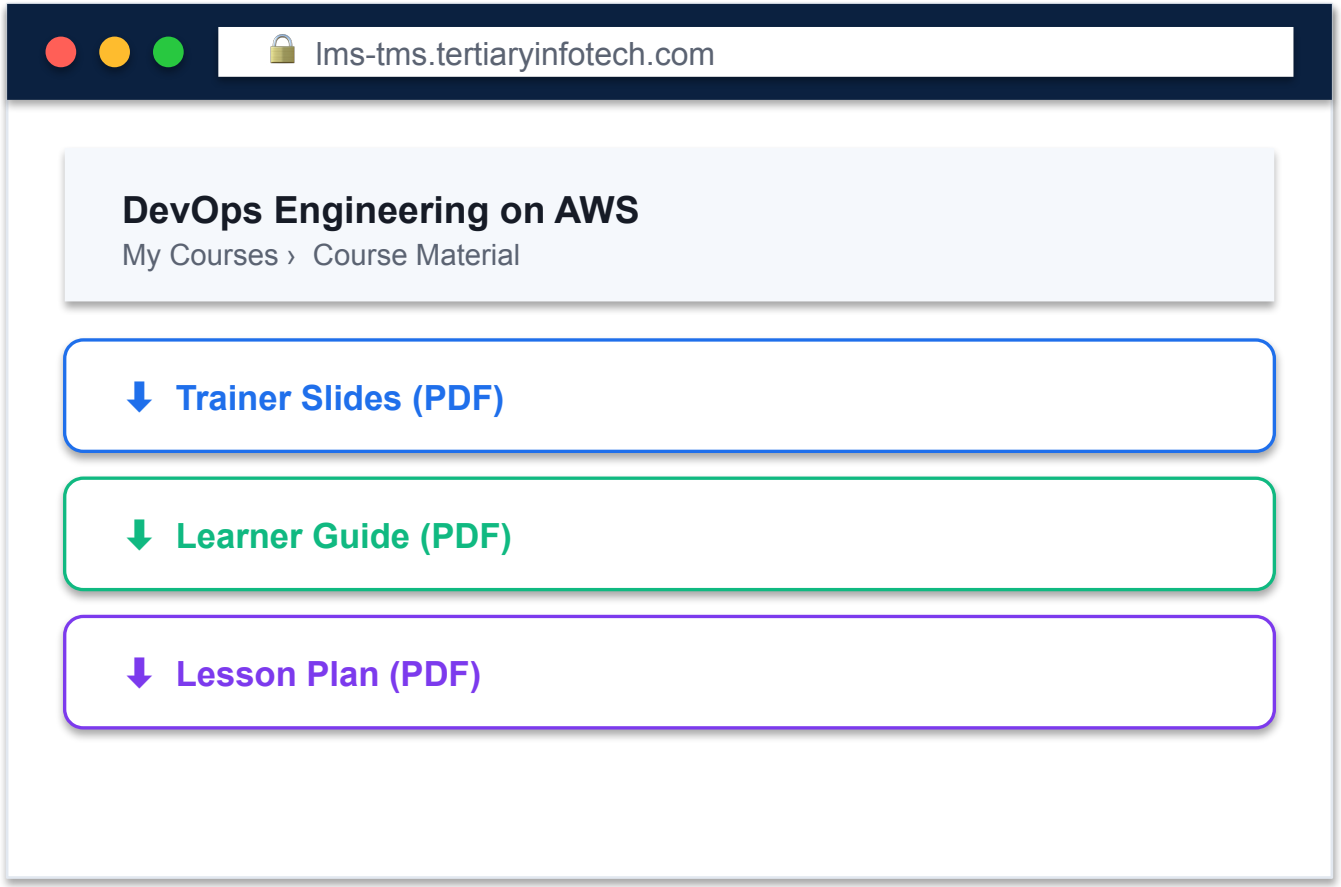
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required for funding.

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COURSE PORTAL

Download Your Course Material



1 Sign in
Log in to lms-tms.tertiaryinfotech.com with your enrolled account.

2 Open your course
Go to My Courses and select DevOps Engineering on AWS.

3 Course Material
Open the Course Material tab for this course.

4 Download
Download the Trainer Slides, Learner Guide and Lesson Plan for the open-book assessment.

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CRITERIA FOR FUNDING

Funding & Eligibility

1

Attendance

Minimum 75% attendance on the SSG Digital Attendance record.

2

Assessment

Complete the assessment and be assessed as ‘Competent’.

3

Open book

The slides and Learner Guide may be used during the assessment.

4

Course portal

All materials, attendance and assessment are on lms-tms.tertiaryinfotech.com.

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COURSE SCHEDULE · 40 HOURS (37 INSTRUCTIONAL + 3 ASSESSMENT)

Lesson Plan — 5 Days · 5 Hands-On Labs per Day (25 Labs)

1

Day 1
Foundations & SDLC Automation (Labs 1–5)

2

Day 2
Configuration Management & IaC (Labs 6–10)

3

Day 3
Resilient Cloud Solutions (Labs 11–15)

4

Day 4
Monitoring, Logging & Incident Response (Labs 16–20)

5

Day 5
Security, Compliance & Capstone (Labs 21–25) + Final Assessment

6

Daily timing
9:30 am – 6:00 pm · 30-minute lunch · tea breaks within

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1

Develop an AWS CI/CD pipeline proof of concept.

2

LO2

Identify the technical and practical requirements in an AWS cloud infrastructure setup.

3

LO3

Develop a resilient cloud solution based on AWS cloud components.

4

LO4

Implement regular AWS monitoring and perform necessary troubleshooting.

5

LO5

Demonstrate solutions to incidents and business problems on the AWS cloud.

6

LO6

Prepare a technical blueprint to address security and compliance.

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TSC MAPPING · K1–K7 · A1–A6 · LO1–LO6

Skills Framework Alignment (ICT-DES-4006-1.1)

Domain	Learning Outcome	Knowledge	Ability
01 SDLC Automation	LO1	K1	A1
02 Configuration Management and IaC	LO2	K3	A3
03 Resilient Cloud Solutions	LO3	K2, K4	A2
04 Monitoring and Logging	LO4	K5	A6
05 Incident and Event Response	LO5	K7	A5
06 Security and Compliance	LO6	K6	A4

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BEFORE THE ASSESSMENT

Briefing for Assessment

- Place phones and other materials under the table or on the floor unless the assessment is open book.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

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FINAL ASSESSMENT

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on AWS DevOps tasks, 2 hours, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Conducted on Day 5 after the training content (3 assessment hours in total — WA 1 hour + PP 2 hours).
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- An appeal process is available if required.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

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FOUNDATIONS

DevOps Engineering on AWS

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THE MINDSET

What is DevOps?

1

Culture + practice

DevOps unites development and operations to deliver value faster and more reliably.

2

Automate everything

Build, test, deploy and infrastructure become code — repeatable and reviewable.

3

Fast, safe feedback

Small, frequent changes with monitoring make failures cheap and recoverable.

4

Shared ownership

Teams own their service end to end — you build it, you run it.

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CULTURE FRAMEWORK

The CALMS Model

1

Culture

Collaboration and shared responsibility across dev and ops.

2

Automation

Remove manual, error-prone steps from the delivery pipeline.

3

Lean

Small batches, fast flow, eliminate waste.

4

Measurement

Metrics and feedback drive continuous improvement.

5

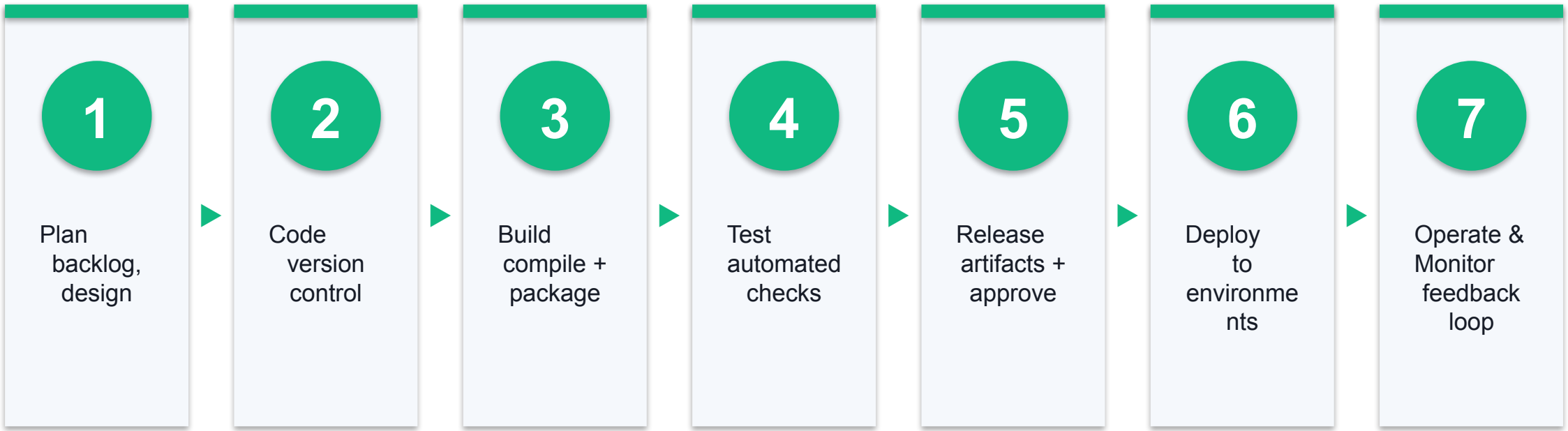
Sharing

Transparency, blameless learning and reusable patterns.

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A CONTINUOUS LOOP

The DevOps Lifecycle



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OUTCOMES THAT MATTER

Measuring DevOps — The DORA Metrics

1

Deployment frequency

How often you ship to production — elite teams deploy on demand.

2

Lead time for changes

Commit → production time — minutes/hours, not weeks.

3

Change failure rate

% of deployments causing a failure — lower is healthier.

4

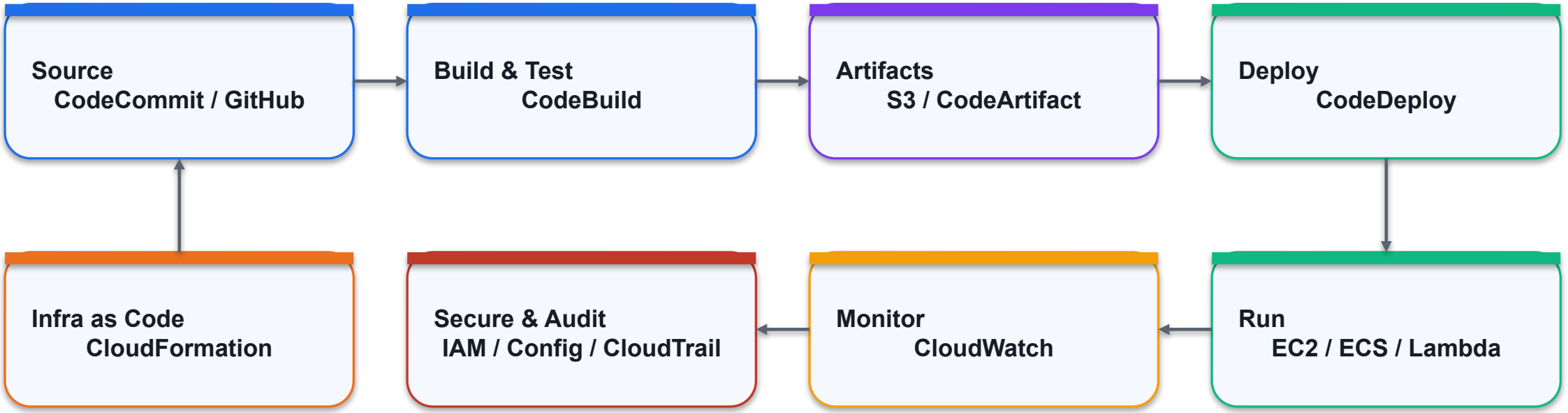
Mean time to recover

How fast you restore service after an incident — minimise it.

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END-TO-END DELIVERY

The AWS DevOps Toolchain



- CodePipeline orchestrates the top row; IaC, monitoring and security wrap the whole lifecycle.

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SIX PILLARS

AWS Well-Architected Framework

1

Operational Excellence

Run and monitor systems; improve processes continuously.

2

Security

Protect data, systems and assets with least-privilege and audit.

3

Reliability

Recover from failure automatically; meet availability targets.

4

Performance Efficiency

Use resources efficiently as demand and technology change.

5

Cost Optimisation

Deliver business value at the lowest price point.

6

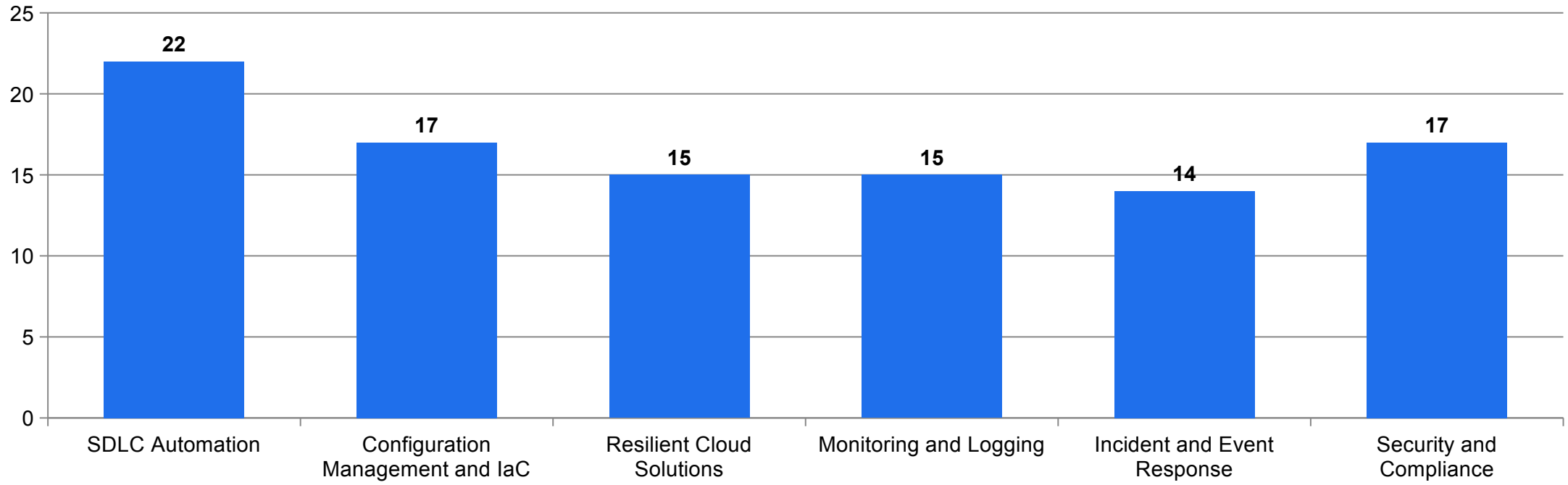
Sustainability

Minimise the environmental impact of running workloads.

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HOW THE COURSE IS WEIGHTED

Exam / Skills Domain Weightings



WHAT THE DATA SHOWS

SDLC Automation (22%) and the two 17% domains — IaC and Security & Compliance — carry the most weight; plan revision time accordingly.

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WHO SECURES WHAT

AWS Shared Responsibility Model

AWS responsibility

- AWS — security OF the cloud
  - Physical data centres & hardware
  - Global network & virtualisation layer
  - Managed-service infrastructure
  - Availability Zones and Regions

Your responsibility

- You — security IN the cloud
  - IAM, identities and permissions
  - OS, network & firewall configuration
  - Application code & data encryption
  - Patching guest OS and dependencies

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THE SERVICES YOU'LL USE

Core AWS Building Blocks

1

Compute

EC2, ECS/EKS, Lambda, Auto Scaling.

2

Networking

VPC, subnets, security groups, ELB, Route 53.

3

Storage & data

S3, EBS, RDS, DynamoDB.

4

Identity & security

IAM, KMS, Secrets Manager, Config.

5

Developer tools

CodePipeline, CodeBuild, CodeDeploy, CloudFormation.

6

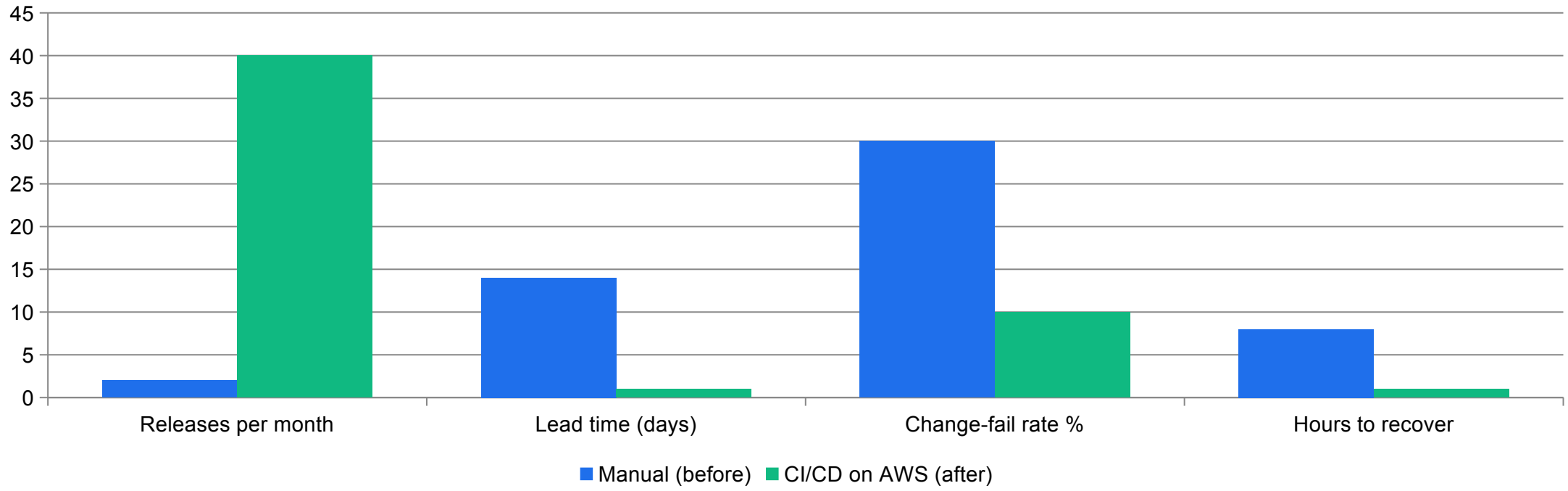
Observability

CloudWatch, X-Ray, CloudTrail, EventBridge.

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THE BUSINESS CASE

Why CI/CD — Release Frequency & Lead Time



WHAT THE DATA SHOWS

Automating the pipeline turns a few risky monthly releases into many small, low-risk ones that recover in hours, not days.

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WHY IT MATTERS

Build, test, deploy — automatically and safely.

The six AWS DevOps domains follow the delivery lifecycle: automate the SDLC, codify infrastructure, make it resilient, observe it, respond to events, and secure it.

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DORA Metrics — The Delivery/Stability Balance

Move faster

- Throughput
  - Deployment frequency — how often you ship
  - Lead time for changes — commit to production
  - Elite: on-demand deploys, <1 day lead time

Stay stable

- Stability
  - Change failure rate — % of deploys that hurt
  - Time to restore service — MTTR after failure
  - Elite: <15% CFR, restore in <1 hour

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HOW TEAMS BRANCH

Version Control Strategy — Trunk-Based vs Long-Lived Branches

1

Trunk-based

Short-lived branches merged to trunk daily — keeps integration continuous.

2

Feature flags

Ship unfinished work dark; decouple deploy from release.

3

Gitflow (heavier)

Release/develop branches — more ceremony, slower integration.

4

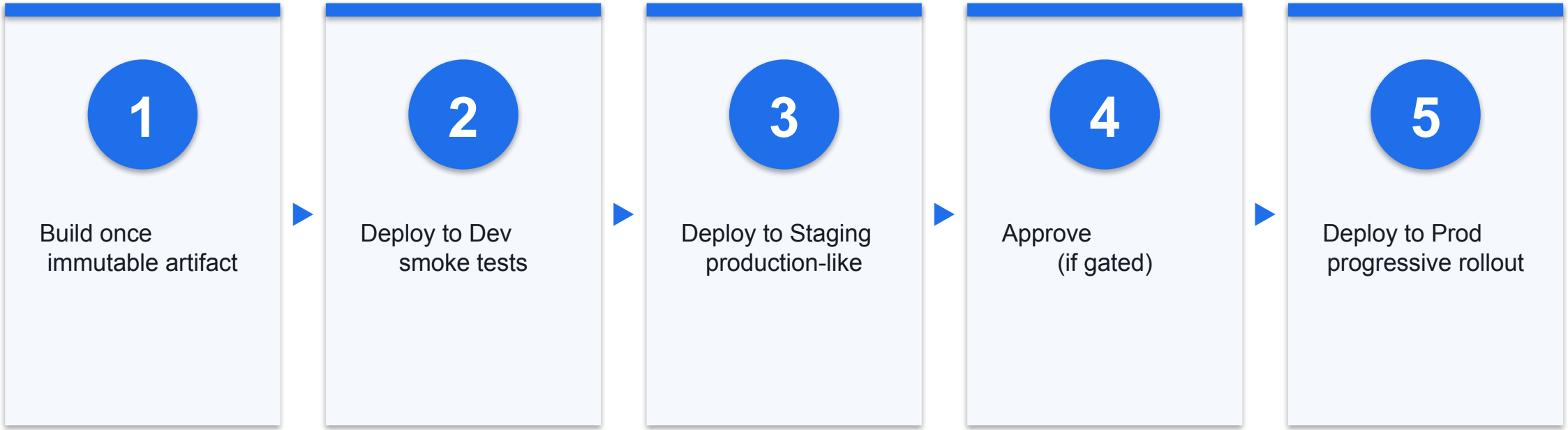
Branch protection

Required reviews and green checks before merge, enforced in GitHub.

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ONE ARTIFACT, MANY ENVIRONMENTS

Promotion Across Environments



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SOURCE INTEGRATION

GitHub via AWS CodeConnections — The Modern Source Path

1

Managed connection

Connect GitHub/GitHub Enterprise/Bitbucket/GitLab without long-lived Git credentials.

2

Webhook triggers

A push starts the pipeline instantly — no polling latency.

3

Keep your workflow

Existing pull requests, reviews and branch protection still apply.

4

CodeCommit optional

AWS-native Git remains available, but is not required for new work.

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A CLOUD-NATIVE MINDSET

Immutable Infrastructure — Cattle, Not Pets

1

Replace, don't patch

Deploy a new, tested image instead of mutating running servers.

2

No configuration drift

Every instance is built the same way from the same artifact.

3

Fast, safe rollback

Roll back by launching the previous known-good image.

4

Bake with a pipeline

EC2 Image Builder / golden AMIs produce reproducible images.

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DEVOPS MEETS FINOPS

Cost Awareness in DevOps — Build It In

1

Right-size & auto-scale

Match capacity to demand; scale to zero where you can.

2

Tag everything

Cost allocation tags make spend visible per team/service.

3

Clean up as code

Tear-down steps in every lab/pipeline stop orphaned resources.

4

Budgets & alarms

AWS Budgets alert before spend surprises the finance team.

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SECURITY IS EVERYONE'S JOB

DevSecOps — Shift Security Left

1

Scan in the pipeline

SAST/SCA on code and dependencies; scan IaC templates.

2

Least-privilege pipelines

Scope build and deploy roles; no secrets in source.

3

Policy as code

Guardrails as SCPs and Config rules, versioned like code.

4

Continuous compliance

Detect and remediate drift automatically, always-on.

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MATCH THE RUNTIME TO THE WORKLOAD

Choosing Compute — EC2 vs ECS/EKS vs Lambda

Option	Best for	Ops overhead	Scaling
EC2 + ASG	Lift-and-shift, full OS control	Higher (patching, AMIs)	Instances via ASG
ECS / Fargate	Containerised services	Medium (Fargate = serverless)	Tasks/services
EKS	Kubernetes-standard workloads	Higher (cluster ops)	Pods via HPA
Lambda	Event-driven, spiky, short tasks	Lowest	Automatic per request

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01

DOMAIN 01

SDLC Automation

Exam weighting 22% · CI/CD pipelines · automated testing · artifacts · deployment strategies

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DELIVERY AUTOMATION

SDLC Automation



- Automating the software-delivery lifecycle on AWS

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DOMAIN 01 · WEIGHTING 22%

Key Concepts — SDLC Automation

1

Continuous Integration

Developers push often; CodeBuild runs the build and tests on every commit so bugs are found early, not at release.

2

Continuous Delivery / Deployment

Every green build is releasable; CodePipeline and CodeDeploy move it through dev → test → prod with approvals.

3

The AWS developer tools

CodeCommit / CodeBuild / CodeDeploy / CodePipeline / CodeArtifact / CodeGuru compose the managed CI/CD toolchain.

4

buildspec & appspec

buildspec.yml tells CodeBuild what to build and test; appspec.yml tells CodeDeploy what and how to deploy.

5

Deployment strategies

In-place, rolling, blue/green and canary trade release speed against blast radius and rollback safety.

6

Artifacts

Build outputs are versioned in S3 / CodeArtifact so any release is reproducible and traceable.

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THE MANAGED CI/CD TOOLCHAIN

The AWS Developer Tools

1

CodeCommit

Managed, private Git repositories (or bring GitHub).

2

CodeBuild

Fully managed build & test service driven by buildspec.yml.

3

CodeDeploy

Automated deployment to EC2, on-prem, ECS and Lambda.

4

CodePipeline

Orchestrates Source → Build → Deploy with approvals.

5

CodeArtifact

Secure artifact & package repository.

6

CodeGuru

ML-powered code reviews and profiler.

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KNOW THE DIFFERENCE

CI vs CD vs Continuous Deployment

Integrate

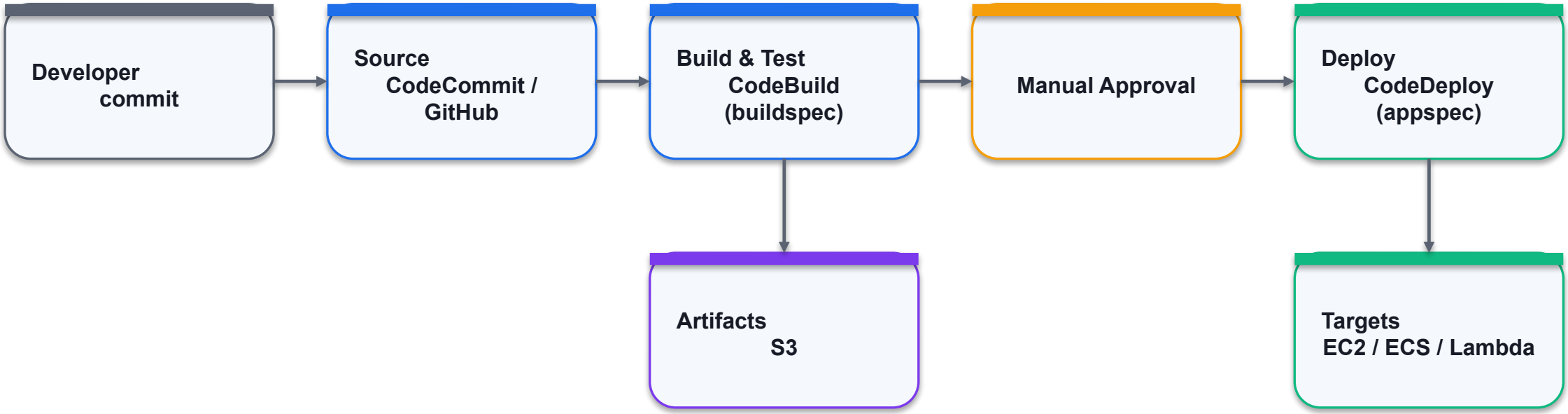
- Continuous Integration
  - Merge & test every commit automatically
  - CodeBuild runs unit tests on push
  - Fast feedback, fewer merge conflicts

Deliver / Deploy

- Continuous Delivery / Deployment
  - Every green build is releasable
  - Delivery = one approval to promote to prod
  - Deployment = fully automatic to prod

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CI/CD Pipeline on AWS



▶ A commit flows Source → Build → Approve → Deploy; a failed test stops the pipeline before production.

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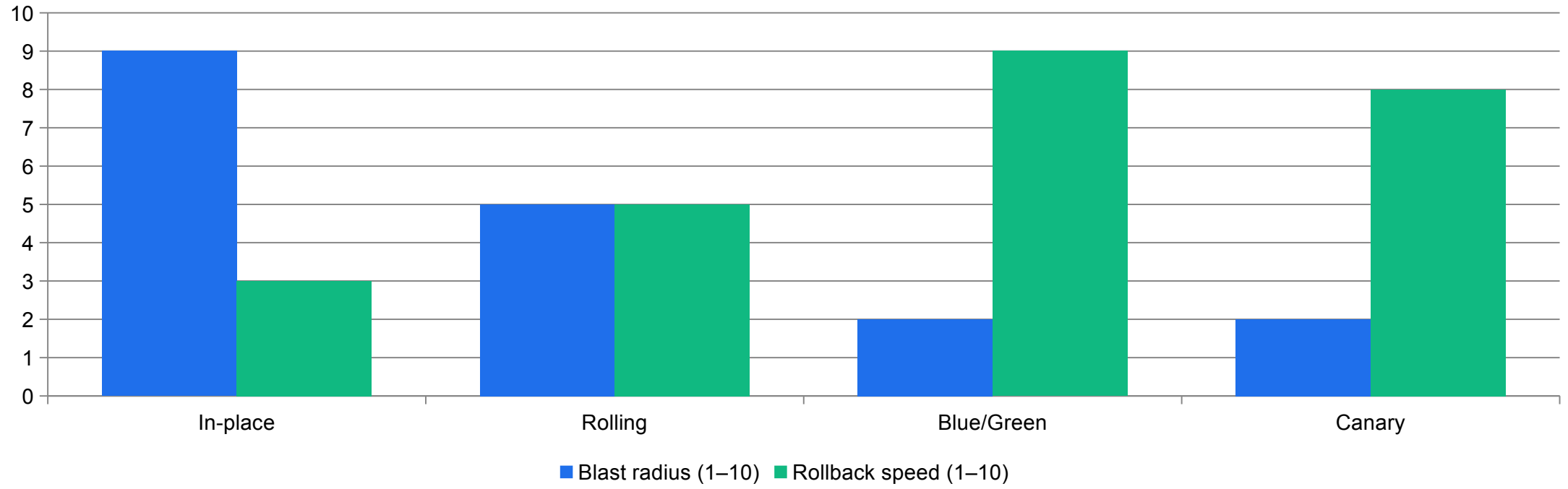
Deployment Strategies Compared

Strategy	How it works	Rollback	Best for
In-place	Update the same instances	Redeploy previous	Dev / simple apps
Rolling	Update in batches	Batch by batch	Steady, low-risk updates
Blue/Green	Shift traffic to a new fleet	Instant switch-back	Zero-downtime prod
Canary	Shift a small % first	Stop & revert	High-risk changes

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TRADE-OFFS

Deployment Strategy — Blast Radius vs Rollback Speed



WHAT THE DATA SHOWS

Blue/green and canary minimise how many users a bad release can affect and let you roll back fastest.

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SHIFT LEFT

Automated Testing in the Pipeline

1

Unit tests

Run in CodeBuild on every commit — fail fast, block the merge.

2

Integration tests

Verify services work together before promotion to staging.

3

Security scans

SAST/SCA on code and dependencies inside the build phase.

4

Quality gates

CodeGuru / linters and coverage thresholds gate the release.

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HOW CODEBUILD KNOWS WHAT TO DO

Anatomy of buildspec.yml

1

version / env

Declare the buildspec version and environment variables / secrets.

2

phases: install

Set up the runtime and install dependencies.

3

phases: pre_build

Log in to registries, run linters.

4

phases: build

Compile, run unit tests, package the app.

5

phases: post_build

Push images, generate reports.

6

artifacts

Declare the versioned output CodePipeline carries forward.

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IN DEPTH

Continuous Integration — mechanics, not just merging

IN DEPTH

- Every commit to the mainline triggers an automated build and test run, so integration problems surface within minutes of the change that caused them.
- CodeBuild runs the build in an ephemeral, reproducible container defined by buildspec.yml — the same steps run for every developer.
- A red build blocks the merge; the team's working agreement is that a broken mainline is fixed before any new work continues.

WHY IT MATTERS

Small, frequent integrations keep the merge delta tiny, so defects are cheap to locate and revert — the opposite of a painful 'big-bang' integration before release.

FAILURE MODE

Long-lived feature branches that integrate weekly recreate the very merge-hell CI exists to prevent; the build stays green only because nothing is really being integrated.

TRADE-OFF / DECISION

Faster feedback needs fast builds — split slow suites into parallel CodeBuild batches or later pipeline stages so the commit-stage stays under ~10 minutes.

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IN DEPTH

Continuous Delivery vs Continuous Deployment

IN DEPTH

- Continuous Delivery keeps every green build in a releasable state; a human approval action gates the final promotion to production.
- Continuous Deployment removes that gate — every change that passes the pipeline goes to production automatically.
- Both rely on the same pipeline; the only difference is whether a person presses the button.

WHY IT MATTERS

Delivery gives regulated or high-blast-radius teams a control point; Deployment gives high-trust teams maximum flow once automated tests and progressive rollout make failures cheap.

FAILURE MODE

Adopting Continuous Deployment before you have strong automated tests and fast rollback turns the pipeline into an automated way to ship bugs to customers.

TRADE-OFF / DECISION

An approval gate adds safety but also queueing delay and 'rubber-stamp' risk — reserve manual approval for genuinely irreversible or compliance-driven steps.

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IN DEPTH

Source control as the pipeline trigger — GitHub via CodeConnections

IN DEPTH

- The modern AWS path connects a GitHub (or GitHub Enterprise / Bitbucket / GitLab) repository through AWS CodeConnections, so a push starts the pipeline over a managed, OAuth/app-based connection.
- CodeCommit remains an AWS-native Git option, but new work standardises on external Git providers via CodeConnections.
- The Source stage stores the commit as an artifact in an encrypted S3 bucket that later stages consume.

WHY IT MATTERS

CodeConnections removes long-lived Git credentials from the pipeline and lets teams keep their existing GitHub workflow, reviews and branch protection.

FAILURE MODE

Polling a repository instead of using the connection's webhook adds minutes of latency and can miss rapid successive commits.

TRADE-OFF / DECISION

Managed connections centralise trust in one integration — scope its permissions and monitor CodeConnections access as part of your security posture.

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IN DEPTH

buildspec.yml — teaching CodeBuild your build

IN DEPTH

- buildspec.yml declares phases (install, pre_build, build, post_build), the runtime versions, environment variables and the artifacts to export.
- Secrets are pulled at build time from Secrets Manager or SSM Parameter Store, never hard-coded in the file.
- The build produces a single versioned artifact that every downstream stage reuses — build once, deploy many times.

WHY IT MATTERS

A declarative build spec makes the build reproducible and reviewable in code review, and identical across every developer and the pipeline.

FAILURE MODE

Baking credentials or environment-specific values into buildspec.yml couples the artifact to one environment and leaks secrets into logs.

TRADE-OFF / DECISION

Rich caching (dependencies, layers) speeds builds but can hide dependency drift — pin versions and periodically build clean.

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IN DEPTH

appspec + deployment lifecycle hooks

IN DEPTH

- appspec.yml tells CodeDeploy what to copy and which lifecycle-hook scripts to run (BeforeInstall, AfterInstall, ApplicationStart, ValidateService).
- The ValidateService hook is where a health check decides whether the new version is actually serving traffic.
- For blue/green, CodeDeploy shifts traffic between target groups instead of updating files in place.

WHY IT MATTERS

Lifecycle hooks turn 'copy files' into a safe, verifiable release with an explicit go/no-go check before the deployment is declared successful.

FAILURE MODE

A ValidateService hook that always exits 0 defeats automated rollback — a broken release is reported as healthy.

TRADE-OFF / DECISION

Longer validation windows catch more problems but slow each deployment and hold the fleet in a mixed-version state.

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IN DEPTH

Artifacts, versioning and traceability

IN DEPTH

- Build outputs are versioned in S3 and packages in CodeArtifact, each tied to the source commit that produced it.
- Any production release can be traced back to an exact commit, build log and test result.
- Rollback becomes 'redeploy the previous known-good artifact', not 'rebuild and hope'.

WHY IT MATTERS

Immutable, versioned artifacts make releases reproducible and auditable — essential for incident forensics and compliance evidence.

FAILURE MODE

Rebuilding from source on rollback can pull newer dependencies and reintroduce or mask the very fault you are rolling back from.

TRADE-OFF / DECISION

Retaining every artifact aids traceability but grows storage cost — apply S3 lifecycle rules to age out old, unreferenced builds.

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IN DEPTH

Automated testing and quality gates in the pipeline

IN DEPTH

- Unit tests run in the commit stage; integration, security (SAST/SCA) and performance checks run in later stages before promotion.
- Quality gates — coverage thresholds, CodeGuru findings, linters — can fail the build automatically.
- Test data and environments are provisioned as code so tests are repeatable.

WHY IT MATTERS

Shifting testing left catches defects when they are cheapest to fix and keeps the mainline releasable at all times.

FAILURE MODE

Flaky tests train the team to ignore red builds; a pipeline everyone overrides provides no protection at all.

TRADE-OFF / DECISION

More gates raise confidence but lengthen lead time — order them fast-to-slow and fail fast on the cheapest checks.

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IN DEPTH

Choosing a deployment strategy by blast radius

IN DEPTH

- In-place is simplest but updates live instances and risks downtime; rolling limits impact to a batch at a time.
- Blue/green stands up a parallel fleet and switches traffic — instant rollback by switching back.
- Canary shifts a small percentage first, watches metrics, then completes or aborts the rollout.

WHY IT MATTERS

Matching the strategy to the change's risk lets you trade release speed against how many users a bad release can touch.

FAILURE MODE

Blue/green without draining connections or warming caches can still cause a visible blip at cut-over.

TRADE-OFF / DECISION

Blue/green needs roughly double capacity during the shift; canary needs good metrics and automated alarms to be safe.

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IN DEPTH

Pipeline structure — stages, actions and promotion

IN DEPTH

- A pipeline is stages (Source → Build → Test → Deploy-to-Staging → Approve → Deploy-to-Prod), each containing actions that can run in parallel.
- The same immutable artifact is promoted across environments; only configuration changes between them.
- Cross-account deployment uses assumed roles so the production account trusts the pipeline without sharing credentials.

WHY IT MATTERS

Modelling promotion explicitly gives a single, auditable path to production and prevents 'works in staging, different build in prod' drift.

FAILURE MODE

Rebuilding per environment breaks the guarantee that what you tested is what you shipped.

TRADE-OFF / DECISION

More environments raise confidence but add cost and lead time — keep pre-prod as production-like as budget allows.

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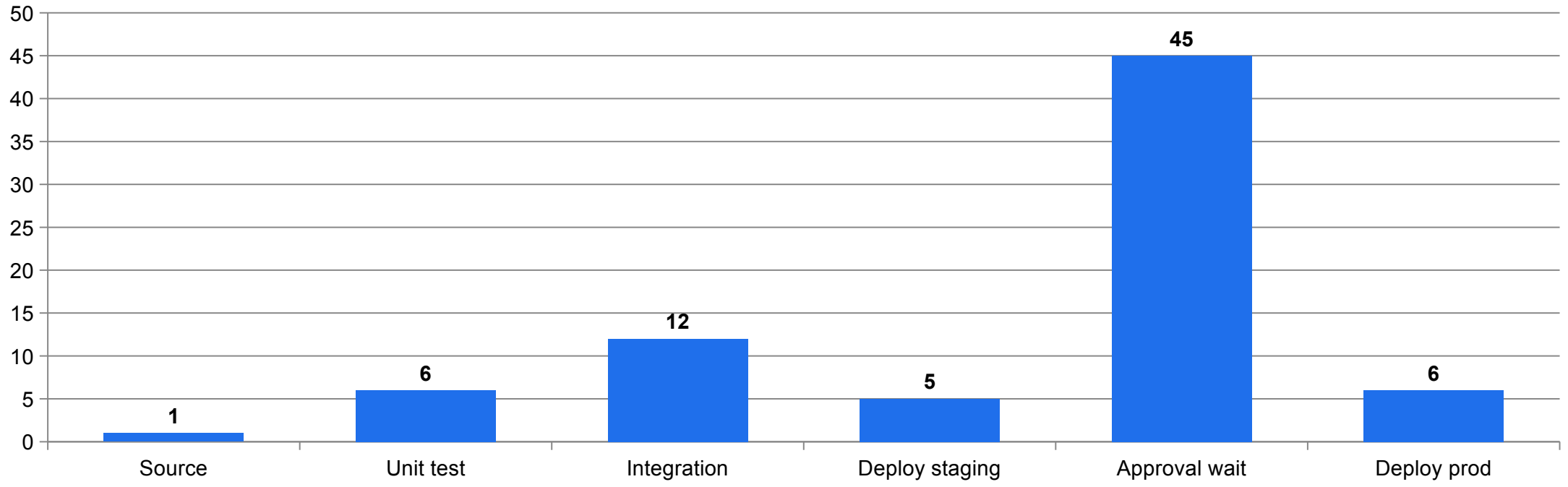
CI/CD Toolchain — AWS-native vs external

Concern	GitHub via CodeConnections	AWS CodeCommit
Where new work goes	Primary, modern path	AWS-native option; not required
Credentials	Managed connection (no static keys)	IAM / Git credentials
Team workflow	Existing GitHub PRs & branch rules	Native Git, AWS console/IAM
Trigger	Connection webhook	CloudWatch Events / EventBridge

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OPTIMISE THE BOTTLENECK

Pipeline Lead Time by Stage — Where the Minutes Go



WHAT THE DATA SHOWS

Manual approval dominates lead time here — automate the checks it waits on so approval is a formality, not a bottleneck.

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WHAT BREAKS PIPELINES

CI/CD Failure Modes & Anti-Patterns

1

Snowflake build agents

Builds pass locally, fail in CodeBuild → pin runtime-versions and build in clean containers.

2

Secrets in source/logs

Keys committed or echoed → pull from Secrets Manager/SSM at build time; mask in logs.

3

Flaky tests ignored

Red builds routinely overridden → quarantine and fix flaky tests; a gate you bypass is no gate.

4

Rebuild on promotion

Different artifact per environment → build once, promote the same artifact.

5

No automated rollback

Bad deploy needs a human at 2am → alarm-based rollback on CodeDeploy.

6

Polling instead of webhooks

Slow, missed triggers → use the CodeConnections webhook.

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PROVEN PRACTICES

Delivery Patterns Worth Adopting

1

Trunk-based development

Short-lived branches merged to trunk daily keep integration continuous.

2

Build once, deploy many

One immutable artifact promoted across environments.

3

Progressive delivery

Canary/blue-green with automated metric analysis before full rollout.

4

Pipeline as code

Define the pipeline in CloudFormation/CDK so it is versioned and reviewable.

5

Fail fast

Cheapest, fastest checks first; stop the line on first failure.

6

Immutable infrastructure

Replace, don't patch, running servers.

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Architecture Decision — Approval Gate or Not?

A — Manual approval before prod

Regulated change, irreversible action, or immature test suite. Adds a control point and audit record — at the cost of queueing delay.

B — Automated to prod

High test coverage, fast rollback and progressive rollout in place. Maximises flow — requires strong alarms and canary analysis.

C — Hybrid

Auto-deploy low-risk services; gate high-blast-radius or data-migration changes. The common enterprise middle ground.

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REASONING THROUGH THE RESPONSE

Scenario — A bad release reaches production

SITUATION

A deployment passes all pipeline stages, but error rates climb 3 minutes after it reaches production.

ANALYSIS

- The build and unit tests were green, so the fault is environment- or scale-specific — exactly what canary analysis is designed to catch.
- Because releases use immutable, versioned artifacts, the last known-good build is one redeploy away.
- A CloudWatch alarm on error rate can trigger CodeDeploy's automatic rollback without waiting for a human.

DECISION

Wire an error-rate alarm to CodeDeploy auto-rollback and move this service to a canary configuration so the next bad change is caught at 5% of traffic, not 100%.

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Scenario — Builds are too slow, developers merge less

SITUATION

The commit-stage build takes 25 minutes; developers batch changes to avoid the wait, so integrations get large and risky.

ANALYSIS

- Slow feedback is pushing the team back towards big-bang integration — the DORA lead-time and batch-size signals both degrade.
- Test suites can be split and run in parallel CodeBuild projects; dependency caching removes repeated downloads.
- Slow end-to-end tests can move to a later stage so the commit stage stays fast.

DECISION

Target a sub-10-minute commit stage: parallelise tests, cache dependencies, and defer slow suites downstream so small, frequent merges become painless again.

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FROM PRACTICE TO NUMBERS

Worked Analysis — Sizing the Impact of CI/CD

Before: 2 releases/month, 14-day lead time	Large batches; each release bundles ~150 changes, so failures are hard to isolate.
After: 40 releases/month, ~1-day lead time	Small batches; each release is ~7 changes, so a regression is trivially bisected.
Change-failure rate 30% → 10%	Automated gates catch defects pre-prod; progressive rollout limits blast radius.
MTTR 8h → 1h	Versioned artifacts + alarm-based rollback make recovery a redeploy, not a rebuild.

TAKEAWAY

The gain is not just speed — smaller batches make every DORA metric move together: deploy more often, recover faster, fail less.

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Set Up a Safe AWS & Git Working Environment and Verify Identity

LAB 1

Day 1 · LO1 · K1, A1

Before automating anything you need a workspace you can trust. In this lab you install and configure AWS CLI v2 with a named least-privilege 'devops' profile (or an IAM Identity Center session), prove exactly who you are with STS, pin a Region, attach a permission boundary so the profile can never exceed its remit, and lock down your Git repository so plaintext secrets …

YOU'LL BUILD

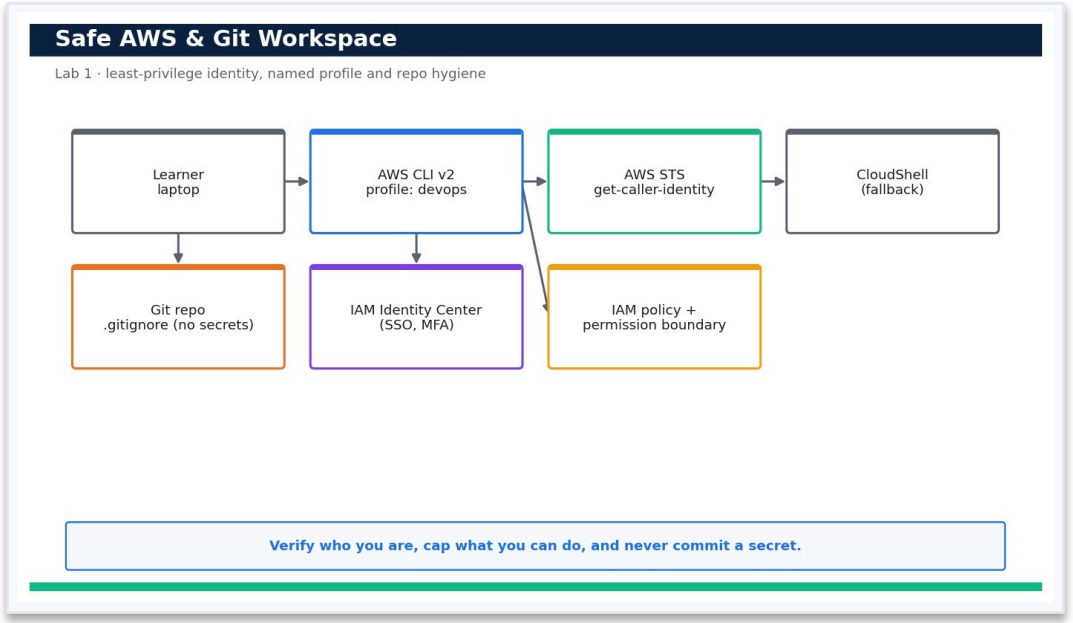
A named, MFA-aware, least-privilege 'devops' CLI profile plus a Git repo with a secrets-safe.gitignore, verified with an AWS caller-identity check and a clean secret scan.

Tools: AWS IAM, AWS CLI v2, IAM Identity Center, CloudShell, Git

OUTCOME & EVIDENCE

Outcome: You can run identity-check.sh and get 'PROFILE OK' with the correct ARN and Region, the permission boundary provably blocks …

Evidence: Screenshot of showing your own Account and ARN.



Reference architecture

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Establish a Source-Control Workflow and Branching for the Sample App

LAB 2

Day 1 · LO1 · K1, A1

Good delivery starts with disciplined source control. In this lab you fork and clone the sample web app, adopt a trunk-based workflow, cut a short-lived feature branch, commit a change, push it, open a pull request and protect the main branch. You …

YOU'LL BUILD

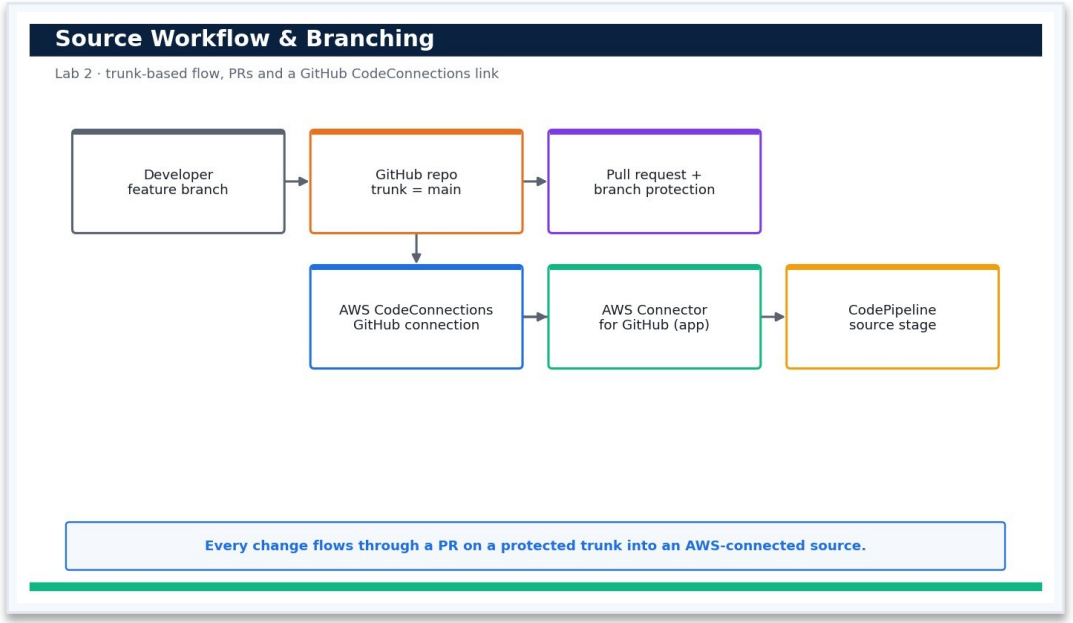
A protected trunk-based GitHub repo with an open/merged PR, plus an AVAILABLE AWS CodeConnections connection to GitHub ready to drive CodeBuild and CodePipeline.

Tools: GitHub, AWS CodeConnections, Git, AWS CodePipeline (source stage)

OUTCOME & EVIDENCE

Outcome: You can push a feature branch, open and merge a PR into a protected main, and prove an AWS CodeConnections connection to …

Evidence: The GitHub PR page for feature/bump-version showing the diff and an approving review.



Reference architecture

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Run Automated Tests on Every Commit with AWS CodeBuild

LAB 3

Day 1 · LO1 · K1, A1

Automated tests are the gate that keeps broken code out of production. In this lab you author a buildspec.yml with install/pre_build/build phases that install dependencies, run a linter and run pytest, then create a CodeBuild project that pulls the source through your CodeConnections link. You start a build, read the phase-by-phase logs in CloudWatch, deliberately …

YOU'LL BUILD

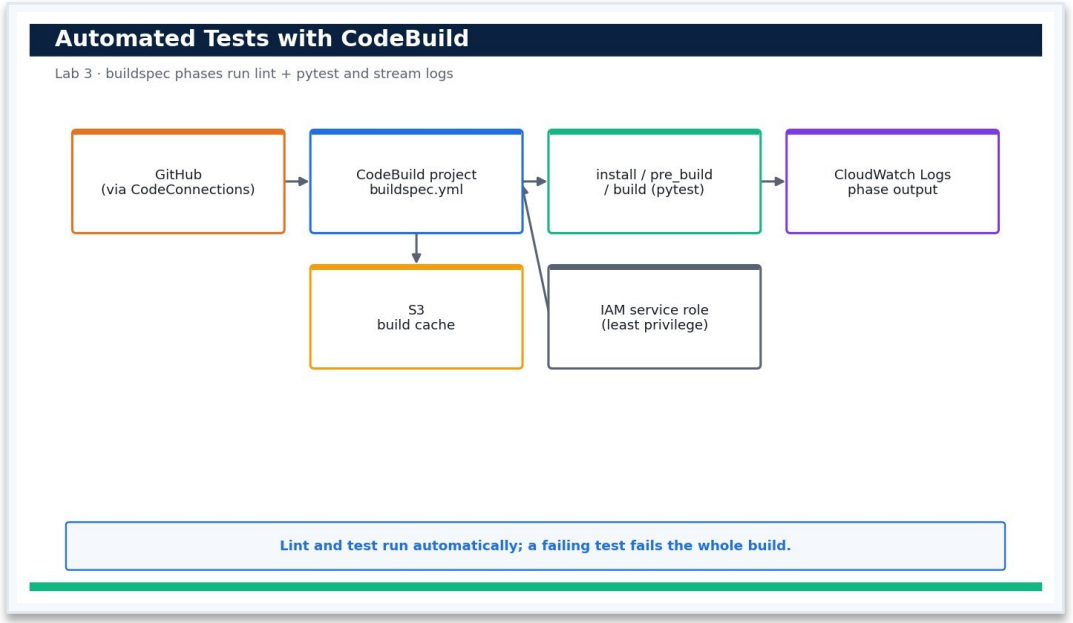
A CodeBuild project that runs the linter and, streams phase logs to CloudWatch, and fails loudly when a test is broken.

Tools: AWS CodeBuild, Amazon S3, AWS IAM, Amazon CloudWatch Logs

OUTCOME & EVIDENCE

Outcome: You can start a build that runs the linter and, read the phase logs, break a test to see the build FAIL, and fix it to see the build SUCCEED …

Evidence: The CodeBuild build page showing buildStatus SUCCEEDED with INSTALL/PRE_BUILD/BUILD green.



Reference architecture

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Produce and Store Versioned Build Artifacts with a Buildspec

LAB 4

Day 1 · LO1 · K1, A1

A pipeline is only trustworthy if you can identify and reproduce exactly what it built. In this lab you extend the buildspec to package the app into a versioned artifact named by commit and semantic version, then upload it to an encrypted S3 artifact bucket with Block Public Access and object versioning enabled. You verify each object version, and prove you can retrieve …

YOU'LL BUILD

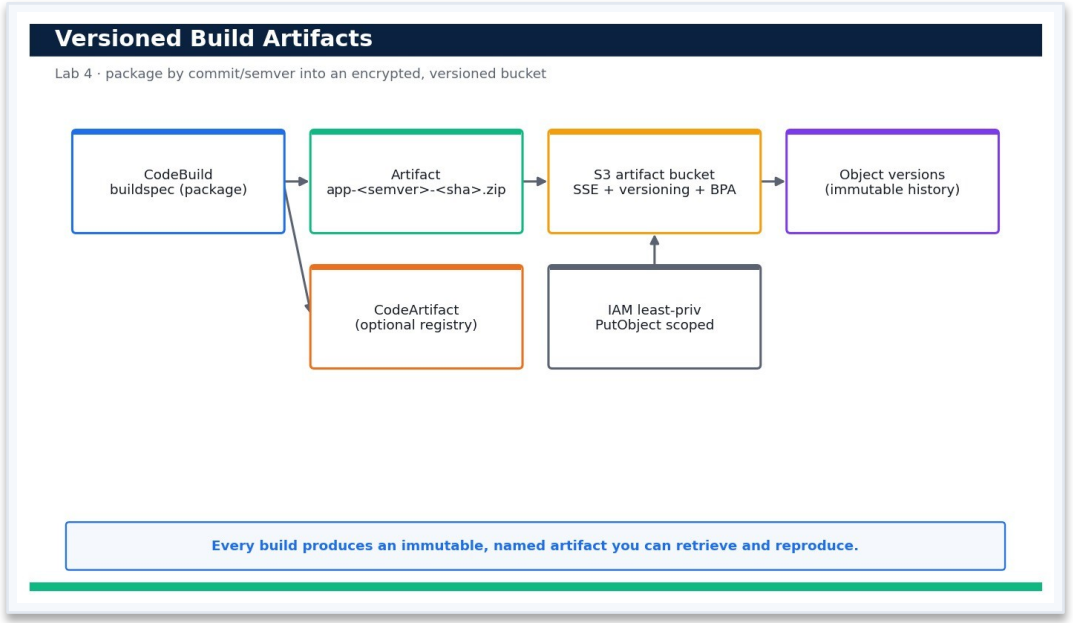
An encrypted, version-enabled S3 artifact bucket holding immutable, commit/semver-named build artifacts that you can list, retrieve and reproduce.

Tools: AWS CodeBuild, Amazon S3, AWS CodeArtifact, AWS IAM

OUTCOME & EVIDENCE

Outcome: You can produce a commit/semver-named artifact, store it in an encrypted, versioned, private bucket, list its version history, and download …

Evidence: output showing Status: Enabled.



Reference architecture

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Assemble a CodePipeline with Manual Approval and a Deployment Strategy

LAB 5

Day 1 · LO1 · K1, A1

Now you connect everything into one governed release path. Using a CloudFormation template plus a pipeline definition, you build Source (CodeConnections) -> Build (CodeBuild) -> Manual Approval -> Deploy (CodeDeploy). You choose between …

YOU'LL BUILD

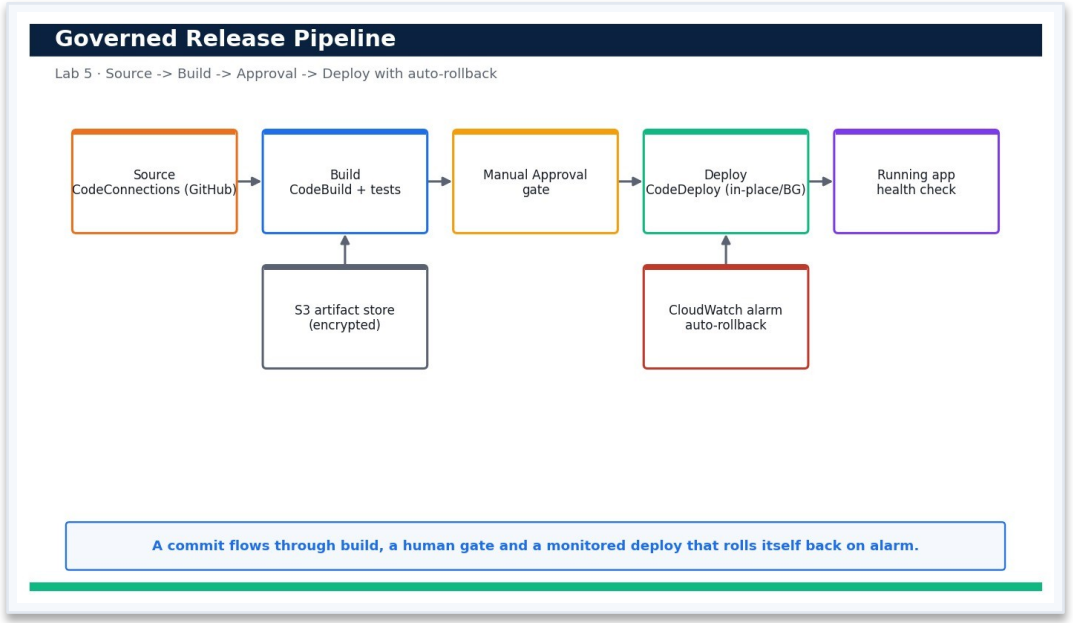
A four-stage CodePipeline (Source -> Build -> Approve -> Deploy) with a manual approval gate, a chosen deployment strategy and alarm-based automatic rollback.

Tools: AWS CodePipeline, CodeBuild, CodeDeploy, CodeConnections, Amazon S3, Amazon CloudWatch, AWS CloudFormation

OUTCOME & EVIDENCE

Outcome: You can run the pipeline end to end, be blocked at a manual approval until you approve, see the app deployed and healthy, and watch …

Evidence: The CodePipeline view showing Source, Build, Approve and Deploy, paused at the Approve gate.



Reference architecture

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DOMAIN RECAP

Recap — SDLC Automation

- You can now: Stand up a least-privilege AWS CLI/Git workspace and confirm your identity
- You can now: Adopt a trunk-based Git workflow and connect the repository to AWS with CodeConnections
- You can now: Author a buildspec and a CodeBuild project that automatically lints and tests the app
- You can now: Extend the buildspec to package versioned artifacts and store them in an encrypted, versioned bucket
- You can now: Wire an end-to-end pipeline with a manual approval gate, a deployment strategy and auto-rollback
- Continuous Integration — Developers push often; CodeBuild runs the build and tests on every commit so bugs are found early, not at release.
- Continuous Delivery / Deployment — Every green build is releasable; CodePipeline and CodeDeploy move it through dev → test → prod with approvals.
- The AWS developer tools — CodeCommit / CodeBuild / CodeDeploy / CodePipeline / CodeArtifact / CodeGuru compose the managed CI/CD toolchain.

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02

DOMAIN 02

Configuration Management and IaC

Exam weighting 17% · Infrastructure as Code · reusable templates · multi-account & multi-Region

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DOMAIN 02 · WEIGHTING 17%

Key Concepts — Configuration Management and IaC

1

Infrastructure as Code

Declare infrastructure in version-controlled templates so environments are repeatable, reviewable and disposable.

2

CloudFormation

Declarative JSON/YAML stacks with parameters, mappings, outputs, change sets, drift detection and automatic rollback.

3

Reusable components

Nested stacks and modules parameterise a pattern once and reuse it across teams and Regions.

4

CDK vs CloudFormation vs Terraform

CDK generates CloudFormation from real code; Terraform is multi-cloud — all describe desired state.

5

Configuration management

Systems Manager (State Manager, Automation, Parameter Store) keeps running instances in their desired configuration.

6

Multi-account & multi-Region

Organizations, StackSets and Control Tower create, onboard and secure accounts at scale.

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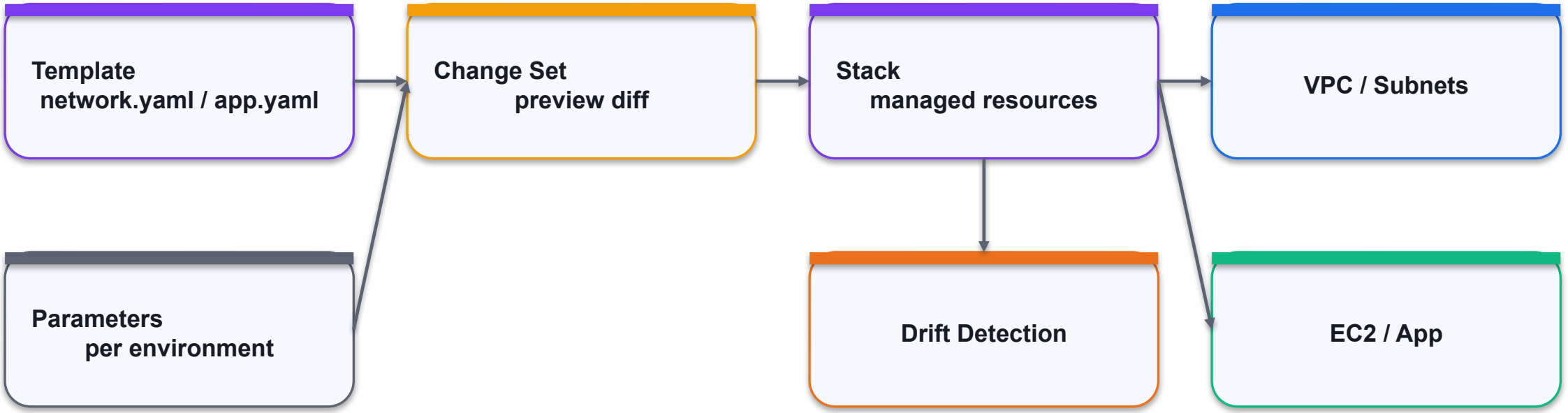
DECLARE DESIRED STATE

IaC Options on AWS

Tool	Language	Scope	When to use
CloudFormation	YAML / JSON	AWS-native declarative	Standard AWS IaC, StackSets
AWS CDK	Python / TS / Java	Code → CloudFormation	Reusable, testable constructs
Terraform	HCL	Multi-cloud	Mixed-cloud estates
SAM	YAML	Serverless-focused	Lambda / API / event apps

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CloudFormation — Reusable Stacks



- One parameterised template deploys identical dev, test and prod environments; change sets preview, drift detection audits.

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MULTI-ACCOUNT / MULTI-REGION

Configuration Management & Governance at Scale

1

Nested stacks / modules

Package a pattern once and reuse it across teams.

2

Systems Manager

State Manager, Automation and Parameter Store keep instances in desired state.

3

StackSets

Deploy the same stack across many accounts and Regions.

4

Organizations & Control Tower

Create, onboard and secure accounts with a landing zone.

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TEMPLATE STRUCTURE

Anatomy of a CloudFormation Template

1

Parameters

Inputs that make the template reusable across environments.

2

Mappings

Static lookup tables (e.g. AMI by Region).

3

Conditions

Create resources only when a condition is met.

4

Resources

The AWS resources to provision (the only required section).

5

Outputs

Values to export and import into other stacks.

6

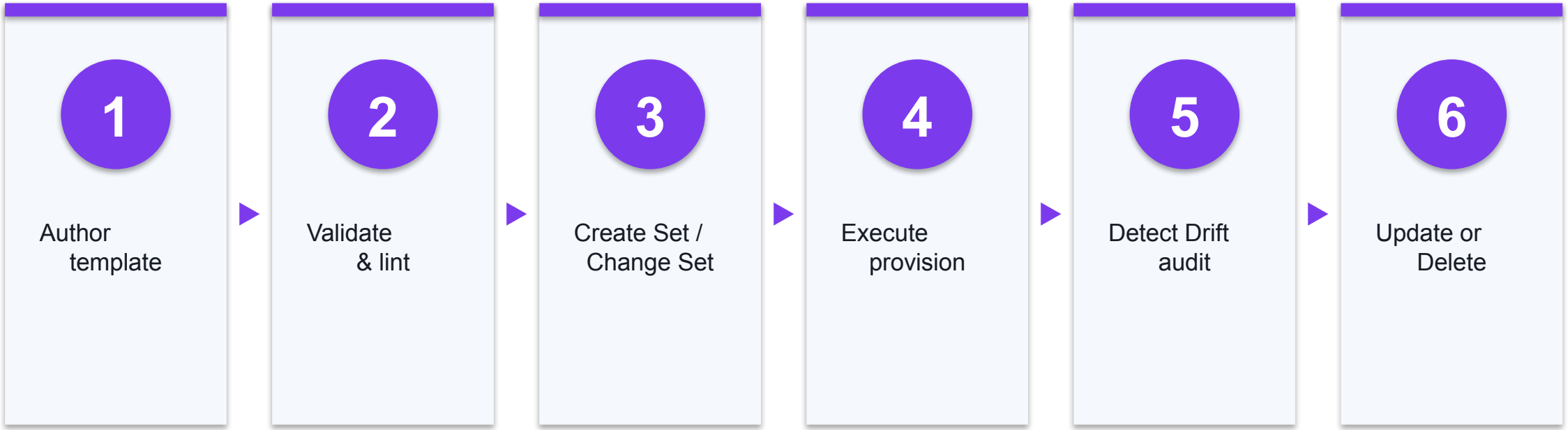
Metadata

Extra structured information about the template.

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SAFE CHANGE MANAGEMENT

The Stack Lifecycle



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Parameter Store vs Secrets Manager

Store	Best for	Rotation	Cost
SSM Parameter Store	Config values & non-secret parameters	Manual	Free (standard)
Secrets Manager	Passwords, API keys, DB credentials	Automatic	Per-secret

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IN DEPTH

Infrastructure as Code — declarative desired state

IN DEPTH

- You describe the target state of infrastructure in version-controlled templates; the service computes and applies the difference.
- The same template deploys identical dev, test and prod environments — parameters, not copy-paste, capture the differences.
- Infrastructure changes go through code review and the pipeline, just like application code.

WHY IT MATTERS

Repeatable, reviewable, disposable environments eliminate configuration drift and the 'it works on that server' class of failure.

FAILURE MODE

Mixing manual console changes with IaC creates drift the template no longer describes — the next deploy can revert or clash with them.

TRADE-OFF / DECISION

Declarative IaC is safe and auditable but less flexible than imperative scripts — use change sets and modules to regain expressiveness.

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IN DEPTH

CloudFormation stacks, change sets and rollback

IN DEPTH

- A stack is a unit of resources created, updated and deleted together from one template.
- A change set previews exactly what an update will add, modify or replace before you execute it.
- On failure, CloudFormation automatically rolls the stack back to its last known-good state.

WHY IT MATTERS

Preview-then-apply plus automatic rollback make infrastructure changes as safe and reversible as a code deployment.

FAILURE MODE

A resource that must be replaced (not updated in place) can cause downtime or data loss if you don't read the change set's Replacement column.

TRADE-OFF / DECISION

Automatic rollback protects you but can mask the real error — always read the stack Events for the first failing resource.

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IN DEPTH

Template anatomy — parameters, mappings, conditions, outputs

IN DEPTH

- Parameters make a template reusable across environments; Mappings hold static lookups such as AMI-by-Region.
- Conditions create resources only when appropriate (e.g. a NAT gateway only in prod); Outputs export values for other stacks.
- Only the Resources section is mandatory — everything else exists to make templates reusable and composable.

WHY IT MATTERS

A well-structured template is a reusable product, not a one-off script — one artifact serves many teams and Regions.

FAILURE MODE

Hard-coded AMI IDs, account numbers or Region strings make a template that only works in the account it was written in.

TRADE-OFF / DECISION

Heavy parameterisation increases flexibility but reduces readability — prefer sensible defaults and SSM public parameters.

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IN DEPTH

Reuse — nested stacks, modules and StackSets

IN DEPTH

- Nested stacks and CloudFormation modules package a pattern once and reuse it across many stacks.
- StackSets deploy the same stack across many accounts and Regions from a single operation.
- Cross-stack references and exports let a network stack publish IDs that app stacks import.

WHY IT MATTERS

Reuse turns a proven design into an organisational standard, so every team gets the same secure, compliant baseline.

FAILURE MODE

Over-nesting creates tightly coupled stacks where one change forces wide, slow updates and hard-to-debug failures.

TRADE-OFF / DECISION

Central StackSets enforce consistency but need governance so account teams aren't blocked waiting on the platform team.

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IN DEPTH

CDK, SAM and Terraform — when to reach for each

IN DEPTH

- The AWS CDK lets you express infrastructure in a general-purpose language and synthesises CloudFormation.
- AWS SAM is a CloudFormation extension optimised for serverless apps.
- Terraform describes desired state in HCL across many clouds and providers.

WHY IT MATTERS

Choosing the right tool matches the team's skills and estate — real code for testable constructs, SAM for serverless, Terraform for multi-cloud.

FAILURE MODE

Adopting CDK without CloudFormation literacy hides what is really deployed and makes drift and failures harder to reason about.

TRADE-OFF / DECISION

Higher-level abstractions boost productivity but add a layer to debug when the generated template misbehaves.

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IN DEPTH

Configuration management with Systems Manager

IN DEPTH

- State Manager keeps instances in a defined configuration; Automation documents run repeatable operational runbooks.
- Parameter Store holds configuration and (with Secrets Manager) secrets, versioned and access-controlled.
- Patch Manager and Inventory keep fleets patched and auditable without SSH access.

WHY IT MATTERS

Config management closes the gap between 'provisioned' and 'correctly configured and maintained over time'.

FAILURE MODE

Treating servers as pets you SSH into and hand-tune reintroduces drift that no template can track.

TRADE-OFF / DECISION

Agent-based management adds an agent to run and secure, but removes bastion hosts and standing SSH access.

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IN DEPTH

Drift detection and remediation

IN DEPTH

- Drift detection compares live resources against the template and reports what changed out-of-band.
- Config rules can watch for the same drift continuously and trigger remediation.
- The fix is to update the template to match intent or revert the resource to the template.

WHY IT MATTERS

Detecting drift early prevents the slow divergence that eventually makes a stack un-updatable.

FAILURE MODE

Ignoring drift until an urgent change is needed means your 'safe' update now collides with undocumented manual edits.

TRADE-OFF / DECISION

Strict no-drift policies improve auditability but can slow emergency fixes — provide a documented break-glass path.

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IN DEPTH

Multi-account governance — Organizations, Control Tower, SCPs

IN DEPTH

- AWS Organizations groups accounts; Service Control Policies set the maximum permissions any account can grant.
- Control Tower stands up a landing zone with guardrails, a shared logging account and account factory.
- New accounts are vended pre-configured to the security and network baseline.

WHY IT MATTERS

Account-level isolation plus org-wide guardrails give strong blast-radius control and a consistent, auditable baseline.

FAILURE MODE

A single shared account for everything makes least-privilege impossible and one mistake blast-radius huge.

TRADE-OFF / DECISION

More accounts improve isolation but add networking and operational overhead — automate account vending to keep it manageable.

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IN DEPTH

Secrets & configuration — Parameter Store vs Secrets Manager

IN DEPTH

- Parameter Store holds configuration values (and SecureString secrets) with a generous free tier.
- Secrets Manager adds built-in, scheduled rotation and tight integration with RDS and other services.
- Both keep sensitive values out of templates, code and logs, retrieved at deploy/run time by IAM-scoped principals.

WHY IT MATTERS

Externalising configuration and secrets lets one immutable artifact run safely in every environment.

FAILURE MODE

Baking a database password into a template or AMI leaks it into every copy and every log line.

TRADE-OFF / DECISION

Automatic rotation is safest but must be tested — a rotation that apps don't reload can cause outages.

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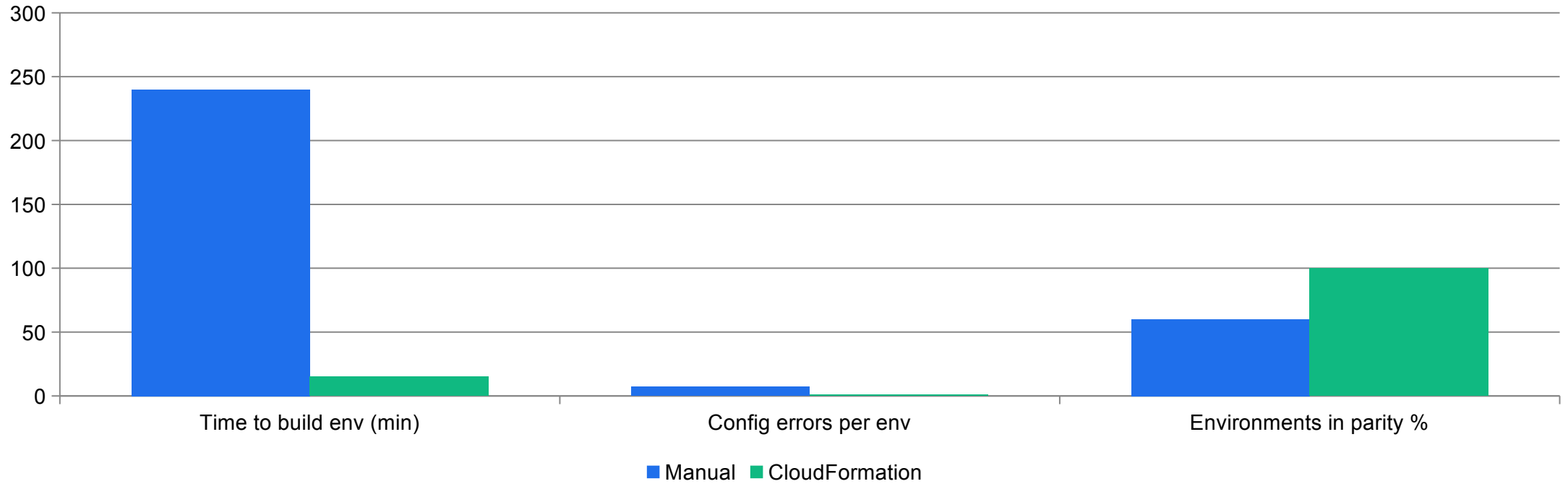
Change Set Replacement — Read Before You Execute

Change type	CloudFormation action	Operational risk
Update in place	Modify the existing resource	Low — no replacement
Replacement	Create new, delete old	High — new ID, possible data loss
Add / remove	Create or delete a resource	Medium — check dependencies
No interruption	Property change, resource kept	Low

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REPEATABILITY PAYS OFF

Environment Provisioning — Manual vs IaC



WHAT THE DATA SHOWS

IaC collapses environment build time and errors and — most importantly — makes every environment identical, so tests actually predict production.

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WHERE IaC GOES WRONG

IaC Failure Modes & Anti-Patterns

1

Out-of-band console edits

Drift the template can't see → detect drift; make console read-only in prod.

2

Hard-coded IDs/Regions

Template only works in one account → parameters + SSM public parameters.

3

Giant monolith stack

Slow, risky updates → split by lifecycle and blast radius.

4

No change-set review

Surprise replacements → always preview before execute.

5

Secrets in templates

Leaked credentials → Secrets Manager / SSM SecureString.

6

Copy-paste per environment

Divergent envs → one template, many parameter sets.

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PROVEN PRACTICES

IaC Patterns Worth Adopting

1

Layered stacks

Separate network, platform and app stacks by change rate.

2

Landing zone

Control Tower baseline + guardrails for every account.

3

Immutable AMIs

Bake with a pipeline (e.g. EC2 Image Builder); replace, don't patch.

4

Public SSM parameters

Reference latest AMIs without hard-coding IDs.

5

Policy as code

Guardrails via SCPs and Config rules, versioned like IaC.

6

Pipeline-deployed stacks

No manual stack updates in production.

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STACK DECOMPOSITION

Architecture Decision — One Big Stack or Many?

A Single stack

Small app, one team, one lifecycle. Simple to reason about — but every change risks the whole system and updates get slow as it grows.

B Layered stacks

Separate network / platform / app by change rate and owner. Smaller blast radius and faster updates — needs disciplined exports/imports.

C StackSets across accounts

Same baseline in many accounts/Regions. Enforces consistency at scale — requires central governance and account-team coordination.

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THE COST OF DRIFT

Scenario — A 'quick' console fix breaks the next deploy

SITUATION

An engineer widens a security group in the console during an incident. Two weeks later, a routine stack update fails or silently reverts the fix.

ANALYSIS

- The live resource no longer matches the template, so CloudFormation's view of desired state is wrong.
- Drift detection would have flagged this the same day; Config rules could have alarmed on the open port immediately.
- The durable fix is to encode the change in the template and re-apply, not to keep patching by hand.

DECISION

Run drift detection on a schedule, alarm on security-group drift via Config, and require that emergency console changes are back-ported into the template within the incident's follow-up.

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SCALING GOVERNANCE

Scenario — Standing up a compliant new account fast

SITUATION

A new business unit needs an AWS account that already meets the company's security, logging and network standards on day one.

ANALYSIS

- Hand-building the account is slow and error-prone and guarantees it drifts from the standard.
- Control Tower's account factory vends accounts pre-wired to guardrails and centralised logging.
- StackSets push the standard network and security stacks into the new account automatically.

DECISION

Vend the account through Control Tower and apply the baseline via StackSets, so 'compliant by default' is the path of least resistance rather than a later audit finding.

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ONE ARTIFACT, MANY ENVIRONMENTS

Worked Analysis — Parameterising One Template for Three Environments

Instance size by env	dev t3.micro, staging t3.small, prod m5.large — a single InstanceType parameter.
High availability by env	A Condition creates the Multi-AZ standby and NAT redundancy only when EnvName = prod.
AMI without hard-coding	An SSM public parameter resolves the latest patched Amazon Linux AMI at deploy time.
Secrets by env	Each environment references its own Secrets Manager secret; none appear in the template.

TAKEAWAY

The differences between environments become a short parameter file — the template itself is identical, so staging genuinely predicts production.

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Deploy a Reusable VPC Network Baseline with CloudFormation

LAB 6

Day 2 · LO2 · K3, A3

Your team keeps building VPCs by hand and no two look alike. In this lab you codify a golden network baseline as a single CloudFormation template — a VPC with two public and two private subnets spread across two Availability Zones, an Internet Gateway and route tables — and publish its identifiers as …

YOU'LL BUILD

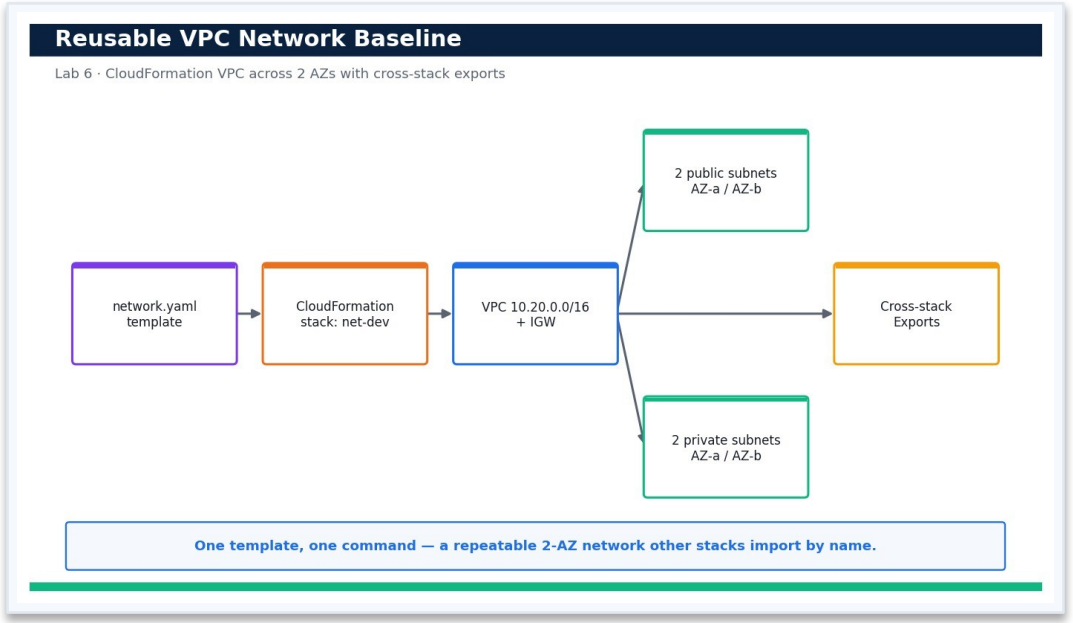
A version-controlled network.yaml that deploys a 2-AZ VPC (2 public + 2 private subnets, IGW, route tables) and publishes VpcId, subnet IDs and the VPC CIDR as reusable cross-stack Exports.

Tools: AWS CloudFormation, Amazon VPC, Amazon EC2

OUTCOME & EVIDENCE

Outcome: Deploying network.yaml with parameters-dev.json produces a reusable 2-AZ VPC whose IDs are published as cross-stack Exports …

Evidence: describe-stacks output showing net-dev at CREATE_COMPLETE with the Outputs tab populated.



Reference architecture

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Deploy a Parameterized Application Stack that Imports Network Exports

LAB 7

Day 2 · LO2 · K3, A3

With the network baseline from Lab 6 in place, you now deploy the application tier on top of it — without copy-pasting a single subnet or VPC ID. The app template uses Fn::ImportValue to …

YOU'LL BUILD

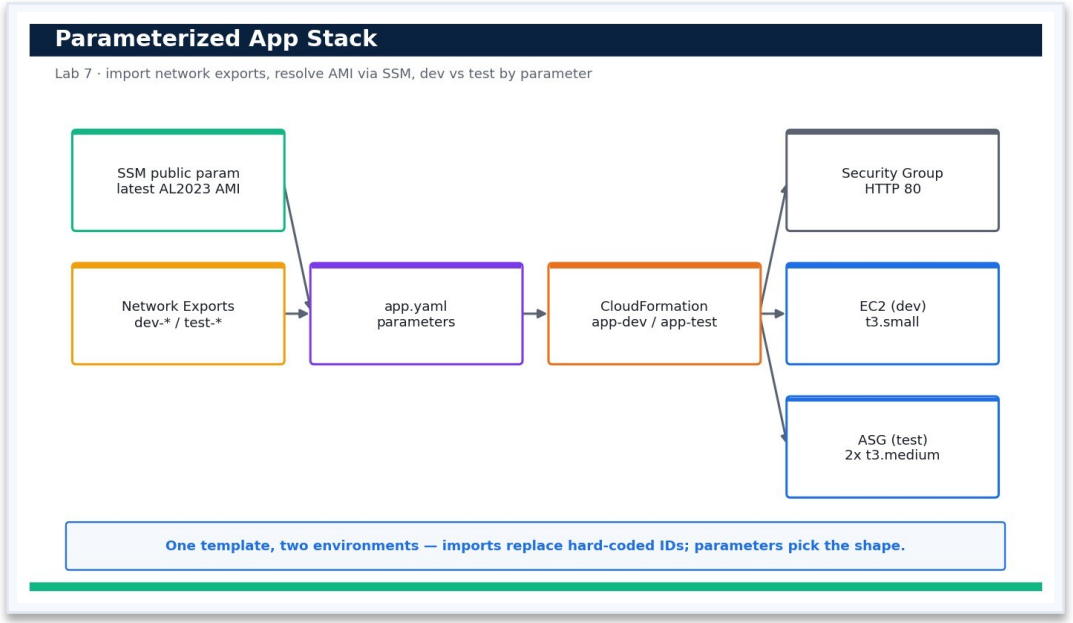
A parameterized app.yaml that imports the network exports, resolves the latest AMI via SSM and deploys a security group plus either a single EC2 instance (dev) or an Auto Scaling group (test) from the same template.

Tools: AWS CloudFormation, Amazon EC2, AWS Systems Manager (SSM public parameters)

OUTCOME & EVIDENCE

Outcome: The same app.yaml, deployed with parameters.json, yields a dev single instance, and with params-test.json yields a multi-AZ Auto Scaling …

Evidence: describe-stacks output showing app-dev and app-test both at a *_COMPLETE status.



Reference architecture

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Preview Changes with Change Sets and Detect & Reconcile Drift

LAB 8

Day 2 · LO2 · K3, A3

Production changes are risky when you cannot see them coming, and infrastructure silently drifts when people click in the console. In this lab you take the app stack from Lab 7, generate a change set to preview a before-you-execute diff, execute it, then deliberately make an out-of-band change directly on a …

YOU'LL BUILD

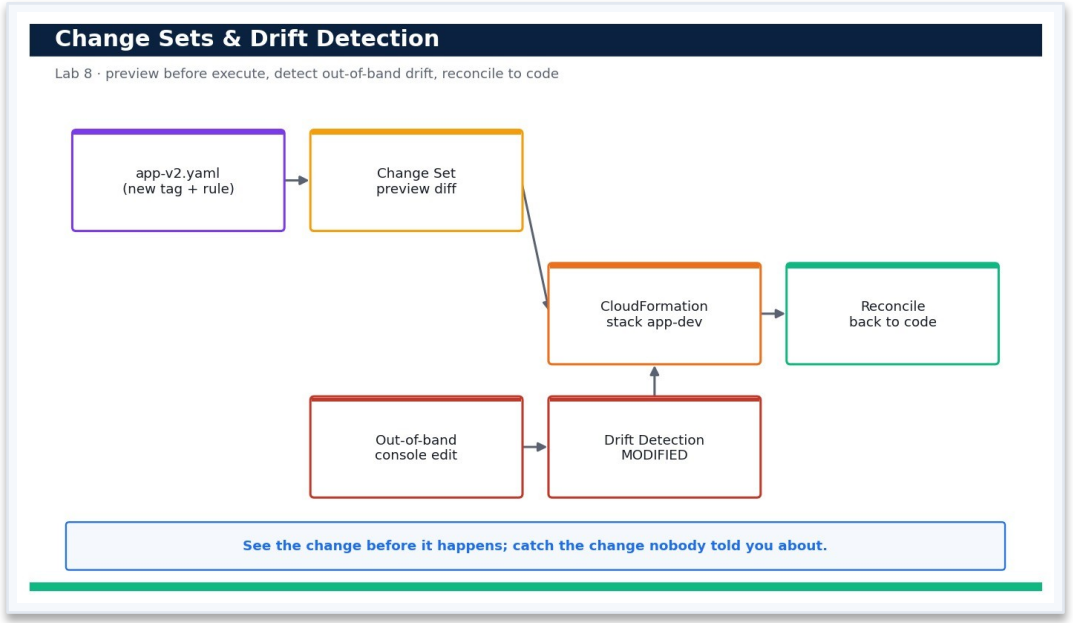
A repeatable workflow: create-change-set.sh renders a reviewable preview of a stack change, and drift-check.sh detects and reports out-of-band drift so you can reconcile infrastructure back to code.

Tools: AWS CloudFormation

OUTCOME & EVIDENCE

Outcome: A change set previews the modification (added tag + narrowed rule) with no unexpected replacements before it is executed; an …

Evidence: describe-change-set output showing the previewed changes with Replacement=False (no destructive changes).



Reference architecture

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Externalize Configuration with SSM Parameter Store

LAB 9

Day 2 · LO2 · K3, A3

Hard-coding configuration and credentials into code and templates is a recipe for leaks and rebuilds. In this lab you externalize your application's configuration into AWS Systems Manager Parameter Store — plain String and String List …

YOU'LL BUILD

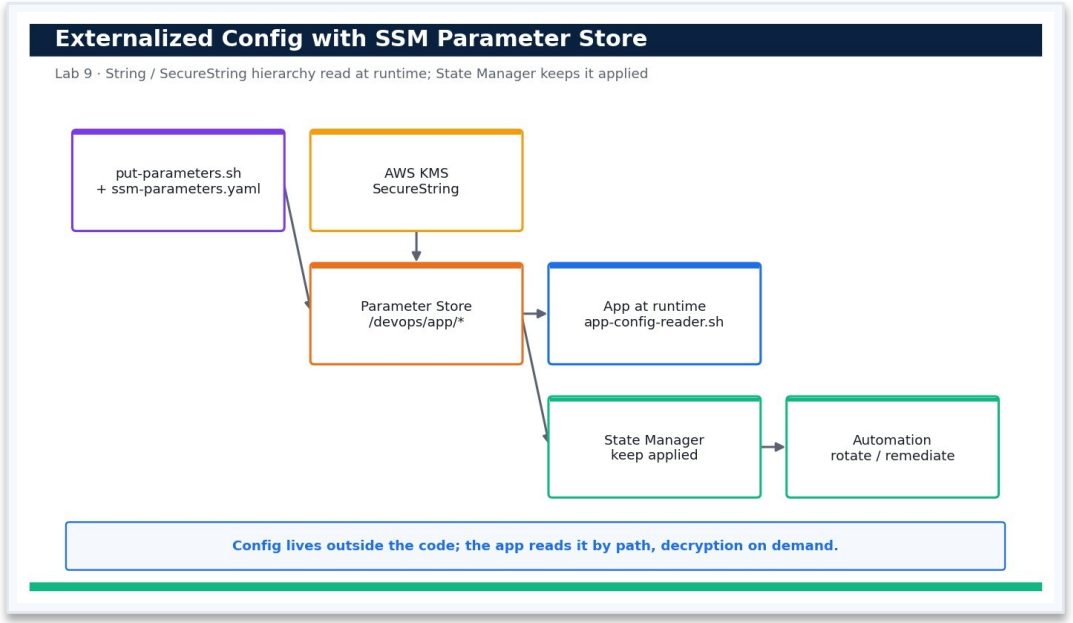
A /devops/app/* parameter hierarchy in SSM (String, StringList and KMS-encrypted SecureString) plus an app-config-reader.sh that pulls the whole configuration at runtime with decryption.

Tools: AWS Systems Manager (Parameter Store, State Manager, Automation), AWS KMS, AWS CloudFormation

OUTCOME & EVIDENCE

Outcome: An application reads all of its configuration from the /devops/app/* Parameter Store hierarchy at runtime — plain values in the clear and the …

Evidence: get-parameters-by-path output showing the /devops/app hierarchy with String, StringList and SecureString types.



Reference architecture

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LAB 10 · DAY 2 · LO2 · HANDS-ON

Multi-Account, Multi-Region Guardrails with CloudFormation StackSets

LAB 10

Day 2 · LO2 · K3, A3

Scaling infrastructure to many accounts by hand does not work. In this lab you build a governance blueprint: using CloudFormation StackSets with AWS Organizations service-managed permissions, you deploy a detective …

YOU'LL BUILD

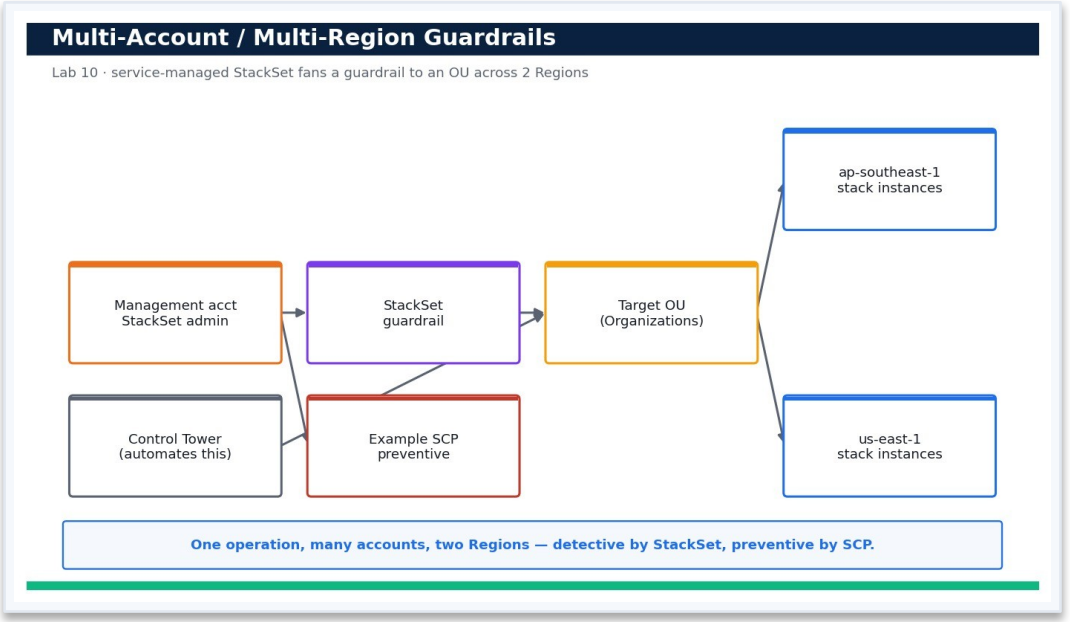
A StackSet (service-managed) that fans a guardrail stack out to an OU across two Regions in one operation, plus an example SCP that enforces the same guardrail preventively.

Tools: AWS CloudFormation StackSets, AWS Organizations, AWS Control Tower (concept)

OUTCOME & EVIDENCE

Outcome: From the management account you deploy one guardrail stack to an entire OU across two Regions in a single StackSet operation (detective …

Evidence: list-stack-instances output showing the guardrail CURRENT in each account across both Regions.



Reference architecture

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DOMAIN RECAP

Recap — Configuration Management and IaC

- You can now: Author, validate and deploy a reusable VPC network baseline with CloudFormation and publish cross-stack exports (LO2, K3, A3)
- You can now: Deploy a parameterized application stack that consumes cross-stack exports and reuse it for dev and test via parameter files (LO2, K3, A3)
- You can now: Preview a stack modification safely with a change set before executing, then detect out-of-band drift and reconcile it (LO2, K3, A3)
- You can now: Externalize application configuration into SSM Parameter Store (String and SecureString), read it at runtime, and compare Parameter Store with Secrets Manager (LO2, K3, A3)
- You can now: Design and deploy a multi-account, multi-Region guardrail using CloudFormation StackSets with AWS Organizations service-managed permissions, and express a preventive guardrail as an SCP (LO2, K3, A3)
- Infrastructure as Code — Declare infrastructure in version-controlled templates so environments are repeatable, reviewable and disposable.
- CloudFormation — Declarative JSON/YAML stacks with parameters, mappings, outputs, change sets, drift detection and automatic rollback.
- Reusable components — Nested stacks and modules parameterise a pattern once and reuse it across teams and Regions.

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03

DOMAIN 03

Resilient Cloud Solutions

Exam weighting 15% · High availability · scalability · automated recovery (RTO/RPO)

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DOMAIN 03 · WEIGHTING 15%

Key Concepts — Resilient Cloud Solutions

1

High availability

Spread compute across multiple Availability Zones behind an Elastic Load Balancer so one AZ failure is survivable.

2

Elasticity & Auto Scaling

Auto Scaling groups add and remove capacity on demand using target-tracking and step scaling policies.

3

Decoupling

SQS, SNS and EventBridge decouple producers from consumers so components fail and scale independently.

4

RTO & RPO

Recovery Time / Point Objectives drive the DR pattern: backup & restore, pilot light, warm standby or multi-site active/active.

5

Data durability

Multi-AZ RDS, DynamoDB global tables, S3 cross-Region replication and automated snapshots protect state.

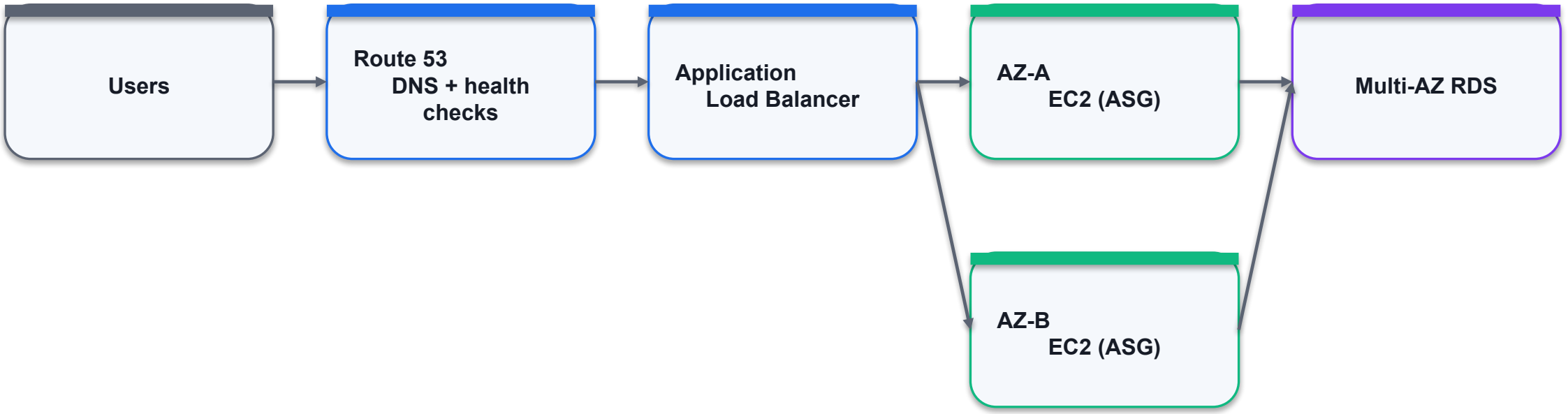
6

Automated recovery

EC2 auto-recovery, ASG health checks and Route 53 health-check failover restore service without human action.

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Highly Available, Auto-Scaling Architecture



- Traffic spreads across two AZs behind the ALB; the Auto Scaling group self-heals and Multi-AZ RDS protects data.

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MEETING RTO / RPO

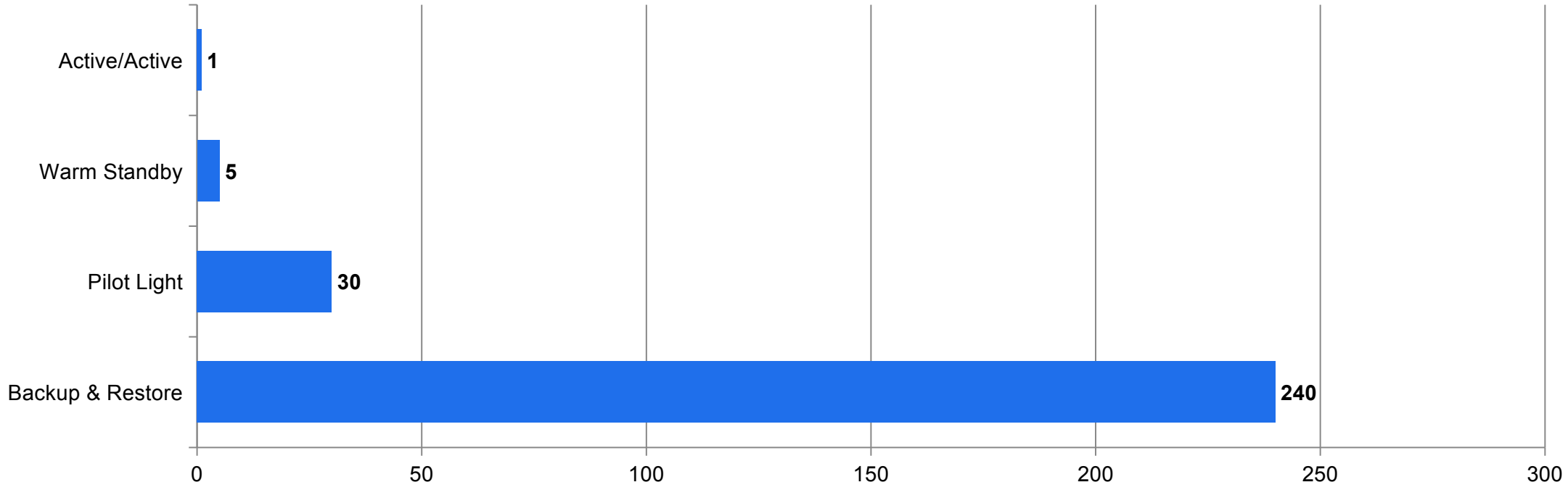
Disaster Recovery Strategies

Pattern	RTO	RPO	Cost
Backup & Restore	Hours	Hours	$
Pilot Light	10s of min	Minutes	$$
Warm Standby	Minutes	Seconds	$$$
Multi-site Active/Active	Near-zero	Near-zero	$$$$

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LOWER IS BETTER

DR Patterns — Approximate Recovery Time (RTO)



WHAT THE DATA SHOWS

As you move from backup/restore to active/active, RTO falls from hours to near-zero — at rising cost.

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DESIGN PRINCIPLES

Building for Resilience

1

Multi-AZ by default

Never depend on a single Availability Zone.

2

Elastic Load Balancing

Distribute traffic and route around unhealthy targets.

3

Auto Scaling

Match capacity to demand; replace failed instances automatically.

4

Decouple with queues

SQS / SNS / EventBridge let components fail independently.

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MATCHING CAPACITY TO DEMAND

Auto Scaling Policy Types

Policy	How it decides	Use when
Target tracking	Keep a metric at a target (e.g. 50% CPU)	Default, simplest
Step scaling	Add/remove capacity by alarm breach size	Fine-grained control
Scheduled	Scale at known times	Predictable traffic patterns
Predictive	ML forecasts demand ahead of time	Recurring cyclical load

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Elastic Load Balancer Types

Type	Layer	Best for
Application LB (ALB)	HTTP/HTTPS (L7)	Web apps, path/host routing
Network LB (NLB)	TCP/UDP (L4)	Ultra-low latency, static IP
Gateway LB (GWLB)	L3 gateway	Inline security appliances

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DURABILITY & RECOVERY

Protecting Your Data

1

Multi-AZ RDS

Synchronous standby with automatic failover.

2

DynamoDB global tables

Multi-Region, active-active replication.

3

S3 cross-Region replication

Durable object copies in another Region.

4

Automated snapshots

Point-in-time recovery for RDS/EBS.

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IN DEPTH

High availability across Availability Zones

IN DEPTH

- Availability Zones are physically isolated data centres within a Region with independent power, cooling and networking.
- Spreading compute across at least two AZs behind an Elastic Load Balancer lets the system survive a full AZ failure.
- Stateless application tiers make AZ-level redundancy trivial — any instance can serve any request.

WHY IT MATTERS

Multi-AZ is the cheapest, highest-leverage resilience decision — most real outages are single-AZ or single-instance, both survivable.

FAILURE MODE

A single-AZ deployment, or a stateful session pinned to one instance, turns a routine hardware failure into a customer-visible outage.

TRADE-OFF / DECISION

Cross-AZ traffic incurs latency and data-transfer cost — a fair price for surviving an AZ loss.

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IN DEPTH

Elasticity and Auto Scaling

IN DEPTH

- Auto Scaling groups add and remove instances to match demand and replace unhealthy ones automatically.
- Target tracking holds a metric (e.g. 50% CPU) at target; step and scheduled policies handle spikes and known patterns; predictive scaling forecasts demand.
- Health checks (EC2 and ELB) let the group cull and replace bad instances without human action.

WHY IT MATTERS

Elasticity matches cost to demand and provides self-healing — a terminated instance is silently replaced.

FAILURE MODE

Scaling on the wrong metric (CPU for an I/O-bound app) leaves you slow under load and over-provisioned when idle.

TRADE-OFF / DECISION

Aggressive scaling reacts fast but can thrash — tune cooldowns and warm-up so the group settles.

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IN DEPTH

Load balancer types and when to use them

IN DEPTH

- Application Load Balancer routes HTTP/HTTPS with host/path rules and works well with containers and Lambda targets.
- Network Load Balancer handles TCP/UDP at ultra-low latency with static IPs.
- Gateway Load Balancer inserts inline security appliances transparently.

WHY IT MATTERS

Choosing the right layer-7 or layer-4 balancer shapes routing, TLS termination, latency and cost.

FAILURE MODE

Forcing an L7 ALB where you need raw TCP throughput or a static IP adds latency and misses the requirement.

TRADE-OFF / DECISION

ALB gives rich routing at higher per-request processing; NLB gives raw speed with fewer features.

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IN DEPTH

Decoupling with queues, topics and events

IN DEPTH

- SQS buffers work between producers and consumers so a slow or failed consumer doesn't drop requests.
- SNS fans a message out to many subscribers; EventBridge routes events by content to many targets.
- Decoupled components fail and scale independently and absorb spikes as backlog rather than errors.

WHY IT MATTERS

Loose coupling converts synchronous, cascading failures into asynchronous backlogs the system can recover from.

FAILURE MODE

A synchronous call chain means one slow downstream service ties up threads and cascades into a full outage.

TRADE-OFF / DECISION

Asynchronous designs add eventual-consistency and idempotency concerns — consumers must handle duplicates and out-of-order delivery.

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IN DEPTH

RTO and RPO drive the DR pattern

IN DEPTH

- Recovery Time Objective is how quickly you must restore service; Recovery Point Objective is how much data you can afford to lose.
- Backup & restore, pilot light, warm standby and multi-site active/active trade rising cost for falling RTO/RPO.
- The right pattern is a business decision priced against the cost of downtime.

WHY IT MATTERS

Naming RTO/RPO turns 'make it resilient' into a concrete, testable and budgetable design target.

FAILURE MODE

Buying active/active for a back-office app wastes money; choosing backup-and-restore for a payment system risks the business.

TRADE-OFF / DECISION

Lower RTO/RPO means more running (and paid-for) standby capacity — match spend to the revenue at risk.

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IN DEPTH

Data durability and multi-Region protection

IN DEPTH

- Multi-AZ RDS keeps a synchronous standby with automatic failover; read replicas add scale and cross-Region copies.
- DynamoDB global tables replicate active-active across Regions; S3 offers cross-Region replication and eleven-nines durability.
- Automated snapshots and point-in-time recovery protect against data corruption and mistakes.

WHY IT MATTERS

Compute is replaceable; data is not — durable, recoverable state is the foundation everything else rebuilds on.

FAILURE MODE

A single-AZ database or un-tested backups mean the app tier can be perfectly redundant yet still lose the business's data.

TRADE-OFF / DECISION

Synchronous replication protects data but adds write latency; cross-Region copies add cost and lag.

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IN DEPTH

Automated recovery and self-healing

IN DEPTH

- ASG health checks replace failed instances; EC2 auto-recovery restores impaired instances.
- Route 53 health checks fail traffic over to a healthy Region or endpoint.
- The goal is that routine failures are handled by the system, not by a pager.

WHY IT MATTERS

Automating recovery shrinks MTTR toward zero for the common failures and frees humans for the novel ones.

FAILURE MODE

Manual failover runbooks that are never rehearsed fail exactly when you need them, under incident pressure.

TRADE-OFF / DECISION

Automated failover must guard against flapping and split-brain — tune thresholds and use health checks carefully.

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IN DEPTH

Designing for graceful degradation

IN DEPTH

- Under stress a resilient system sheds load, serves cached or default responses and protects its core function.
- Timeouts, retries with backoff and jitter, and circuit breakers stop failures from amplifying.
- Bulkheads isolate resources so one hot tenant or feature can't starve the rest.

WHY IT MATTERS

Graceful degradation keeps the most important user journeys working even when parts of the system are unhealthy.

FAILURE MODE

Unbounded retries during an incident create a retry storm that turns a brownout into a full outage.

TRADE-OFF / DECISION

Serving degraded responses reduces functionality temporarily to preserve availability — usually the right call.

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PRICING RESILIENCE

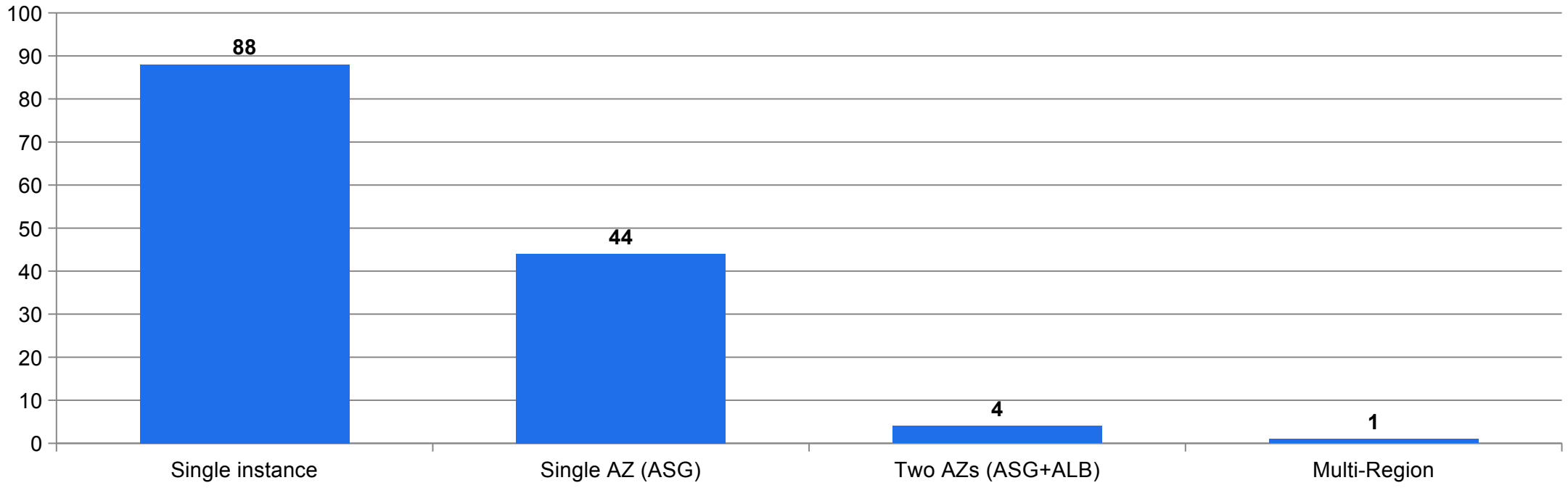
Disaster Recovery Patterns — Cost vs Recovery

Pattern	RTO	RPO	Relative cost
Backup & Restore	Hours	Hours	$
Pilot Light	Tens of minutes	Minutes	$$
Warm Standby	Minutes	Seconds	$$$
Multi-site Active/Active	Near-zero	Near-zero	$$$$

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WHY MULTI-AZ

Availability Math — Adding a Second AZ



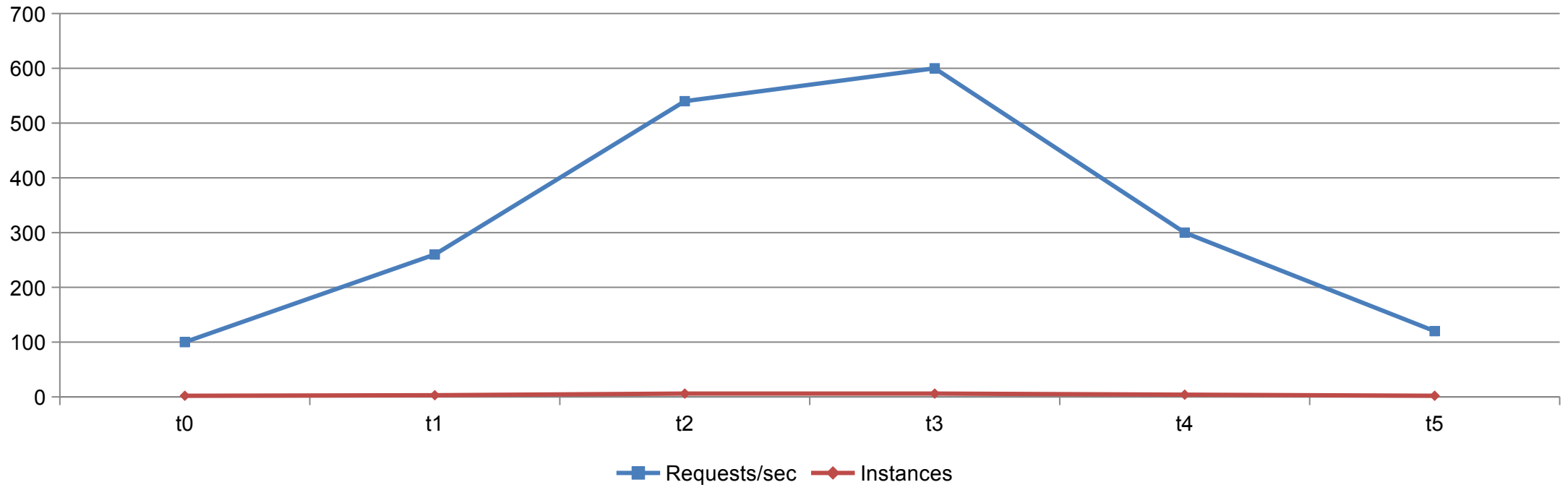
WHAT THE DATA SHOWS

Most availability gain comes from the first jump to multi-AZ; multi-Region adds resilience against rare Region events at much higher cost.

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ELASTICITY IN ACTION

Auto Scaling Under a Load Spike



WHAT THE DATA SHOWS

Target tracking follows demand up and back down, so capacity tracks load — you pay for peak only while the peak lasts.

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HOW 'HA' SYSTEMS STILL FALL OVER

Resilience Failure Modes & Anti-Patterns

1

Single-AZ database

App tier redundant, data tier isn't → Multi-AZ RDS / global tables.

2

Sticky in-memory sessions

One instance loss logs everyone out → externalise session state.

3

Retry storms

Unbounded retries amplify an outage → backoff + jitter + circuit breakers.

4

Untested failover

Runbook fails under pressure → game-days and automated failover.

5

Scaling on wrong metric

Slow under load, wasteful when idle → scale on the true constraint.

6

No health-check depth

LB routes to a process that's up but broken → deep health checks.

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PROVEN PRACTICES

Resilience Patterns Worth Adopting

1

Multi-AZ by default

Never depend on a single Availability Zone.

2

Stateless tiers

Push state to RDS/DynamoDB/ElastiCache so any instance is disposable.

3

Backoff with jitter

Spread retries to avoid synchronised load.

4

Circuit breaker

Stop calling a failing dependency; fail fast and recover.

5

Health-check failover

Route 53 / ELB route around unhealthy targets.

6

Game-days

Rehearse failure so recovery is routine, not heroic.

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MATCH SPEND TO RISK

Architecture Decision — Which DR Pattern?

A Backup & Restore

Non-critical or easily rebuilt workloads. Cheapest — but recovery takes hours and loses recent data.

B Pilot Light / Warm Standby

Important workloads needing minutes-scale recovery. A scaled-down copy runs ready to promote — moderate standing cost.

C Multi-site Active/Active

Revenue-critical systems that can't be down. Near-zero RTO/RPO — at roughly double the running cost and real complexity.

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Scenario — Surviving an Availability Zone outage

SITUATION

An AZ becomes impaired. Half the fleet is there, and the primary database node runs in it.

ANALYSIS

- Because compute is stateless behind an ALB across two AZs, the load balancer simply stops routing to the failed AZ.
- The Auto Scaling group launches replacement instances in the healthy AZ to restore capacity.
- Multi-AZ RDS fails over to the synchronous standby in the other AZ automatically.

DECISION

No human action is required for the common case — the design absorbs an AZ loss. Confirm capacity headroom so the healthy AZ can carry full load during the event.

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ELASTICITY UNDER PRESSURE

Scenario — A Black-Friday traffic spike

SITUATION

Traffic will rise 6x for a few hours during a sale, then return to normal.

ANALYSIS

- Fixed capacity sized for the peak wastes money the other 360 days; sized for normal, it falls over during the sale.
- Scheduled scaling pre-warms capacity just before the known event; target tracking then follows real demand.
- Decoupling checkout with SQS lets bursts queue instead of failing when a downstream service saturates.

DECISION

Combine scheduled pre-scaling for the known start time with target-tracking for the actual curve, and buffer spiky writes through a queue so load is absorbed as backlog.

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FROM SLA TO ARCHITECTURE

Worked Analysis — Reading an RTO/RPO Requirement

'Restore within 15 minutes' (RTO)	Rules out backup & restore (hours). Warm standby or pilot light needed.
'Lose at most a few seconds' (RPO)	Requires synchronous or near-real-time replication — Multi-AZ RDS or global tables.
'Survive a Region event'	Forces a cross-Region strategy, not just multi-AZ.
Budget is limited	Pilot light: minimal always-on footprint, scaled up on failover — a pragmatic middle.

TAKEAWAY

Each phrase in the requirement eliminates patterns until one fits — resilience design is requirement-driven, not taste-driven.

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Deploy a Multi-AZ Application Load Balancer with an Auto Scaling Group

LAB 11

Day 3 · LO3 · K2, K4, A2

A single instance in a single Availability Zone is a single point of failure. In this lab you codify a highly available web tier as one CloudFormation template: an internet-facing Application Load …

YOU'LL BUILD

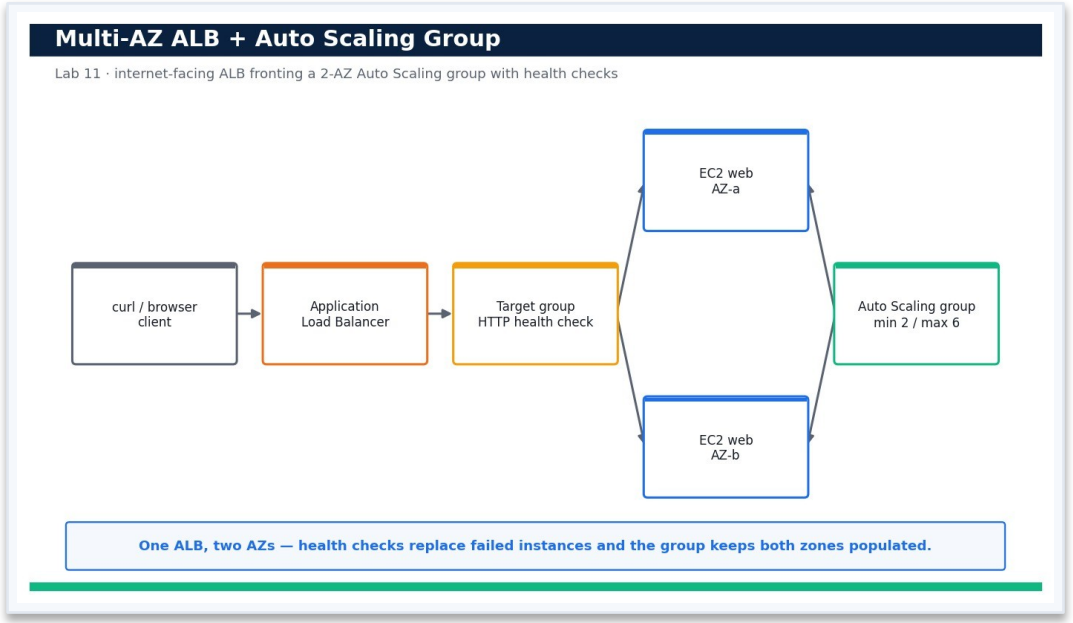
A running multi-AZ stack — an ALB, a target group with health checks, a launch template and an Auto Scaling group (min 2 / max 6) spread across two AZs — whose DNS name serves traffic from instances in both AZs.

Tools: Amazon EC2, Elastic Load Balancing (ELBv2 / ALB), AWS Auto Scaling, AWS CloudFormation

OUTCOME & EVIDENCE

Outcome: Curling the ALB DNS name 20 times returns responses from two distinct instances located in two different Availability Zones, target health …

Evidence: describe-stacks output showing stack ha-web at CREATE_COMPLETE with AlbDnsName and AsgName in the Outputs.



Reference architecture

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Attach Target-Tracking Auto Scaling and Validate Scale-Out Under Load

LAB 12

Day 3 · LO3 · K2, K4, A2

A static two-instance web tier survives an AZ failure but not a traffic spike. In this lab you attach a target-tracking scaling policy that keeps average CPU at 50% to the Auto Scaling group you built in Lab 11, generate CPU load on the instances, and watch CloudWatch alarms trip and the group scale out. When the …

YOU'LL BUILD

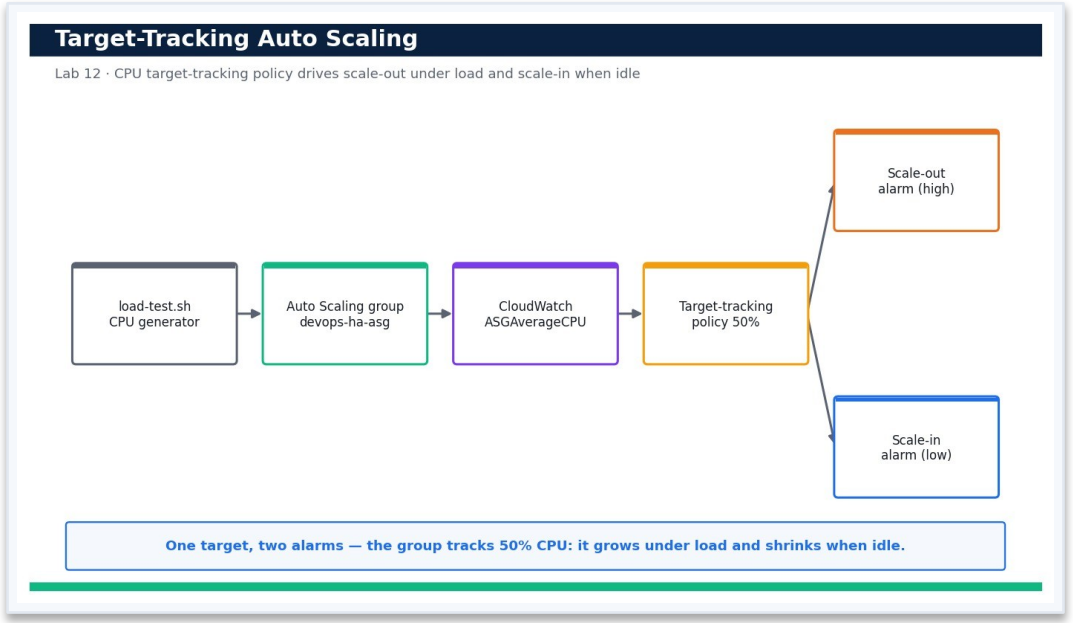
A target-tracking scaling policy (50% average CPU) on devops-ha-asg with its two auto-created CloudWatch alarms, plus a repeatable load test that drives observable scale-out and scale-in.

Tools: AWS Auto Scaling, Amazon CloudWatch, Amazon EC2

OUTCOME & EVIDENCE

Outcome: Under generated CPU load the AlarmHigh trips, the group scales out above two instances, and when the load stops the AlarmLow trips and …

Evidence: describe-policies output showing cpu-target-tracking-50 attached with TargetValue 50.0.



Reference architecture

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Select a DR Strategy by RTO/RPO and Implement a Pilot-Light Skeleton

LAB 13

Day 3 · LO3 · K2, K4, A2

Not every workload needs active-active — and not every business can accept a day-long restore. In this lab you complete an RTO/RPO worksheet, compare the four AWS DR …

YOU'LL BUILD

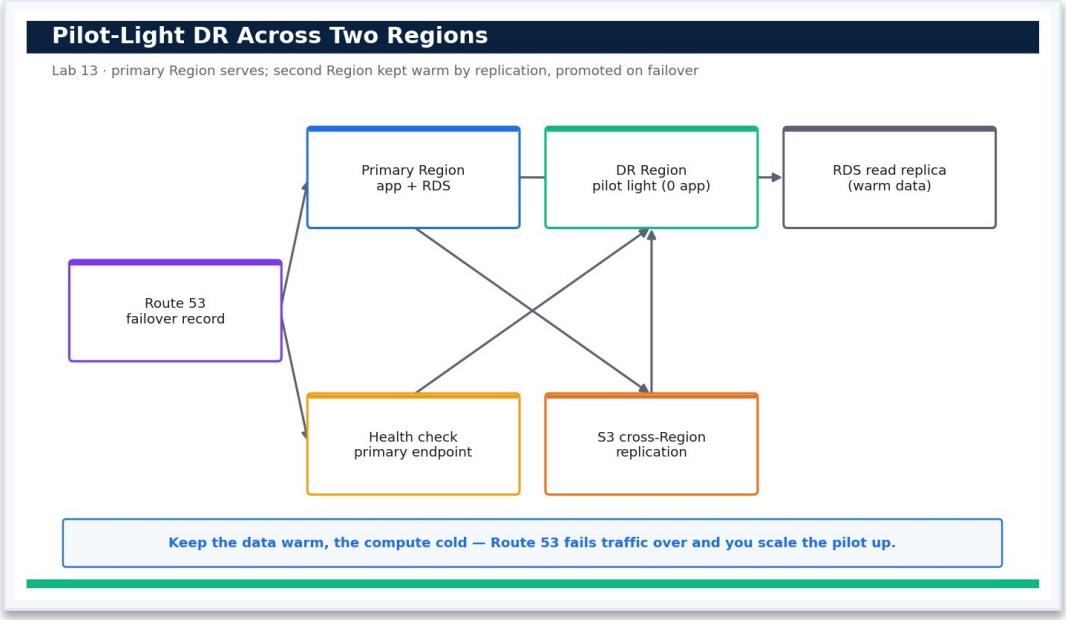
A completed DR selection worksheet plus a deployed pilot-light skeleton in a second Region (minimal VPC, a scaled-to-zero app tier and an S3 cross-Region replication path) fronted by a Route 53 failover record set.

Tools: Amazon Route 53, Amazon RDS, AWS CloudFormation, Amazon S3 (Cross-Region Replication)

OUTCOME & EVIDENCE

Outcome: A test object uploaded to the primary bucket appears in the DR Region bucket (data is warm), and the DR Auto Scaling group can be …

Evidence: The completed dr-plan.md worksheet showing the workload RTO/RPO and the justified strategy choice.



Reference architecture

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Automate Backup, Restore and Data Protection with AWS Backup

LAB 14

Day 3 · LO3 · K2, K4, A2

Resilience is not only about staying up — it is about getting your data back. In this lab you build a centralized AWS Backup plan with a scheduled rule and a tag-based resource selection so every tagged EBS volume and RDS database is backed up automatically. You then run an on-demand backup, restore …

YOU'LL BUILD

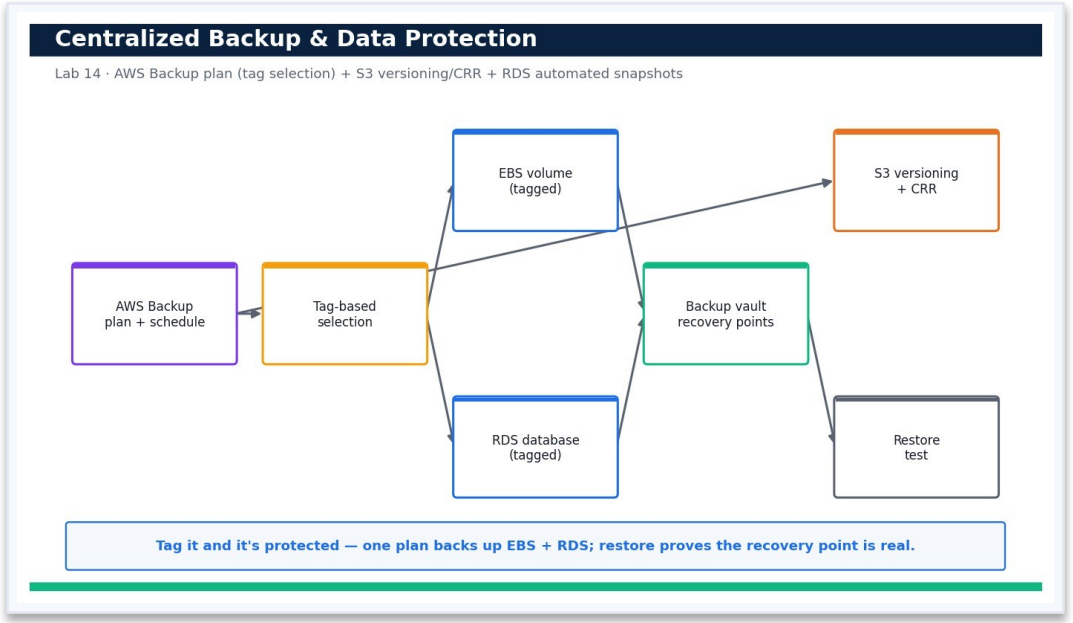
An AWS Backup plan with a tag-based selection protecting EBS and RDS, a verified restore from a recovery point, plus S3 versioning + cross-Region replication and RDS automated snapshots enabled.

Tools: AWS Backup, Amazon RDS, Amazon S3, Amazon EBS

OUTCOME & EVIDENCE

Outcome: An on-demand backup produces a COMPLETED recovery point that restores cleanly into a new EBS volume, S3 retains multiple object …

Evidence: get-backup-plan output showing devops-daily-plan with its daily rule and 35-day lifecycle.



Reference architecture

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Inject Failure and Verify Self-Healing with AWS FIS

LAB 15

Day 3 · LO3 · K2, K4, A2

A resilient design is only proven when you break it on purpose. In this lab you first terminate an instance manually and watch the Auto Scaling group launch a replacement to restore …

YOU'LL BUILD

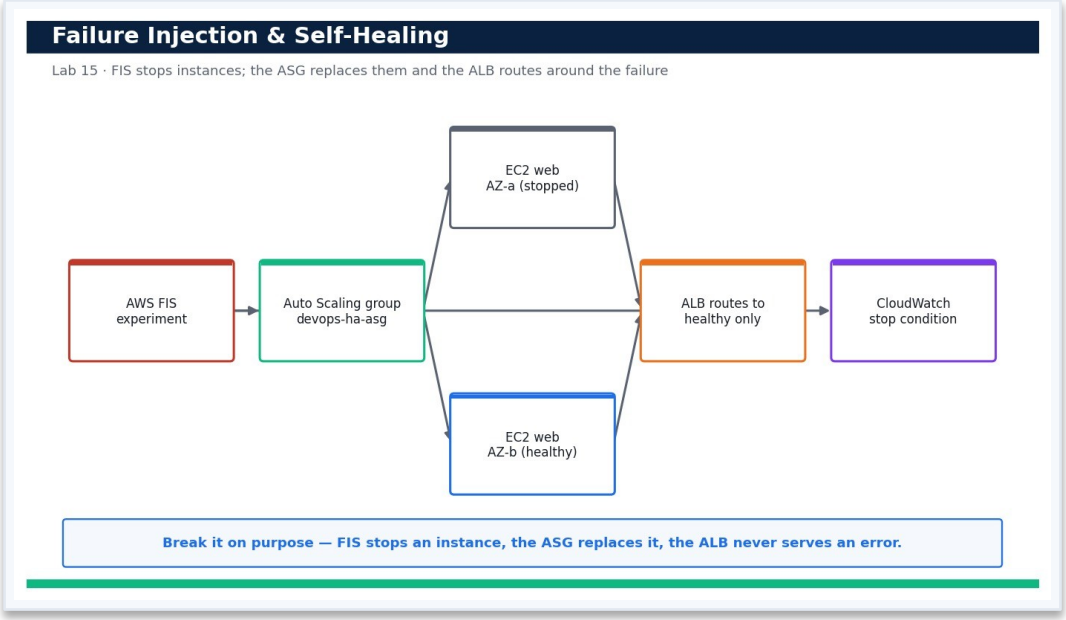
A verified self-healing web tier: a manual instance termination is auto-replaced by the Auto Scaling group, and an AWS FIS experiment stops instances while the ALB continues to serve traffic with zero errors.

Tools: AWS Fault Injection Service (FIS), AWS Auto Scaling, Amazon EC2, Amazon CloudWatch

OUTCOME & EVIDENCE

Outcome: Terminating an instance and running a FIS stop-instances experiment both leave the ALB serving HTTP 200 with zero sustained 5XX …

Evidence: describe-scaling-activities output showing a successful 'Launching a new EC2 instance' after the manual termination.



Reference architecture

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DOMAIN RECAP

Recap — Resilient Cloud Solutions

- You can now: Deploy a highly available multi-AZ Application Load Balancer fronting an Auto Scaling group and prove requests are served from both Availability Zones (LO3, K2, K4, A2)
- You can now: Attach a target-tracking scaling policy to an Auto Scaling group and validate automatic scale-out under generated load and scale-in when the load stops (LO3, K2, K4, A2)
- You can now: Compare the four AWS disaster-recovery strategies against RTO/RPO targets, select an appropriate pattern, and implement a pilot-light skeleton in a second Region (LO3, K2, K4, A2)
- You can now: Automate centralized backups with AWS Backup (plan + tag-based selection), test a restore, and add S3 versioning with cross-Region replication and RDS automated snapshots for defense in depth (LO3, K2, K4, A2)
- You can now: Prove the multi-AZ tier self-heals by terminating an instance and watching the Auto Scaling group replace it, then run an AWS FIS experiment and confirm the ALB routes around the impaired instances (LO3, K2, K4, A2)
- High availability — Spread compute across multiple Availability Zones behind an Elastic Load Balancer so one AZ failure is survivable.
- Elasticity & Auto Scaling — Auto Scaling groups add and remove capacity on demand using target-tracking and step scaling policies.
- Decoupling — SQS, SNS and EventBridge decouple producers from consumers so components fail and scale independently.

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04

DOMAIN 04

Monitoring and Logging

Exam weighting 15% · Metrics · logs · dashboards · alarms · automated event management

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Monitoring and Logging



- Resilience and observability working together to keep services healthy

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DOMAIN 04 · WEIGHTING 15%

Key Concepts — Monitoring and Logging

1

CloudWatch metrics

Namespaces, dimensions and custom metrics measure system and application health over time.

2

CloudWatch Logs

Log groups, streams, subscription filters and metric filters turn raw logs into searchable, alertable signals.

3

Dashboards

A single pane of glass correlates metrics, logs and alarms for fast troubleshooting.

4

Alarms & composite alarms

Thresholds and anomaly detection notify SNS or trigger auto-remediation before users are affected.

5

Tracing & insights

X-Ray traces distributed requests; CloudWatch Logs Insights and Container/Lambda Insights find the root cause.

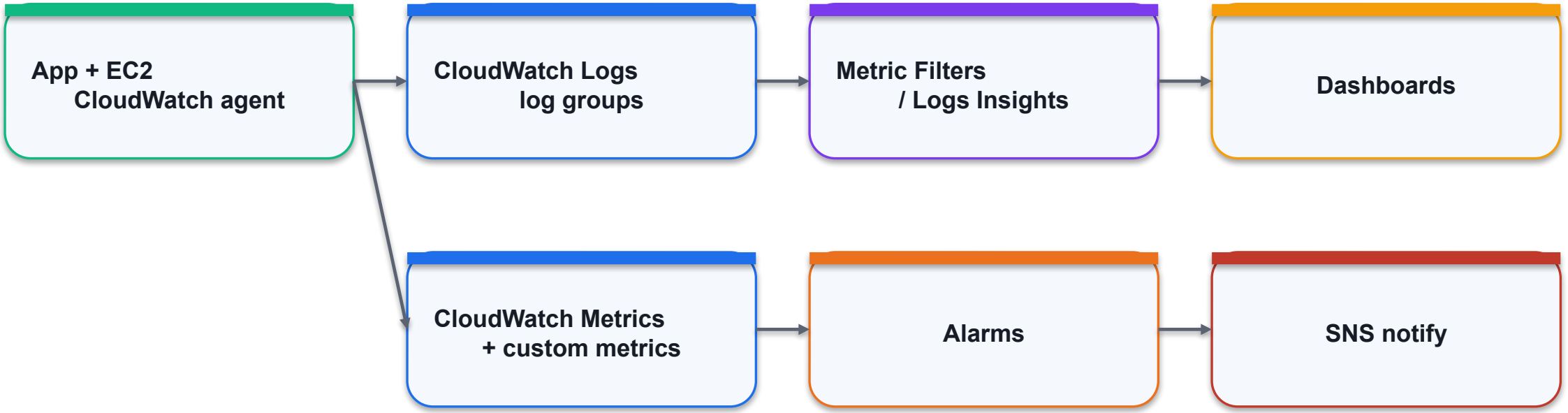
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Aggregation at scale

Cross-account observability, the CloudWatch agent and OpenTelemetry centralise telemetry from many sources.

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Monitoring & Logging Pipeline



- Logs and metrics feed dashboards and alarms; alarms notify SNS and trigger the incident response in Domain 05.

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KNOW YOUR SIGNALS

The Three Pillars of Observability

1

Metrics

Numeric time-series — CPU, latency, error rate, request count.

2

Logs

Event records — application, system and access logs in CloudWatch Logs.

3

Traces

Request paths across services with AWS X-Ray.

4

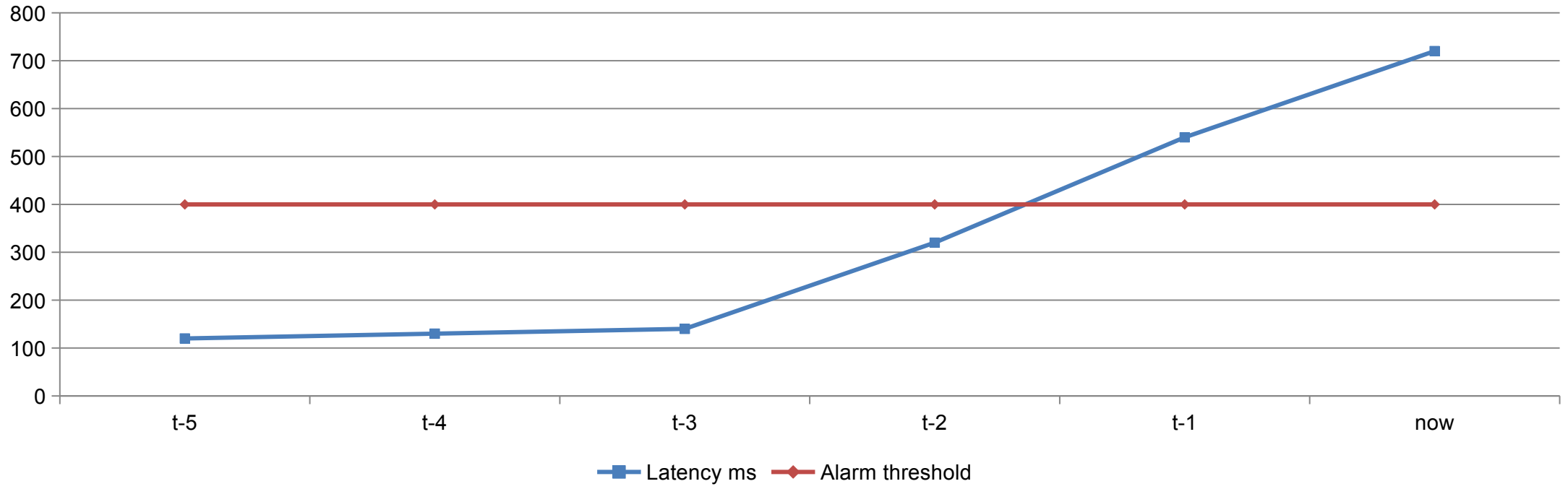
Dashboards & alarms

Correlate all three; alarm on anomalies before users notice.

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READING THE SIGNALS

Alarm Thresholds — Healthy vs Incident



WHAT THE DATA SHOWS

When p90 latency crosses the alarm threshold, CloudWatch fires and the automated response kicks in.

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THE METRICS MODEL

Anatomy of CloudWatch

1

Namespaces

Group metrics by service (AWS/EC2, custom app namespaces).

2

Dimensions

Name/value pairs that identify a metric (InstanceId, etc.).

3

Statistics & periods

Aggregate over time — Average, Sum, p90, p99.

4

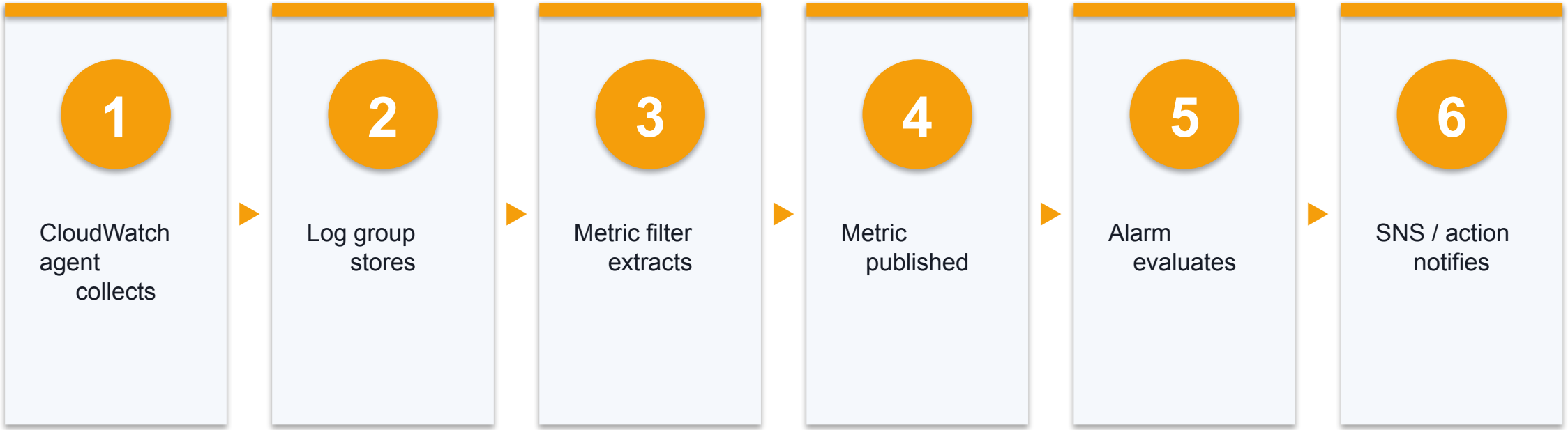
Alarms & composite

Trigger on thresholds or anomaly detection; combine with AND/OR.

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TURNING LOGS INTO SIGNALS

The Logs-to-Action Pipeline



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CloudWatch Insights Tools

Tool	What it shows
Logs Insights	Query and analyse log data with a purpose-built language
Container Insights	Metrics & logs for ECS/EKS/Fargate
Lambda Insights	Performance & cold-start metrics for functions
AWS X-Ray	Distributed traces across microservices

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IN DEPTH

The three pillars — metrics, logs, traces

IN DEPTH

- Metrics are cheap numeric time-series for trends and alarms; logs are detailed event records for context; traces follow one request across services.
- CloudWatch provides metrics and logs; X-Ray provides distributed traces.
- You alarm on metrics, investigate with logs, and localise with traces.

WHY IT MATTERS

Each pillar answers a different question — 'is it broken?', 'why?', and 'where?' — and you need all three to troubleshoot fast.

FAILURE MODE

Logging everything but graphing nothing leaves you searching text during an outage instead of seeing the problem at a glance.

TRADE-OFF / DECISION

High-cardinality metrics and verbose logs cost money and can slow queries — sample and aggregate deliberately.

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IN DEPTH

CloudWatch metrics model — namespaces, dimensions, statistics

IN DEPTH

- Metrics live in namespaces (AWS/EC2, custom app namespaces) and are identified by dimensions such as InstanceId.
- Statistics (Average, Sum, p90, p99) and periods aggregate raw data points over time.
- Custom and high-resolution metrics capture application-level signals AWS can't see.

WHY IT MATTERS

Understanding dimensions and percentiles is what separates 'average looks fine' from seeing the p99 pain real users feel.

FAILURE MODE

Alarming on Average hides tail latency — the average is healthy while your slowest 1% of users are timing out.

TRADE-OFF / DECISION

Percentiles and high-resolution metrics cost more but reveal problems averages mask.

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IN DEPTH

Turning logs into signals

IN DEPTH

- CloudWatch Logs stores log groups and streams; metric filters extract numbers (e.g. ERROR count) from log text.
- Subscription filters stream logs in real time to Lambda, OpenSearch or Kinesis for processing.
- Logs Insights queries large volumes with a purpose-built query language.

WHY IT MATTERS

A metric filter converts unsearchable text into an alarmable metric, so logs drive automated response, not just post-hoc reading.

FAILURE MODE

Relying on humans to eyeball logs means problems are found only after users complain.

TRADE-OFF / DECISION

Extracting many metrics from logs adds cost and complexity — extract the few that drive alarms and dashboards.

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IN DEPTH

Alarms, composite alarms and anomaly detection

IN DEPTH

- Alarms evaluate a metric against a threshold over N periods and change state (OK/ALARM/INSUFFICIENT_DATA).
- Composite alarms combine several alarms with AND/OR to cut noise and express real conditions.
- Anomaly detection bands adapt to normal daily/weekly patterns instead of a fixed line.

WHY IT MATTERS

Well-designed alarms notify on real, actionable conditions and feed the automated incident response in Domain 5.

FAILURE MODE

Too many noisy single-metric alarms cause alert fatigue; responders learn to ignore the pager.

TRADE-OFF / DECISION

Anomaly detection reduces false alarms but needs a learning period and can miss slow, gradual regressions.

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IN DEPTH

Dashboards and the single pane of glass

IN DEPTH

- Dashboards correlate metrics, logs and alarm state so responders see the whole system in one view.
- Service and cross-account dashboards aggregate telemetry from many sources.
- The best dashboards are organised around user journeys and the golden signals, not raw resource lists.

WHY IT MATTERS

During an incident a good dashboard turns minutes of hunting into seconds of seeing — it is a force multiplier for MTTR.

FAILURE MODE

A wall of 200 unlabelled graphs is as useless as none — no one can spot the anomaly under pressure.

TRADE-OFF / DECISION

Curated dashboards need maintenance as the system changes — stale dashboards mislead.

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IN DEPTH

Distributed tracing with X-Ray

IN DEPTH

- X-Ray captures a trace per request and a service map of dependencies and latencies.
- Traces reveal which downstream call in a chain is slow or failing.
- Sampling keeps overhead and cost bounded on high-traffic services.

WHY IT MATTERS

In microservices, a symptom (slow checkout) and its cause (a slow third-party call three hops away) are far apart — tracing connects them.

FAILURE MODE

Without tracing, teams blame each other's services because no one can see the end-to-end request path.

TRADE-OFF / DECISION

Full tracing is expensive at scale — sampling trades completeness for affordable, representative visibility.

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IN DEPTH

Observability at scale — agents, OpenTelemetry, cross-account

IN DEPTH

- The CloudWatch agent and the AWS Distro for OpenTelemetry collect host and application telemetry.
- Container Insights and Lambda Insights add curated metrics for ECS/EKS and functions.
- Cross-account observability centralises telemetry from many accounts into a monitoring account.

WHY IT MATTERS

A consistent collection standard across a large estate makes telemetry comparable and centrally queryable.

FAILURE MODE

Every team inventing its own metrics and log formats makes cross-service correlation impossible.

TRADE-OFF / DECISION

Centralisation eases correlation but concentrates cost and access — govern the monitoring account carefully.

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IN DEPTH

The golden signals and SLOs

IN DEPTH

- Latency, traffic, errors and saturation are the golden signals that describe almost any service's health.
- Service Level Objectives set target thresholds (e.g. 99.9% of requests < 300ms) and an error budget.
- Alarms and dashboards are built to protect the SLO, not to watch every raw resource metric.

WHY IT MATTERS

Tying monitoring to SLOs focuses attention on what users actually experience and gives an objective 'are we healthy?' answer.

FAILURE MODE

Alerting on CPU while the SLO quietly burns means you page on non-issues and miss the user-facing failure.

TRADE-OFF / DECISION

Strict SLOs drive reliability work but can slow feature delivery — the error budget is the negotiated balance.

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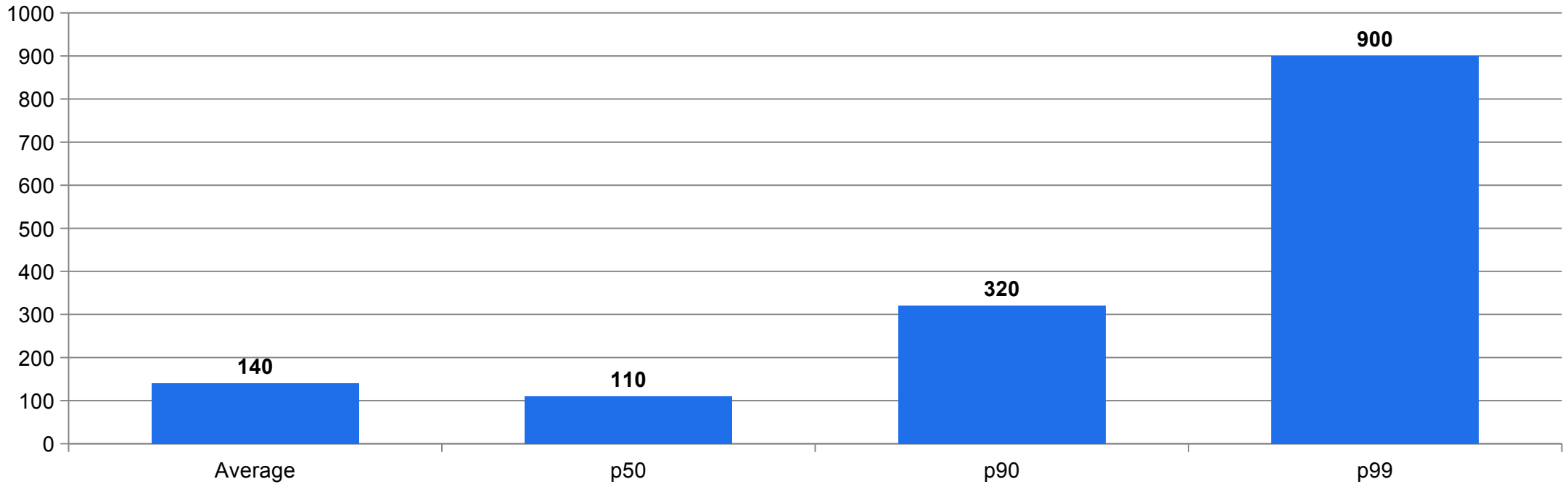
CloudWatch Insight Tools — Pick the Right Lens

Tool	What it shows	Reach for it when
Logs Insights	Ad-hoc queries over log data	You need the 'why' from raw events
Container Insights	Curated ECS/EKS/Fargate metrics	A container platform is slow
Lambda Insights	Function perf & cold starts	Serverless latency/cost issues
X-Ray	End-to-end request traces	Latency spans many services

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WHY p99 MATTERS

Average Hides the Tail — Latency Percentiles



WHAT THE DATA SHOWS

The average looks healthy at 140ms while the p99 user waits nearly a second — always alarm on the percentile your users feel.

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BLIND SPOTS THAT HURT

Monitoring Failure Modes & Anti-Patterns

1

Alarm on average

Tail latency hidden → alarm on p90/p99.

2

Alert fatigue

Noisy alarms ignored → composite alarms, SLO-based alerting.

3

Logs but no metrics

Manual searching in incidents → metric filters + dashboards.

4

No traces in microservices

Cross-team blame → adopt X-Ray/OTel.

5

Vanity dashboards

200 graphs, no signal → golden-signals layout.

6

Unmonitored dependencies

Third-party outage invisible → synthetic canaries.

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PROVEN PRACTICES

Observability Patterns Worth Adopting

1. **Golden-signals dashboards**
   Latency, traffic, errors, saturation per service.

2. **SLOs + error budgets**
   Objective health and a delivery/reliability balance.

3. **Structured logging**
   JSON logs with correlation IDs for Logs Insights.

4. **Synthetic canaries**
   Probe key journeys before real users hit failures.

5. **Metric filters → alarms**
   Turn log events into automated response.

6. **Centralised observability**
   Cross-account monitoring account.

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Architecture Decision — What Should Page a Human?

A — Page immediately

SLO-burning, user-facing symptoms (error rate, latency breach). Actionable now — keep these few and trustworthy.

B — Ticket / review later

Trends and capacity signals (rising saturation). Important but not urgent — dashboards and weekly review.

C — Auto-remediate, don't page

Known, safe fixes (restart, scale, isolate). Let EventBridge + Lambda handle it and notify, not wake, the team.

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Scenario — 'Everything is green' but users complain

SITUATION

Dashboards show CPU and memory healthy, yet support reports slow checkouts.

ANALYSIS

- Resource metrics are averages and miss the user-facing symptom; the SLO is about request latency, not CPU.
- A p99 latency metric and a Logs Insights query on the checkout path would reveal the slow requests.
- An X-Ray service map would localise the slow downstream call.

DECISION

Re-orient alerting around the golden signals and the SLO (p99 latency, error rate), and add tracing so the next 'green but slow' is visible and localisable in seconds.

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REDUCING NOISE

Scenario — Taming an alert storm

SITUATION

A single database blip fires 40 alarms across dependent services and the on-call mutes the channel.

ANALYSIS

- Many single-metric alarms describe the same root cause, creating noise and fatigue.
- A composite alarm can express 'DB unhealthy AND app errors rising' and suppress the downstream chatter.
- SLO-based alerting pages on user impact, not on every correlated resource wobble.

DECISION

Consolidate with composite alarms and alert on SLO burn, so one meaningful page fires instead of forty — restoring trust in the pager.

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THE LOGS-TO-ACTION PATH

Worked Analysis — From Log Line to Alarm

Raw log: 'ERROR checkout timeout'	Unstructured text — invisible to metrics and alarms as-is.
Metric filter extracts AppErrorCount	A numeric metric now exists in a custom namespace.
Alarm: Sum > 5 in 1 minute	The metric becomes an actionable state change.
Alarm → SNS/EventBridge	Notifies responders and can trigger auto-remediation (Domain 5).

TAKEAWAY

Observability is a pipeline: capture → extract → alarm → act. Each stage turns raw data into faster recovery.

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Collect Custom Metrics with the CloudWatch Agent and Build an Operations Dashboard

LAB 16

Day 4 · LO4 · K5, A6

Your team runs a web app on EC2 but only has the default hypervisor CPU metric — you are blind to memory pressure, disk usage and application latency. In this lab you install and configure the CloudWatch agent to publish memory and disk …

YOU'LL BUILD

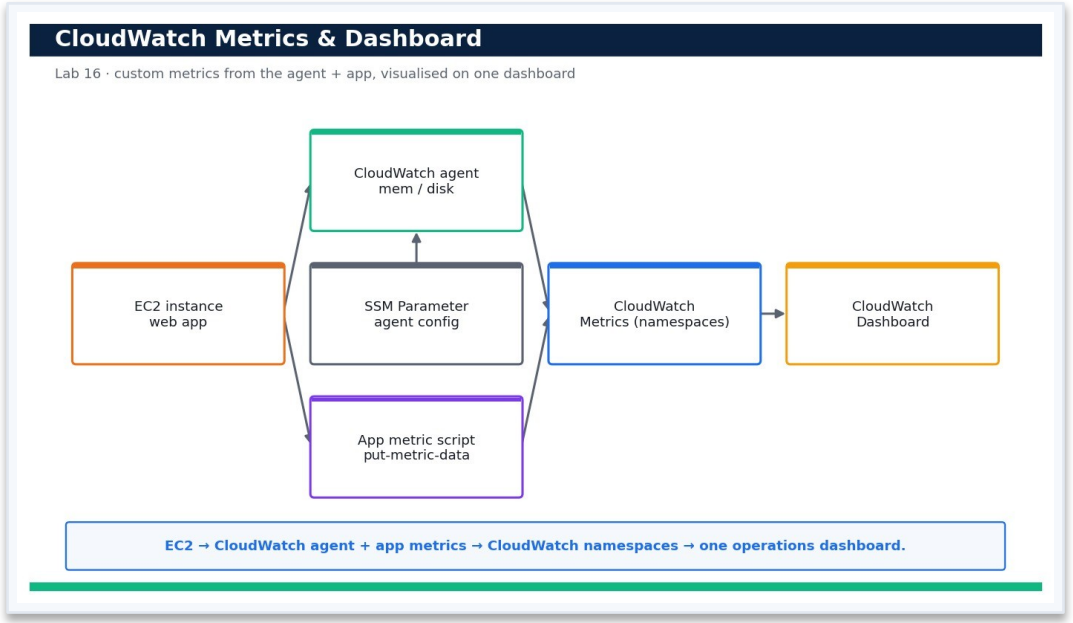
A running CloudWatch agent emitting memory/disk custom metrics, a script publishing app latency/request/error metrics, and a CloudWatch dashboard visualising CPU, p90 latency, errors and request count.

Tools: Amazon CloudWatch, CloudWatch agent, Amazon EC2, AWS Systems Manager (SSM Parameter Store), IAM

OUTCOME & EVIDENCE

Outcome: Publishing a fresh batch of samples with put-metric.sh causes the latency (p90), request-count and error-count widgets to move within one …

Evidence: Output of showing the agent running.



Reference architecture

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Centralise Logs in CloudWatch Logs and Analyse Them with Logs Insights

LAB 17

Day 4 · LO4 · K5, A6

When an incident hits, grepping across scattered instances wastes precious minutes. In this lab you centralise your application and system logs into CloudWatch Logs — creating log groups with a sensible retention policy and extending the …

YOU'LL BUILD

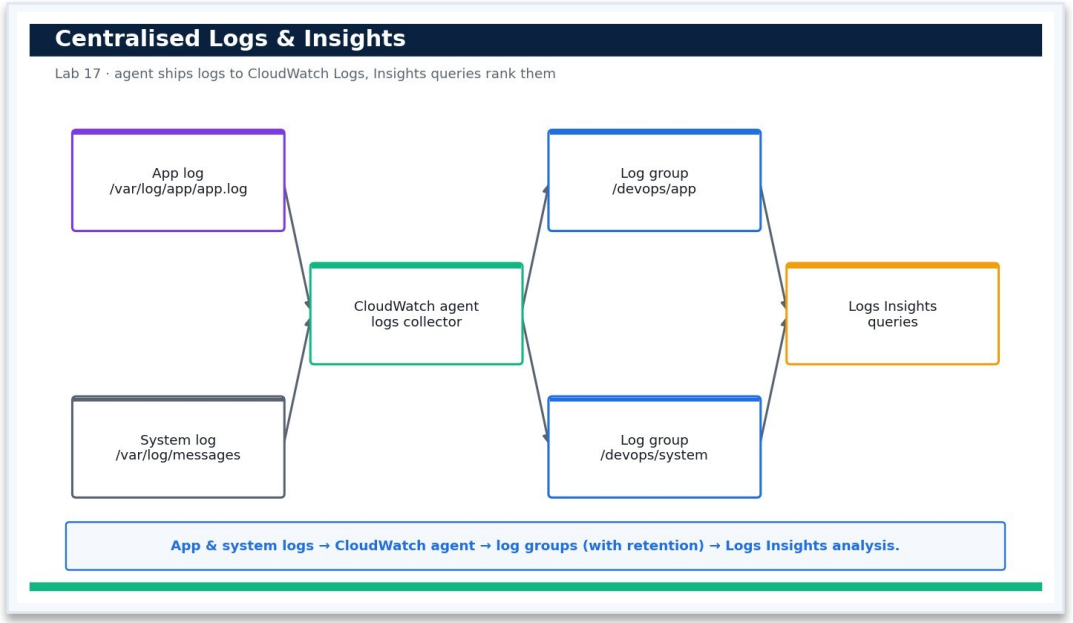
Centralised app and system log groups (with retention) fed by the CloudWatch agent, plus a set of saved Logs Insights queries that surface the slowest and most error-prone requests.

Tools: Amazon CloudWatch Logs, CloudWatch Logs Insights, CloudWatch agent, Amazon EC2, IAM

OUTCOME & EVIDENCE

Outcome: Running the 'slowest requests' and 'errors by path' Insights queries over the last 30 minutes returns concrete endpoints ordered by …

Evidence: output showing /devops/app and /devops/system with retentionInDays 14.



Reference architecture

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Turn Logs into Signals: Metric Filters, CloudWatch Alarms and SNS Notifications

LAB 18

Day 4 · LO4 · K5, A6

Logs are only useful if they page someone when things go wrong. In this lab you turn the centralised app logs from Lab 17 into a signal: a CloudWatch Logs metric filter counts ERROR lines into a metric, then two alarms — one on the error metric …

YOU'LL BUILD

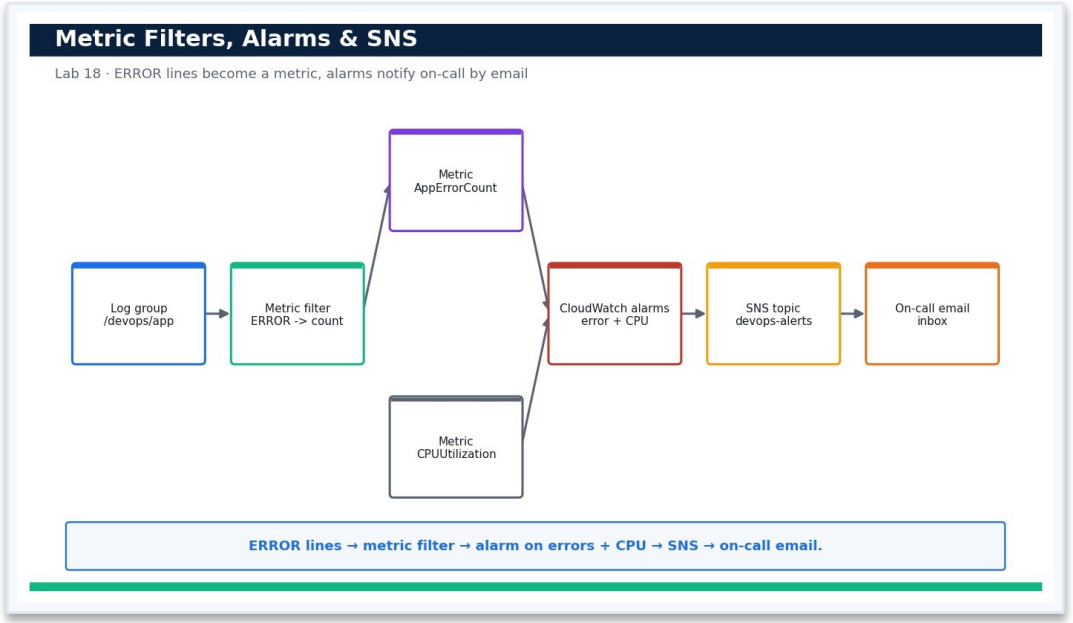
A metric filter converting ERROR log lines into a CloudWatch metric, two alarms (errors and high CPU) wired to an SNS topic with a confirmed email subscription, and a demonstrated ALARM-state email notification.

Tools: Amazon CloudWatch (metric filters + alarms), Amazon CloudWatch Logs, Amazon SNS, AWS CloudFormation, IAM

OUTCOME & EVIDENCE

Outcome: After you generate ERROR log lines, the metric filter increments AppErrorCount, the devops-app-errors alarm transitions to ALARM, and …

Evidence: output showing app-error-count attached to /devops/app.



Reference architecture

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DOMAIN RECAP

Recap — Monitoring and Logging

- You can now: Instrument an EC2 workload with memory/disk and application metrics, then visualise CPU, latency, errors and request count on a CloudWatch dashboard (LO4, K5, A6)
- You can now: Ship application and system logs into CloudWatch Logs with retention, then use Logs Insights queries to find the slowest and most error-prone requests (LO4, K5, A6)
- You can now: Create a metric filter that counts ERROR log lines, alarm on that metric and on high CPU, and notify an on-call engineer by email via SNS (LO4, K5, A6)
- CloudWatch metrics — Namespaces, dimensions and custom metrics measure system and application health over time.
- CloudWatch Logs — Log groups, streams, subscription filters and metric filters turn raw logs into searchable, alertable signals.
- Dashboards — A single pane of glass correlates metrics, logs and alarms for fast troubleshooting.

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05

DOMAIN 05

Incident and Event Response

Exam weighting 14% · Event sources · automated response · configuration change · troubleshooting

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Key Concepts — Incident and Event Response

1

Event-driven operations

EventBridge routes events from AWS services and applications to targets that act automatically.

2

Notify & act

SNS notifies people; Lambda, Systems Manager Automation and Step Functions take the remediating action.

3

Auto-remediation

AWS Config rules and EventBridge patterns detect drift and fix it — tag, isolate, restart or roll back.

4

Runbooks & playbooks

Systems Manager documents codify the response so incidents are handled consistently and fast.

5

Deployment failure response

CloudWatch alarms roll back CodeDeploy; pipeline stages stop and alert on failure.

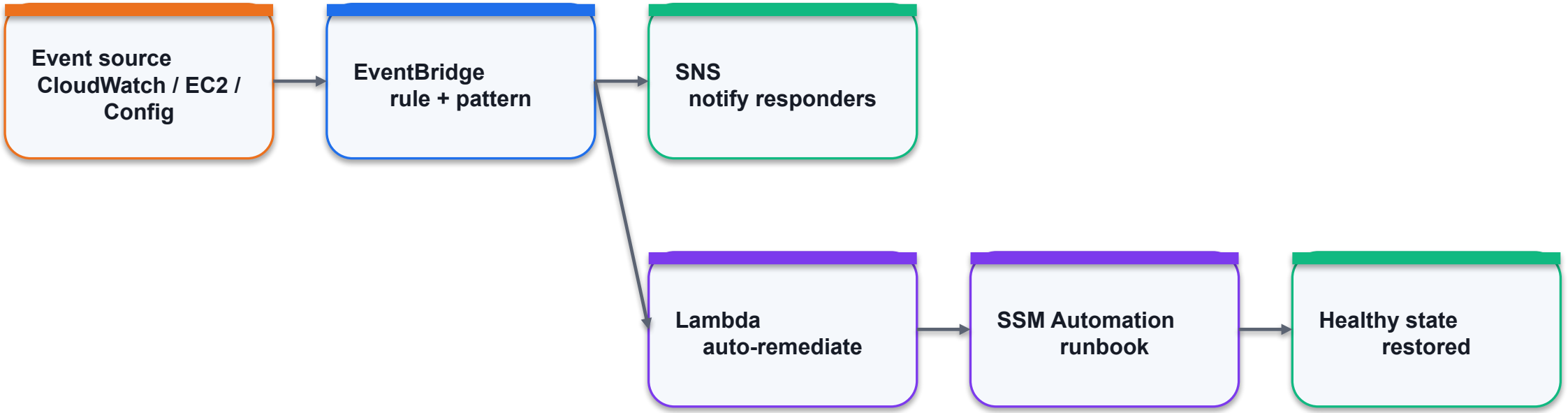
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Troubleshooting

Correlate metrics, logs, traces and CloudTrail events to find and fix system and application failures.

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Event-Driven Incident Response



- EventBridge routes the event to notify people and trigger a Lambda/SSM runbook that fixes the issue automatically.

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THE RESPONSE LOOP

From Detection to Remediation

1

Detect

CloudWatch alarms, Config rules and service events signal a problem.

2

Route

EventBridge matches the event pattern and fans out to targets.

3

Notify

SNS alerts on-call responders by email / chat / SMS.

4

Remediate

Lambda or SSM Automation runs the runbook — restart, isolate, roll back.

5

Roll back

CloudWatch alarms auto-roll-back a bad CodeDeploy deployment.

6

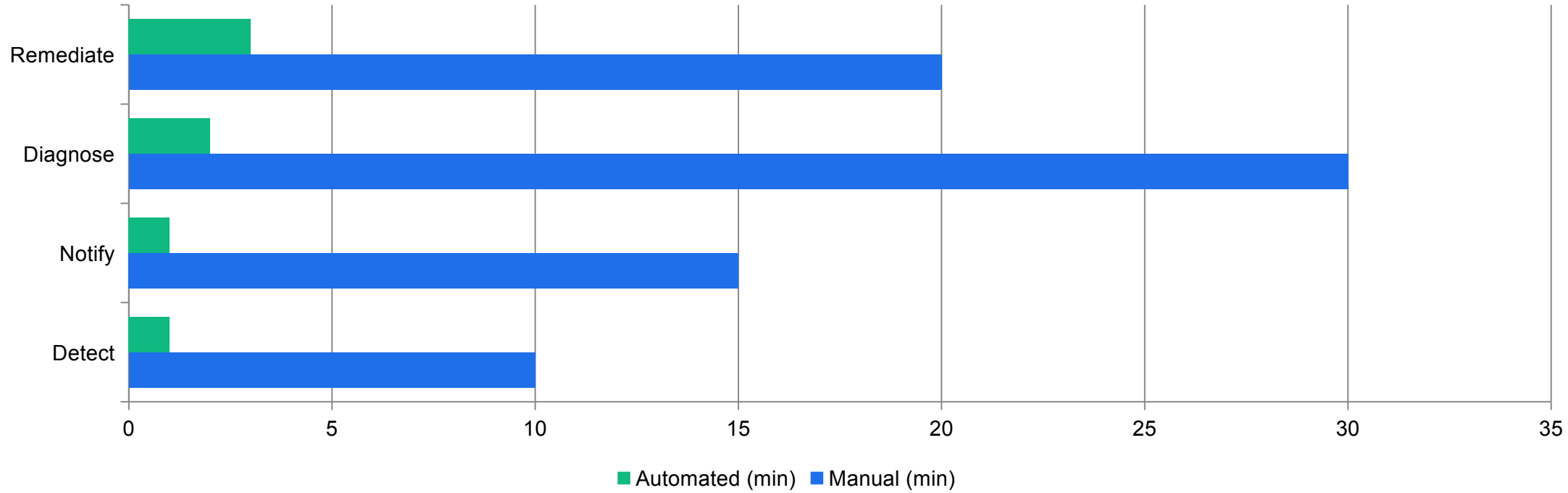
Learn

Blameless review; codify the fix into the runbook.

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WHY AUTOMATE RESPONSE

Automation Cuts Mean Time To Recover



WHAT THE DATA SHOWS

Event-driven automation collapses MTTR from over an hour to a few minutes across every phase.

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THE EVENT ROUTER

Amazon EventBridge Concepts

1

Event bus

Receives events from AWS services, SaaS and your apps.

2

Rules & patterns

Match events by source, type and detail.

3

Targets

Route matched events to Lambda, SNS, SQS, Step Functions…

4

Scheduler

Run tasks on a cron/rate schedule at scale.

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CODIFYING THE RESPONSE

Runbooks vs Playbooks

Automate the known

- Runbook
  - Step-by-step operational procedure
  - Often codified as SSM Automation documents
  - Automates a known, repeatable fix

Guide the unknown

- Playbook
  - Broader incident-response guidance
  - Roles, comms and decision points
  - Used for novel or major incidents

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FAIL SAFELY

Responding to Deployment Failures

1

Alarm-based rollback

CloudWatch alarms auto-roll-back a bad CodeDeploy deployment.

2

Pipeline stop

A failed stage halts CodePipeline and alerts the team.

3

Circuit breaker

ECS deployment circuit breaker reverts failed rollouts.

4

Blue/green switch-back

Instantly shift traffic back to the last good fleet.

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IN DEPTH

Event-driven operations with EventBridge

IN DEPTH

- EventBridge receives events from AWS services, SaaS partners and your applications on event buses.
- Rules match events by source, detail-type and content pattern and route them to targets.
- Targets include Lambda, SNS, SQS, Step Functions and Systems Manager Automation.

WHY IT MATTERS

Events let operations react the instant something changes, instead of polling or waiting for a human to notice.

FAILURE MODE

Over-broad event patterns match far more than intended and trigger remediation on benign events.

TRADE-OFF / DECISION

Fine-grained rules are precise but numerous to manage; broad rules are simple but risk false triggers.

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IN DEPTH

Notify people, trigger machines

IN DEPTH

- SNS notifies humans by email, SMS or chat; it also fans out to multiple subscribers.
- Lambda, Systems Manager Automation and Step Functions take the actual remediating action.
- The two paths run in parallel: tell the responders and start the fix simultaneously.

WHY IT MATTERS

Separating 'notify' from 'act' means the automated fix starts immediately while humans stay informed and can intervene.

FAILURE MODE

Notifying without automating leaves recovery gated on human reaction time at 3am.

TRADE-OFF / DECISION

Automating actions demands tight IAM scoping and validation so a bad event can't trigger a harmful action.

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IN DEPTH

Auto-remediation with Config and Automation

IN DEPTH

- AWS Config detects non-compliant or drifted resources and can invoke an SSM Automation remediation.
- Common auto-fixes: re-enable encryption, remove a public S3 setting, isolate an instance, re-tag resources.
- Every remediation is logged for audit and can require approval for higher-risk actions.

WHY IT MATTERS

Continuous detection plus automated remediation keeps the environment compliant without waiting for a human review cycle.

FAILURE MODE

Auto-remediating without guardrails can fight a legitimate change or mask a deeper problem instead of fixing it.

TRADE-OFF / DECISION

Immediate auto-fix is fast but can hide recurring root causes — track remediation frequency as a signal.

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IN DEPTH

Runbooks and playbooks

IN DEPTH

- A runbook is a precise, often-automated procedure for a known problem (SSM Automation documents encode these).
- A playbook is broader human guidance for novel or major incidents — roles, comms and decision points.
- Codifying responses makes them consistent, fast and improvable after every incident.

WHY IT MATTERS

Runbooks turn hard-won incident knowledge into repeatable automation; playbooks coordinate the human response to the unknown.

FAILURE MODE

Tribal knowledge in one engineer's head fails the moment that person is on leave during an incident.

TRADE-OFF / DECISION

Automated runbooks must be maintained as the system changes, or they confidently do the wrong thing.

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IN DEPTH

Deployment-failure response

IN DEPTH

- CloudWatch alarms can trigger CodeDeploy automatic rollback when a new release misbehaves.
- A failed pipeline stage halts CodePipeline and alerts the team rather than continuing to prod.
- The ECS deployment circuit breaker reverts a failed rollout; blue/green switches traffic back instantly.

WHY IT MATTERS

Wiring release health into automated rollback keeps a bad deployment from becoming a prolonged outage.

FAILURE MODE

Deployments with no health signal or rollback path require a manual, stressful, error-prone recovery.

TRADE-OFF / DECISION

Fast auto-rollback is safe but can oscillate if the health signal is noisy — tune alarm sensitivity.

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IN DEPTH

Troubleshooting methodically across signals

IN DEPTH

- Correlate metrics (what changed), logs (what the app said), traces (where it slowed) and CloudTrail (who changed what).
- Start from the user-facing symptom and the SLO, then narrow by time and by service.
- CloudTrail often reveals that an incident began with a configuration or permission change.

WHY IT MATTERS

A disciplined, signal-driven method finds root cause faster than guessing and prevents fixing symptoms only.

FAILURE MODE

Restarting things at random during an incident can clear the symptom while leaving the cause to recur.

TRADE-OFF / DECISION

Thorough diagnosis takes minutes you may feel you don't have — but mis-diagnosis costs more.

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IN DEPTH

Blameless post-incident review

IN DEPTH

- After recovery, a blameless review reconstructs the timeline and identifies systemic causes.
- Action items harden the system: new alarms, a new runbook, a guardrail, or a design change.
- The fix is codified so the same incident cannot recur silently.

WHY IT MATTERS

Treating incidents as learning — not fault — turns each outage into durable resilience and better automation.

FAILURE MODE

Blaming individuals drives problems underground; people stop reporting near-misses and the system stops improving.

TRADE-OFF / DECISION

Reviews cost time and candour — the payoff is compounding reliability over many incidents.

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IN DEPTH

Orchestrating complex responses with Step Functions

IN DEPTH

- Step Functions coordinate multi-step remediations with retries, branching and error handling.
- State machines make a response auditable and resumable rather than a fragile chain of Lambdas.
- Human-approval states can pause automation for a decision on higher-risk actions.

WHY IT MATTERS

Some incidents need a workflow, not a single action — orchestration gives reliable, observable, multi-step recovery.

FAILURE MODE

A tangle of Lambdas invoking Lambdas becomes impossible to trace or safely modify.

TRADE-OFF / DECISION

Orchestration adds a service to design and cost, justified when the response has real branching and steps.

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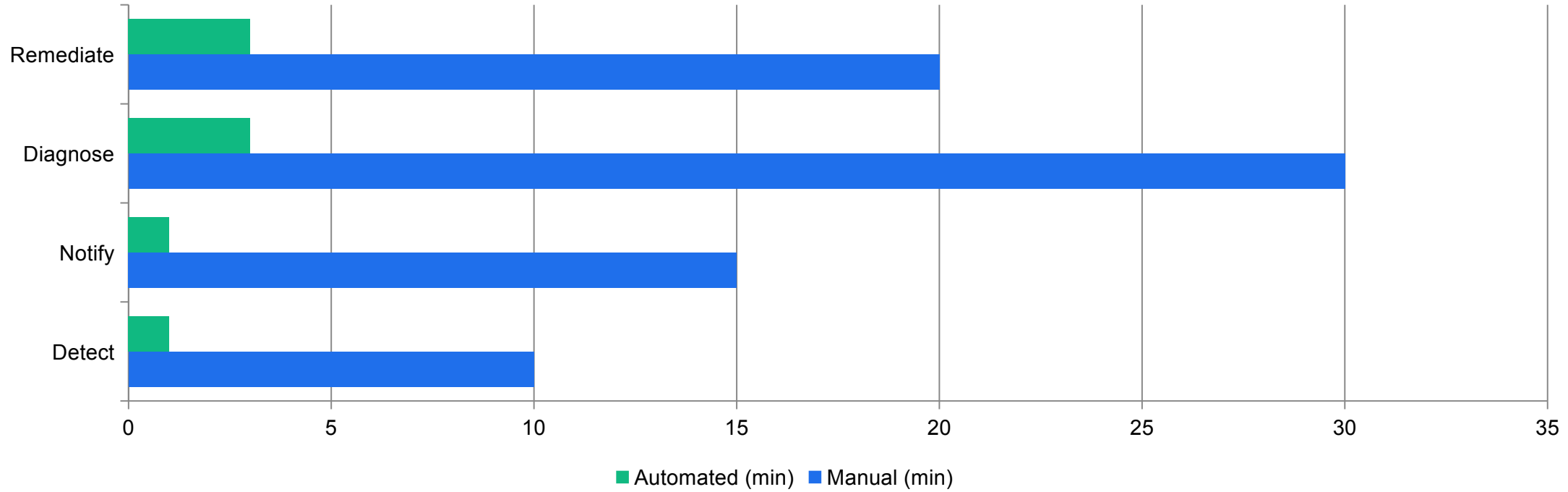
Runbook vs Playbook — Know the Difference

Aspect	Runbook	Playbook
Scope	One known, repeatable fix	Broader incident guidance
Form	Often automated (SSM Automation)	Human roles, comms, decisions
Best for	'Disk full → expand volume'	Novel or major incidents
Outcome	Fast, consistent auto-recovery	Coordinated human response

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WHY AUTOMATE RESPONSE

Automation Cuts MTTR Across Every Phase



WHAT THE DATA SHOWS

Event-driven automation collapses mean-time-to-recover from over an hour to a few minutes — most of the saving is in detection and remediation.

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WHAT SLOWS RECOVERY

Incident-Response Failure Modes & Anti-Patterns

1

Over-broad event patterns

Remediation on benign events → precise patterns, test them.

2

Notify-only, no automation

Recovery gated on human reaction → wire the fix.

3

Unscoped remediation IAM

A bad event does damage → least-privilege action roles.

4

Tribal knowledge

Fix lives in one head → codify runbooks.

5

Random restarts

Symptom cleared, cause remains → signal-driven diagnosis.

6

No post-incident learning

Same outage recurs → blameless review + action items.

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PROVEN PRACTICES

Response Patterns Worth Adopting

1. **Detect → route → notify+act**
   EventBridge fans out to people and automation.
2. **Alarm-based rollback**
   Release health auto-reverts bad deploys.
3. **Automated runbooks**
   SSM Automation for known fixes.
4. **Guarded auto-remediation**
   Config → remediation with approval for risky actions.
5. **Orchestrated recovery**
   Step Functions for multi-step responses.
6. **Blameless reviews**
   Every incident hardens the system.

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RESPONSE STRATEGY

Architecture Decision — Automate the Fix or Page a Human?

A **Fully automate**

Known cause, safe and reversible fix, frequent occurrence. Fastest recovery — needs tight IAM and validation.

B **Automate with approval**

Higher-risk action or ambiguous trigger. Step Functions pauses for a human yes/no before acting.

C **Notify only**

Novel or unclear incidents. Page the on-call with rich context; automation isn't safe until the pattern is understood.

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DETECT AND AUTO-REMEDIATE

Scenario — A public S3 bucket appears at 2am

SITUATION

Someone changes a bucket to public. Data exposure risk is immediate; no engineer is awake.

ANALYSIS

- AWS Config's s3-bucket-public-read-prohibited rule detects the non-compliant state within minutes.
- An EventBridge rule on the Config compliance-change event can trigger an SSM Automation remediation.
- The remediation re-applies Block Public Access and SNS notifies the security channel.

DECISION

Wire Config → EventBridge → SSM Automation to auto-remediate and notify, so the exposure window is minutes and unattended — with the action logged for the morning review.

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RELEASE-HEALTH AUTOMATION

Scenario — A deploy degrades error rate

SITUATION

Minutes after a rollout, the error-rate alarm crosses threshold.

ANALYSIS

- The alarm can be an automatic rollback trigger for the CodeDeploy deployment, no human needed.
- EventBridge can also notify the team and open an incident record for context.
- A post-incident action is to move the service to canary so 5% catches this next time.

DECISION

Bind the error-rate alarm to CodeDeploy auto-rollback and notify in parallel; recovery becomes automatic and the follow-up hardens the pipeline.

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EVENT TO RECOVERY

Worked Analysis — Anatomy of an Automated Response

Signal	CloudWatch alarm / Config finding / service event fires.
Route	EventBridge rule matches the event pattern precisely.
Notify	SNS informs responders with context — they watch, not scramble.
Act & verify	Lambda/SSM runbook remediates; a follow-up check confirms healthy state, else escalates.

TAKEAWAY

A good response is a small, testable pipeline — precise trigger, parallel notify-and-act, and a verification step that escalates if the fix didn't hold.

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Route Incidents with Amazon EventBridge

LAB 19

Day 4 · LO5 · K7, A5

Your team wants a single 'routing brain' for operational incidents so that a CloudWatch alarm firing, an EC2 instance changing state, or an AWS Config compliance change all land in the …

YOU'LL BUILD

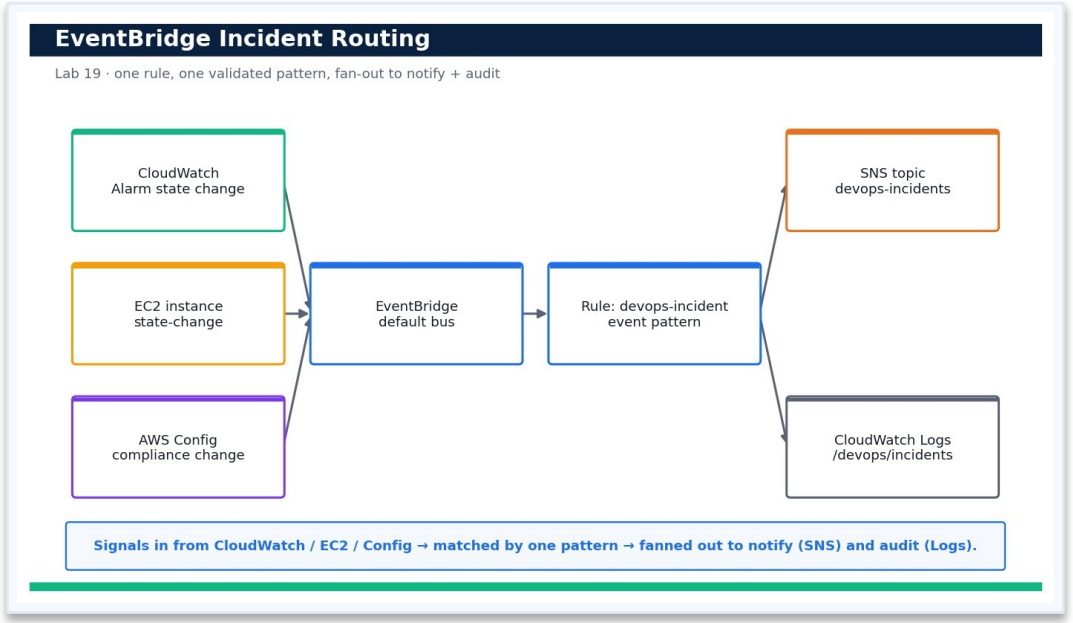
An enabled EventBridge rule on the default bus whose validated event pattern matches CloudWatch Alarm state changes, EC2 state-change notifications and AWS Config compliance-change events, fanning out to an SNS topic (notify) and a CloudWatch …

Tools: Amazon EventBridge, Amazon SNS, Amazon CloudWatch, CloudWatch Logs, IAM

OUTCOME & EVIDENCE

Outcome: You author a precise multi-source event pattern, prove it matches CloudWatch Alarm and EC2 events but rejects an unrelated S3 event …

Evidence: The three results showing true / true / false for the alarm, EC2 and unrelated samples.



Reference architecture

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Automated Remediation with Lambda and Systems Manager

LAB 20

Day 4 · LO5 · K7, A5

Notification alone does not resolve an incident — the on-call engineer still has to act. In this lab you close the loop: an EventBridge rule detects an unhealthy/stopped EC2 instance …

YOU'LL BUILD

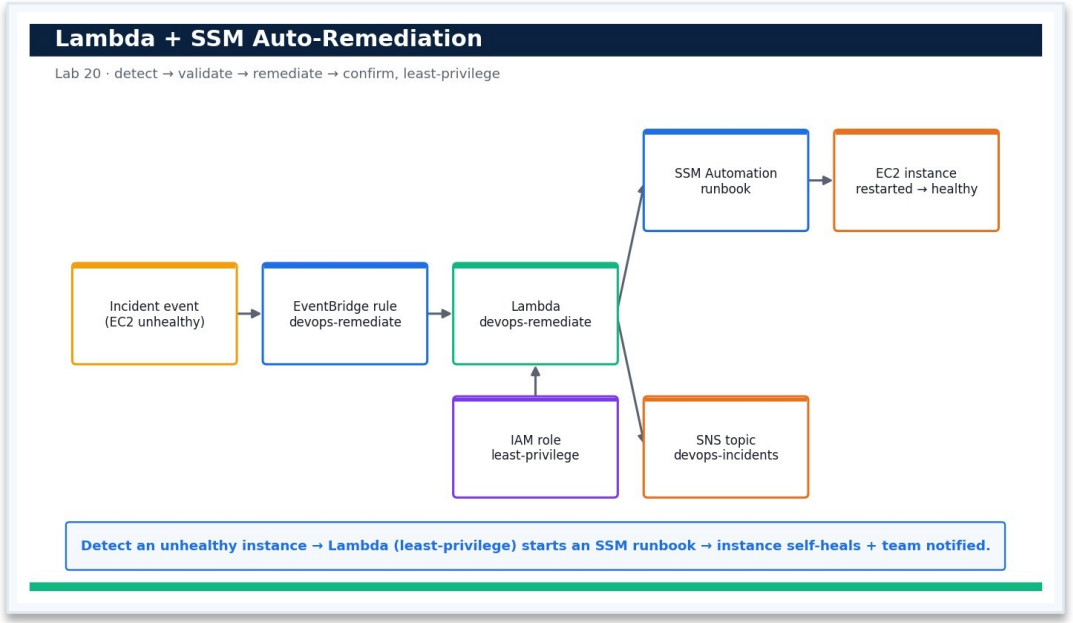
A CloudFormation-deployed auto-remediation stack (EventBridge rule + least-privilege Lambda + SSM Automation runbook + SNS notify) that, on an unhealthy-instance event, restarts the instance automatically and returns it to a running/healthy state — proven …

Tools: AWS Lambda, AWS Systems Manager Automation, Amazon EventBridge, AWS IAM, Amazon CloudWatch Logs, Amazon SNS …

OUTCOME & EVIDENCE

Outcome: You stop the test instance to create an incident, EventBridge invokes the least-privilege Lambda, the Lambda starts the SSM …

Evidence: output proving the Lambda role is least-privilege (scoped runbook + tag-conditioned EC2, no broad wildcards).



Reference architecture

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DOMAIN RECAP

Recap — Incident and Event Response

- You can now: Build an EventBridge rule that matches incident signals and fans them out to notify and log targets (LO5, K7, A5)
- You can now: Wire an EventBridge target to a least-privilege Lambda that triggers an SSM Automation runbook so an unhealthy resource self-heals (LO5, K7, A5)
- Event-driven operations — EventBridge routes events from AWS services and applications to targets that act automatically.
- Notify & act — SNS notifies people; Lambda, Systems Manager Automation and Step Functions take the remediating action.
- Auto-remediation — AWS Config rules and EventBridge patterns detect drift and fix it — tag, isolate, restart or roll back.

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06

DOMAIN 06

Security and Compliance

Exam weighting 17% · Identity & access at scale · data protection · security monitoring & audit

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Security and Compliance



- Security and compliance as continuous, automated guardrails

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DOMAIN 06 · WEIGHTING 17%

Key Concepts — Security and Compliance

1

Identity & access management

IAM users, roles, policies and permission boundaries enforce least privilege; roles beat long-lived keys.

2

IAM at scale

Organizations SCPs, IAM Identity Center and federation manage access across many accounts.

3

Data protection

KMS encryption, Secrets Manager, S3 Block Public Access and TLS protect data at rest and in transit.

4

Detective controls

CloudTrail, AWS Config, GuardDuty and Security Hub record, assess and score the account's posture.

5

Automated compliance

Config rules and conformance packs continuously check resources against a technical blueprint.

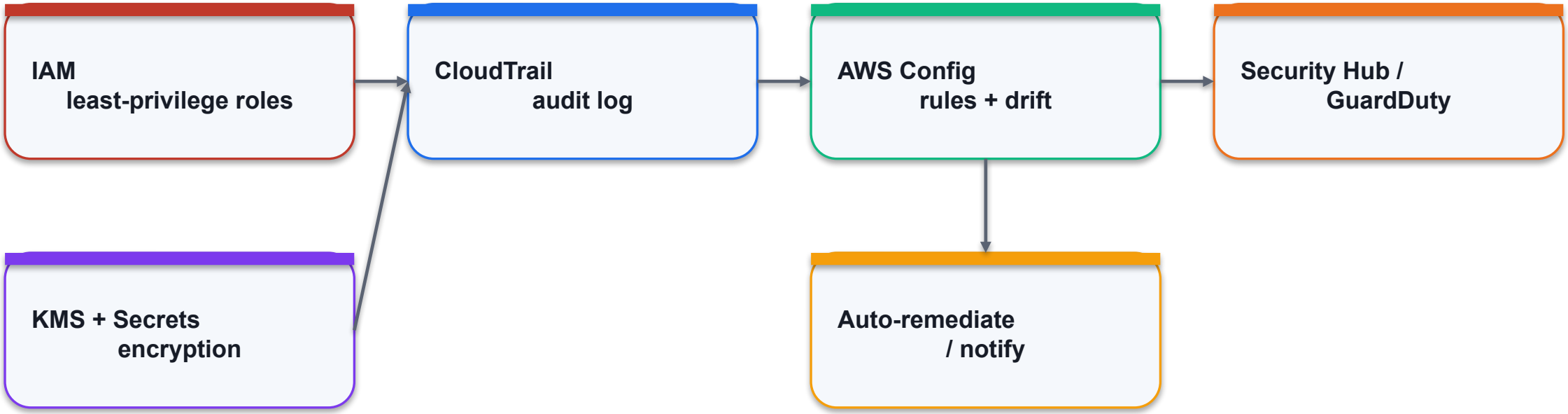
6

Governance blueprint

A landing zone plus guardrails give a documented, auditable security and compliance baseline.

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Security & Compliance Blueprint



- Least-privilege identity + encryption, with CloudTrail, Config and Security Hub providing continuous audit and scoring.

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LAYERED SECURITY

Defence in Depth on AWS

1

Identity

IAM roles, permission boundaries, IAM Identity Center, SCPs.

2

Data protection

KMS encryption, Secrets Manager, S3 Block Public Access, TLS.

3

Detective controls

CloudTrail, AWS Config, GuardDuty, Security Hub.

4

Preventive controls

Security groups, WAF, SCPs and least-privilege policies.

5

Automated compliance

Config rules and conformance packs check continuously.

6

Auditable blueprint

Landing zone + guardrails give a documented baseline.

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LEAST PRIVILEGE

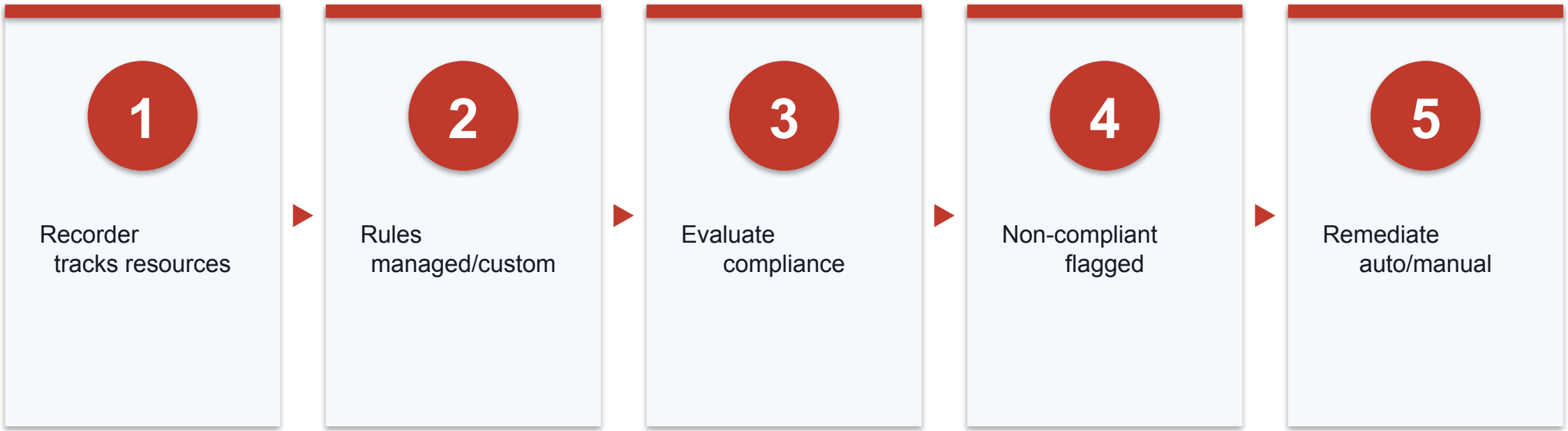
IAM — Users vs Roles vs Policies

Concept	What it is	Best practice
IAM User	Long-lived identity + credentials	Avoid; prefer federation
IAM Role	Temporary, assumable permissions	Use for apps & cross-account
Policy	JSON permissions document	Least privilege; scope by condition
Permission boundary	Max permissions a principal can have	Guardrail for delegated admin

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GUARDRAILS THAT WATCH

AWS Config — Continuous Compliance



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KNOW WHAT HAPPENED

Detective & Governance Services

Service	What it does
CloudTrail	Records every API call for audit and forensics
AWS Config	Tracks resource configuration & compliance over time
GuardDuty	ML threat detection from logs & network traffic
Security Hub	Aggregates findings & scores posture against standards

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PROTECTING DATA

Encryption Everywhere

1

At rest

KMS-managed keys for S3, EBS, RDS, DynamoDB.

2

In transit

TLS for all service and app traffic.

3

Secrets

Secrets Manager with automatic rotation.

4

Key governance

Key policies, grants and rotation with AWS KMS.

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IN DEPTH

Identity and access — least privilege by design

IN DEPTH

- IAM policies grant explicit permissions; roles provide temporary, assumable credentials for apps and cross-account access.
- Permission boundaries cap the maximum permissions a delegated admin can grant.
- Prefer roles and short-lived credentials over long-lived access keys everywhere.

WHY IT MATTERS

Least privilege limits the blast radius of any compromised credential or mistaken policy to exactly what was needed.

FAILURE MODE

A wildcard 'Action: *' policy or shared long-lived keys turn one leaked secret into full account compromise.

TRADE-OFF / DECISION

Tight policies take more effort to author and maintain — Access Analyzer and policy generation reduce that cost.

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IN DEPTH

Identity at scale — Identity Center, federation, SCPs

IN DEPTH

- IAM Identity Center centralises workforce access and single sign-on across many accounts.
- Federation lets your corporate identity provider issue temporary AWS access — no per-user IAM users.
- Service Control Policies set org-wide guardrails no account can exceed.

WHY IT MATTERS

Central identity plus org guardrails make access consistent, auditable and revocable across a whole organisation.

FAILURE MODE

Per-account IAM users multiply credentials to rotate and audit and inevitably drift out of least privilege.

TRADE-OFF / DECISION

Central control needs governance so it doesn't become a bottleneck for teams needing new access.

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IN DEPTH

Data protection — encryption and secrets

IN DEPTH

- KMS manages encryption keys and key policies for data at rest across S3, EBS, RDS and DynamoDB; TLS protects data in transit.
- Secrets Manager stores and rotates credentials; S3 Block Public Access prevents accidental exposure.
- Envelope encryption and key policies control exactly who can use which key.

WHY IT MATTERS

Encrypting everywhere and centralising key and secret governance protect data even if another control fails.

FAILURE MODE

An unencrypted volume or a public bucket exposes data regardless of how strong the network perimeter is.

TRADE-OFF / DECISION

Customer-managed keys give control and auditability but add key-management responsibility and cost.

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IN DEPTH

Detective controls — the audit trail

IN DEPTH

- CloudTrail records every API call for audit and forensics; AWS Config tracks configuration state over time.
- GuardDuty applies ML threat detection to logs and network flows; Security Hub aggregates and scores findings.
- Together they answer 'what happened, when, by whom, and are we compliant?'

WHY IT MATTERS

Detective controls provide the evidence for incident forensics, audits and continuous compliance scoring.

FAILURE MODE

No CloudTrail (or a trail without log-file validation) means an incident can't be reconstructed and audits fail.

TRADE-OFF / DECISION

Full logging and threat detection cost money and generate findings to triage — tune and prioritise by severity.

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IN DEPTH

Automated compliance — Config rules and conformance packs

IN DEPTH

- Config rules continuously evaluate resources against a technical blueprint (managed or custom rules).
- Conformance packs bundle rules and remediations to a standard (e.g. CIS, PCI) and deploy org-wide.
- Non-compliant resources are flagged and can be auto-remediated.

WHY IT MATTERS

Continuous, codified compliance replaces point-in-time audits with always-on assurance and evidence.

FAILURE MODE

Manual quarterly audits miss the days-to-weeks a resource sits non-compliant between checks.

TRADE-OFF / DECISION

Strict conformance packs improve posture but can generate noise — phase in rules and remediations.

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IN DEPTH

Preventive controls and defence in depth

IN DEPTH

- Security groups and network ACLs, WAF, and least-privilege policies stop bad requests before they reach resources.
- SCPs prevent whole classes of action org-wide; VPC design limits lateral movement.
- Layered controls mean no single failure exposes the system.

WHY IT MATTERS

Defence in depth ensures that when one layer fails — and layers do fail — others still protect the data.

FAILURE MODE

Relying on a single perimeter (a firewall) fails completely the moment that one control is bypassed.

TRADE-OFF / DECISION

More layers add complexity and can impede legitimate traffic — design them to be observable and testable.

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IN DEPTH

Governance blueprint — landing zone and guardrails

IN DEPTH

- A landing zone (via Control Tower) gives a documented, auditable security and network baseline.
- Preventive guardrails (SCPs) and detective guardrails (Config rules) are applied to every account.
- New accounts inherit the baseline automatically through account vending.

WHY IT MATTERS

A governance blueprint makes 'secure and compliant' the default state of every account, not a per-project effort.

FAILURE MODE

Ad-hoc, per-project security guarantees nothing is consistent and every audit re-discovers the same gaps.

TRADE-OFF / DECISION

A central blueprint needs a platform team and change process — the payoff is org-wide, provable compliance.

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IN DEPTH

Securing the pipeline itself (DevSecOps)

IN DEPTH

- Scan source and dependencies (SAST/SCA) and infrastructure templates in the build; sign and verify artifacts.
- Scope pipeline and deployment roles to least privilege and isolate environments by account.
- Store no secrets in code; pull them at deploy time from Secrets Manager/SSM.

WHY IT MATTERS

The pipeline has powerful credentials and touches production — a compromised pipeline is a compromised environment.

FAILURE MODE

A pipeline role with broad permissions and secrets in the repo is a direct path from a code leak to full compromise.

TRADE-OFF / DECISION

Security gates add build time — order them fast-to-slow and fail fast to keep the pipeline usable.

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LEAST PRIVILEGE AT SCALE

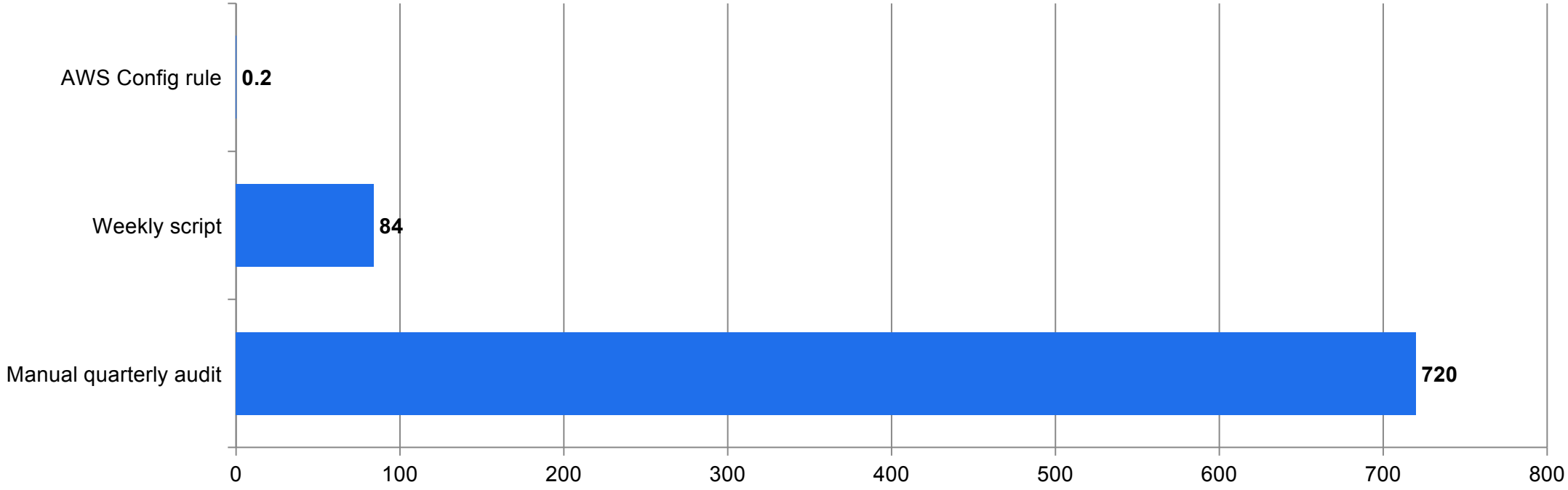
IAM — Users vs Roles vs Federation

Identity	What it is	Best practice
IAM user + keys	Long-lived credential	Avoid; rotate/retire, prefer roles
IAM role	Temporary, assumable	Use for apps and cross-account
Federation / Identity Center	IdP issues short-lived access	Standard for workforce access
Permission boundary	Cap on granted permissions	Guardrail for delegated admins

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CONTINUOUS COMPLIANCE WINS

Time-to-Detect a Public Bucket — Manual vs Continuous



WHAT THE DATA SHOWS

Continuous Config evaluation shrinks the exposure window from months to minutes — the difference between a finding and a breach.

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HOW ACCOUNTS GET COMPROMISED

Security Failure Modes & Anti-Patterns

1

Wildcard IAM policies

One leaked key = full access → least privilege + Access Analyzer.

2

Long-lived access keys

Hard to rotate/audit → roles and federation.

3

Public S3 / open SGs

Accidental exposure → Block Public Access, Config rules.

4

No CloudTrail / validation

Incidents unauditable → org trail + log-file validation.

5

Secrets in code

Leak on repo access → Secrets Manager/SSM.

6

Over-privileged pipeline

Code leak → prod compromise → scope deploy roles.

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PROVEN PRACTICES

Security Patterns Worth Adopting

1. **Least privilege everywhere**
   Roles, boundaries, scoped policies, no wildcards.

2. **Encrypt at rest and in transit**
   KMS + TLS by default on every store.

3. **Continuous compliance**
   Config rules + conformance packs, auto-remediate.

4. **Central logging account**
   Immutable CloudTrail + Security Hub aggregation.

5. **Guardrails by SCP**
   Prevent whole action classes org-wide.

6. **DevSecOps**
   Scan code, deps and IaC in the pipeline.

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PREVENT, DETECT, REMEDIATE

Architecture Decision — Where Do Compliance Controls Live?

A — Preventive (SCP / policy)

Stop the action from ever happening org-wide. Strongest — but blunt; can block legitimate edge cases.

B — Detective (Config / GuardDuty)

Allow, but continuously detect and score. Flexible and auditable — leaves a (short) exposure window.

C — Automated remediation

Detect then auto-fix (Config → SSM). Closes the window fast — needs careful scoping and logging.

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EVIDENCE ON DEMAND

Scenario — Passing an external security audit

SITUATION

An auditor asks you to prove that all volumes are encrypted and no bucket is public, historically and now.

ANALYSIS

- Point-in-time console screenshots don't prove continuous compliance over the audit period.
- AWS Config records configuration history and rule-compliance over time — that history is the evidence.
- CloudTrail with log-file validation proves the audit record itself wasn't tampered with.

DECISION

Present Config compliance history and conformance-pack results as continuous evidence, backed by a validated CloudTrail — turning audit from a scramble into a query.

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LEAST PRIVILEGE IN PRACTICE

Scenario — An app needs one new permission

SITUATION

A service fails with AccessDenied writing to a new S3 prefix. The quick fix is to grant s3:* on all buckets.

ANALYSIS

- A wildcard grant fixes the error but massively widens blast radius if the role is ever compromised.
- IAM Access Analyzer can generate a policy scoped to exactly the actions and resources the app used.
- The correct grant is s3:PutObject on that specific bucket/prefix, nothing more.

DECISION

Grant the single scoped permission the app actually needs (not a wildcard), using Access Analyzer to right-size the policy — keeping least privilege intact under time pressure.

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Worked Analysis — Building a Compliance Blueprint

Requirement: encrypt all data at rest	Config rule encrypted-volumes + S3 default encryption; KMS keys with scoped policies.
Requirement: no public data	S3 Block Public Access org-wide + s3-bucket-public-read-prohibited rule.
Requirement: full audit trail	Org CloudTrail to an encrypted, access-controlled logging account with validation.
Requirement: least-privilege identity	Roles + Identity Center + permission boundaries + Access Analyzer reviews.

TAKEAWAY

A compliance blueprint maps each written requirement to a concrete, testable AWS control — so 'we are compliant' is something you can prove, continuously.

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Enforce IAM Least Privilege and Analyse Policies for Over-Permissive Access

LAB 21

Day 5 · LO6 · K6, A4

Broad permissions are the single most common cloud security failure. In this lab you replace a wildcard grant with a tightly scoped policy for the sample app's role, attach a permission boundary so the role can never exceed its remit, then use IAM Access Analyzer policy validation, and an Access Analyzer …

YOU'LL BUILD

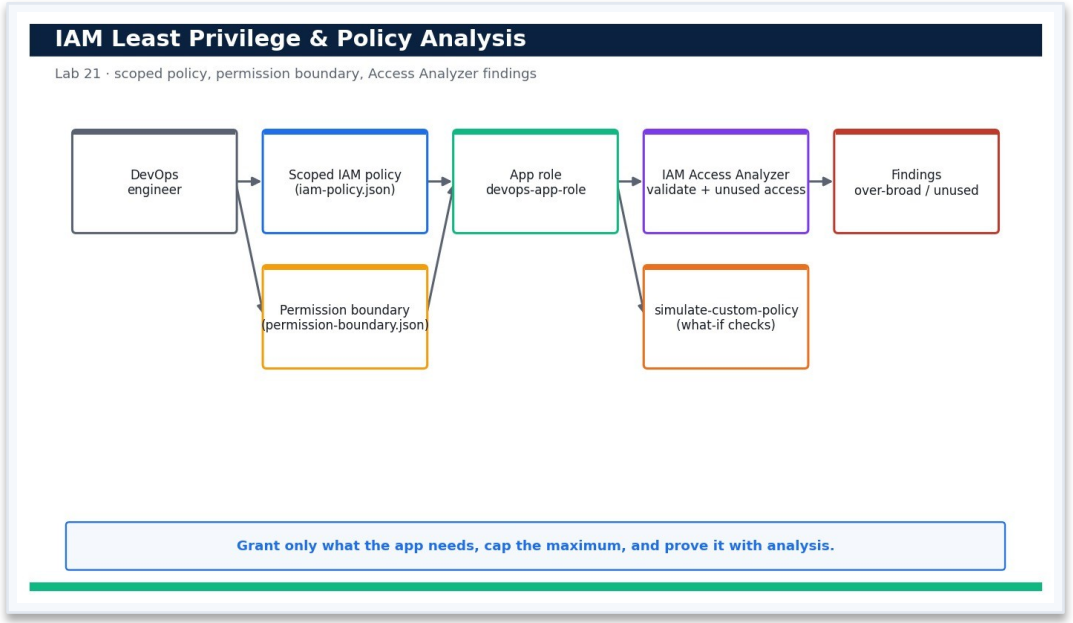
A least-privilege IAM policy for the app role, capped by a permission boundary, that passes Access Analyzer validation with no security warnings and grants nothing beyond the app's real needs.

Tools: AWS IAM, IAM Access Analyzer, AWS STS, AWS CloudFormation

OUTCOME & EVIDENCE

Outcome: You can prove, with Access Analyzer and policy simulation, that the app role grants exactly the actions the app needs and nothing more …

Evidence: The output showing zero SECURITY_WARNING findings for iam-policy.json.



Reference architecture

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Protect Data with Secrets Manager and Customer-Managed KMS Encryption

LAB 22

Day 5 · LO6 · K6, A4

Sensitive data must be encrypted with keys you control and secrets must never live in code. In this lab you create a customer-managed KMS key with a least-privilege key policy, store an application credential in AWS Secrets Manager …

YOU'LL BUILD

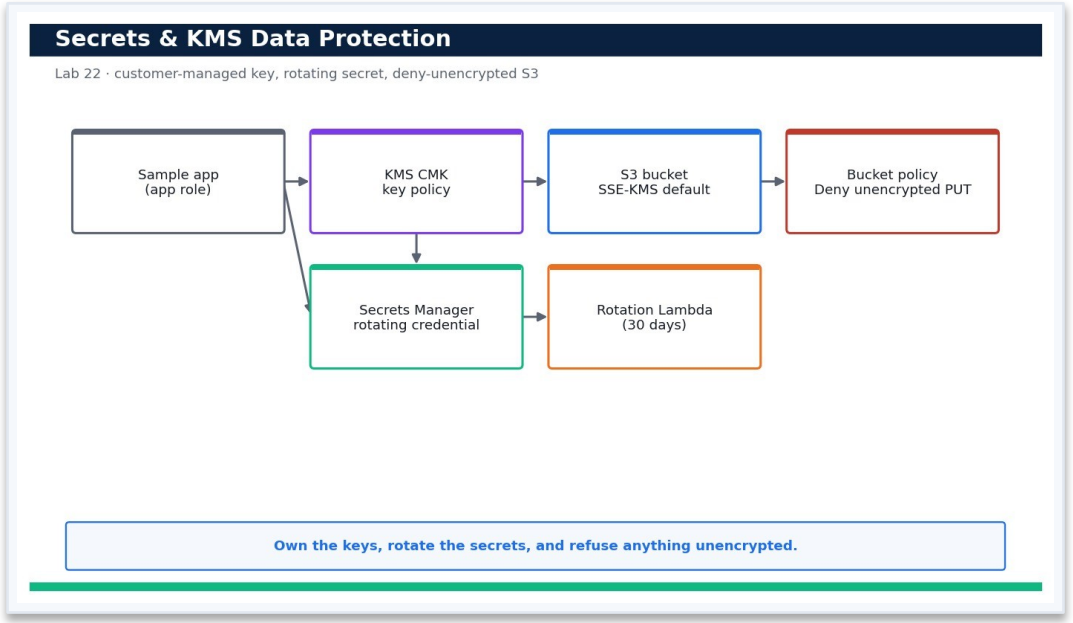
A customer-managed KMS key with a scoped key policy, a rotating Secrets Manager secret, and an S3 bucket that encrypts by default with your key and rejects any unencrypted upload.

Tools: AWS KMS, AWS Secrets Manager, Amazon S3, AWS Lambda (rotation), AWS CloudFormation

OUTCOME & EVIDENCE

Outcome: You can store and rotate a secret in Secrets Manager under a customer-managed KMS key, retrieve it via KMS decrypt, and prove the …

Evidence: describe-key output showing KeyManager CUSTOMER, KeyState Enabled and rotation enabled for alias/devops-app.



Reference architecture

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Produce CloudTrail Audit Evidence and Answer Who-Did-What-When

LAB 23

Day 5 · LO6 · K6, A4

Auditors and incident responders need a tamper-evident record of every API call. In this lab you create a multi-Region CloudTrail with log-file validation that delivers to an encrypted, access-blocked S3 bucket, then query the events two ways …

YOU'LL BUILD

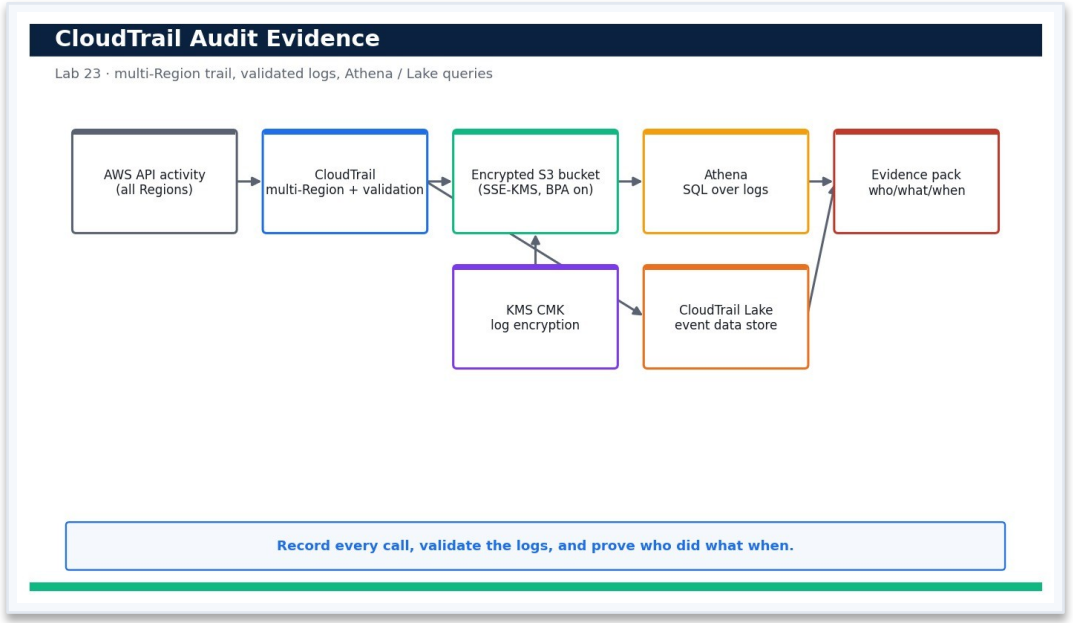
A validated, multi-Region CloudTrail delivering encrypted logs to S3, plus Athena/CloudTrail Lake queries that answer who-did-what-when and an exported evidence pack.

Tools: AWS CloudTrail, Amazon S3, Amazon Athena, CloudTrail Lake, AWS KMS

OUTCOME & EVIDENCE

Outcome: You can answer 'who did what when' from CloudTrail two independent ways (Athena over S3 and CloudTrail Lake SQL), prove the …

Evidence: describe-trails output showing multi-Region true, log-file validation true and a KMS key on devops-trail.



Reference architecture

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Run Continuous Compliance with AWS Config and Security Hub

LAB 24

Day 5 · LO6 · K6, A4

Compliance is not a one-off audit — it must run continuously. In this lab you turn on the AWS Config recorder with managed rules (s3-bucket-public-read-prohibited, encrypted-volumes and more) delivered as a conformance pack, enable AWS …

YOU'LL BUILD

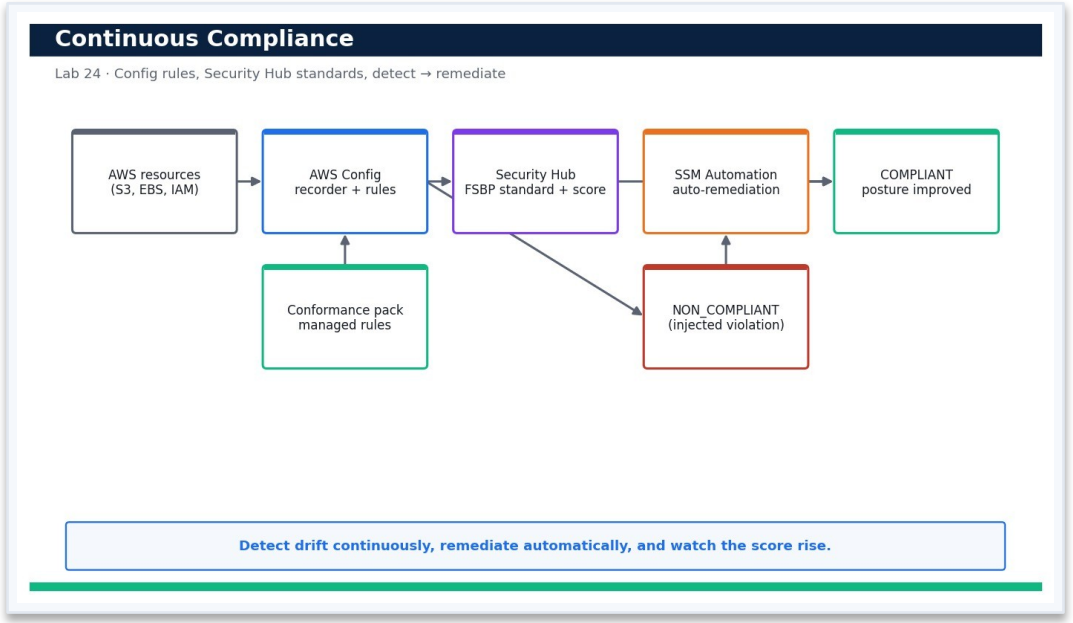
A continuously evaluating AWS Config + Security Hub setup that detects an injected violation as NON_COMPLIANT, auto-remediates it to COMPLIANT, and shows the posture score improve.

Tools: AWS Config, AWS Security Hub, AWS Systems Manager Automation, Amazon S3, AWS CloudFormation, conformance packs

OUTCOME & EVIDENCE

Outcome: You can run a full continuous-compliance loop: inject a real violation, watch AWS Config and Security Hub flag it …

Evidence: describe-configuration-recorder-status showing recording true and get-enabled-standards showing FSBP enabled.



Reference architecture

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Capstone — Blueprint, Well-Architected Review and Evidence Pack for the Whole Delivery System

LAB 25

Day 5 · LO1–LO6 · K1, K2, K3, K4, K5, K6, K7, A1, A2, A3, A4, A5, A6

This capstone ties the entire course together. You produce a single technical blueprint that connects all six domains — SDLC automation (Domain 1), configuration management & IaC (Domain 2), resilient architecture (Domain 3), monitoring & …

YOU'LL BUILD

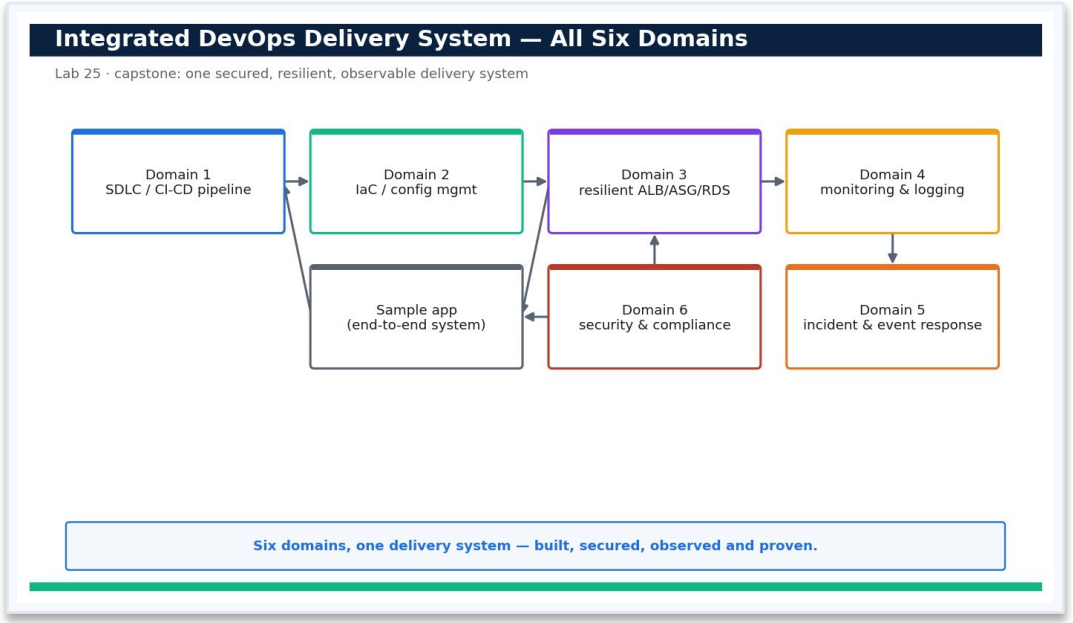
A completed technical blueprint, a Well-Architected pillar review with scores and actions, and an evidence pack that proves the whole secured/resilient/observable delivery system against LO1–LO6.

Tools: AWS CodePipeline/CodeBuild/CodeDeploy, CloudFormation, EC2/ALB/Auto Scaling, RDS, CloudWatch, EventBridge, SNS, Lambda …

OUTCOME & EVIDENCE

Outcome: You can present one coherent delivery system — a commit flowing through CI/CD to a resilient, observed, auto-remediating, secured and …

Evidence: The completed blueprint.md with all six domain sections filled and cross-referenced to the labs.



Reference architecture

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DOMAIN RECAP

Recap — Security and Compliance

- You can now: Write a scoped IAM policy, cap it with a permission boundary, and prove least privilege with IAM Access Analyzer
- You can now: Create a customer-managed KMS key with a key policy, store a secret with rotation, and enforce encryption on S3
- You can now: Stand up a multi-Region CloudTrail with log-file validation and query it to export audit evidence
- You can now: Enable AWS Config managed rules and Security Hub standards, inject a violation and remediate to COMPLIANT
- You can now: Synthesise Labs 1–24 into one secured, resilient, observable delivery system with a blueprint, an operational review and an evidence pack
- Identity & access management — IAM users, roles, policies and permission boundaries enforce least privilege; roles beat long-lived keys.
- IAM at scale — Organizations SCPs, IAM Identity Center and federation manage access across many accounts.
- Data protection — KMS encryption, Secrets Manager, S3 Block Public Access and TLS protect data at rest and in transit.

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PUTTING IT ALL TOGETHER

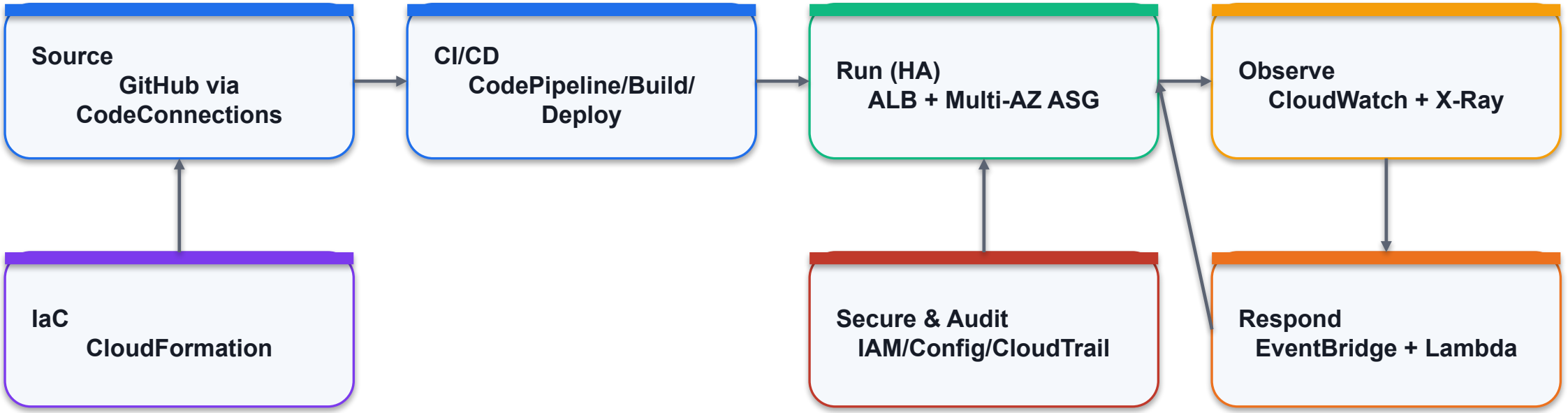
Cross-Domain Integration

The six domains are one system — a single sample application delivered, run and secured end to end

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ALL SIX DOMAINS AS ONE SYSTEM

The Sample Application, End to End



▶ Domains 1–6 map onto one delivery-to-operations loop: build it (1), codify it (2), make it resilient (3), watch it (4), respond (5) and secure it (6).

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ONE CONNECTED SYSTEM · 5 LABS × 5 DAYS

How the 25 Labs Connect

1

Day 1 → Day 2

The CI/CD pipeline built across Labs 1–5 deploys the Infrastructure as Code defined in Labs 6–10.

2

Day 2 → Day 3

The network and app stacks become the highly available, auto-scaling, DR-ready foundation (Labs 11–15).

3

Day 3 → Day 4

The running fleet emits the metrics, logs and events monitored and routed in Labs 16–20.

4

Day 4 → Day 5

Alarms and events drive automated response, then least-privilege, encryption and audit (Labs 21–24).

5

Labs → Capstone

Lab 25 integrates all six domains into one secured, resilient, observable blueprint + evidence pack.

6

One system

Each lab produces an artifact the next builds on — 25 connected labs across 5 days, not isolated exercises.

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TRACING ONE EVENT ACROSS ALL SIX DOMAINS

Capstone Scenario — A change, a failure, a recovery

SITUATION

A developer merges a change that passes tests but degrades checkout latency once at production scale.

ANALYSIS

- Domain 1: the pipeline built and promoted an immutable artifact with a canary stage.
- Domain 3/4: canary p99 latency breaches the SLO; the CloudWatch alarm fires within minutes.
- Domain 5: the alarm triggers CodeDeploy auto-rollback and notifies the on-call via EventBridge/SNS.
- Domain 6: CloudTrail records exactly who changed what, feeding the blameless review.
- Domain 2: the durable fix is encoded in the template/config and re-applied, not hand-patched.

DECISION

No 2am heroics: the system detects, rolls back and records the event automatically. The follow-up hardens the canary threshold and adds a regression test.

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MAPPING A NEED ONTO THE DOMAINS

From Business Requirement to Six-Domain Design

'Ship weekly without downtime'	Domain 1 CI/CD + blue/green; Domain 3 multi-AZ so releases don't drop traffic.
'Recover an AZ loss in minutes'	Domain 3 Multi-AZ ASG + RDS failover; Domain 4 alarms to confirm health.
'Know within a minute when it breaks'	Domain 4 golden-signal alarms; Domain 5 event-driven notify + remediate.
'Prove we are compliant'	Domain 6 Config + CloudTrail; Domain 2 guardrails as code in every account.

TAKEAWAY

Real requirements cut across domains — a competent DevOps engineer composes controls from all six into one coherent design.

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A RECURRING DEVOPS TRADE-OFF

Decision — Build It Yourself or Use a Managed Service?

A — Managed service

Default choice. Less undifferentiated heavy lifting (e.g. RDS vs self-run DB) — at higher unit cost and less control.

B — Self-managed on EC2

Only when you need control the managed option can't give (custom kernel, licensing). More ops burden and patching you own.

C — Serverless-first

For spiky, event-driven work. Pay-per-use and minimal ops — within the service's limits and cold-start profile.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

LO1

Develop an AWS CI/CD pipeline proof of concept.

2

LO2

Identify the technical and practical requirements in an AWS cloud infrastructure setup.

3

LO3

Develop a resilient cloud solution based on AWS cloud components.

4

LO4

Implement regular AWS monitoring and perform necessary troubleshooting.

5

LO5

Demonstrate solutions to incidents and business problems on the AWS cloud.

6

LO6

Prepare a technical blueprint to address security and compliance.

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NEXT STEPS

Continue Your AWS DevOps Journey

1

Rebuild the labs

Redo all six labs from memory until the AWS workflows are automatic.

2

Automate more

Add tests, canary deploys and IaC to your own projects.

3

Well-Architected reviews

Run Well-Architected reviews on real workloads.

4

Keep learning

Explore ECS/EKS, Step Functions, and multi-account governance.

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Assessment

- Written Assessment (SAQ) — 1 hour. Practical Performance (PP) — 2 hours.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers on the LMS at https://lms-tms.tertiaryinfotech.com/.

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then PP — open book
4. Submit your answers on the LMS
5. Sign the Assessment Summary Record

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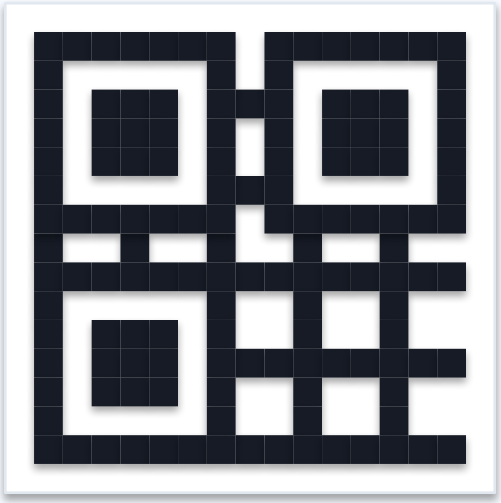
Digital Attendance (Mandatory)

Take the SSG digital attendance at three points on every funded day:

AM	PM	ASSESSMENT
Start of the morning session	Start of the afternoon session	Before the Day 5 assessment

1. The trainer/administrator displays the SSG digital-attendance QR code from the portal.
2. Scan the QR code with your mobile phone camera and submit your attendance.
3. It is mandatory for AM, PM and the Assessment sessions on WSQ-funded courses.

SCAN TO SUBMIT ATTENDANCE



SSG digital-attendance QR — displayed live from the portal by the trainer

Minimum 75% attendance on the SSG Digital Attendance record is required to be eligible for assessment and funding.

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WELL DONE

Thank You!

You can now plan, automate, operate and secure DevOps solutions on AWS across the full delivery lifecycle.

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